16332-24-001

SOW NO. 16332-24-01
29 February 2024

**NOT SENSITIVE**
**IRA-7303**

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STATEMENT OF WORK

TO

SUSTAIN 161kV TRANSFORMERS

ARNOLD ENGINEERING DEVELOPMENT COMPLEX

ARNOLD AIR FORCE BASE, TN 37389-9998

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# 1. SCOPE

1.1. General: The scope of work is to refurbish, rewind, repair, design, supply, and/ or install of the existing 161kV Transformers in multiple locations in the following 161kV substations: Main, PWT, PES, VKF, ETF 1, ETF 2, ETF A, ASTF Airside, and ASTF Exhaust. All work shall be done at Arnold Air Force Base, Tennessee, unless approved by Government. Scope to include, but may not limited to the following:

| Transformer Name | High Level Requirements | Additional Requirements |
|---|---|---|
| ETF-C A2 | New bushings, gaskets, controls, paint, fans, pumps, oil processed | As Needed |
| ETF-C A3 | New bushings, gaskets, controls, paint, fans, pumps, oil processed, rebuild tap changer | As Needed |
| ETF-C A4 | New bushings, gaskets, controls, paint, fans, pumps, oil processed | As Needed |
| ETF-C AA | New bushings, gaskets, controls, paint, oil processed | As Needed |
| ETF-C E1 | New bushings, gaskets, controls, paint, fans, pumps, oil processed | As Needed |
| ETF-C E2 | New bushings, gaskets, controls, paint, fans, pumps, oil processed | As Needed |
| ETF-C E3 | New bushings, gaskets, controls, paint, fans, pumps, oil processed | As Needed |
| ETF-C E4 | New bushings, gaskets, controls, paint, fans, pumps, oil processed | As Needed |
| ETF-C E5 | New bushings, gaskets, controls, paint, fans, pumps, oil processed | As Needed |

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|---|---|---|
| ETF-C E6 | New bushings, gaskets, controls, paint, fans, pumps, oil processed, rebuild tap changer | As Needed |
| ETF-C EA | New bushings, gaskets, controls, paint, oil processed | As Needed |
| PWT1 | Remove pig mat and process oil, internal inspection | As Needed |
| PWT2 | Gaskets, Process Oil and fix oil leaks | As Needed |
| PWT3 | New bushings, gaskets, controls, paint, fans, pumps, oil processed | As Needed |
| ETF2 | Gaskets, pumps, coolers, process oil, internal inspection | As Needed |
| CF1 | Gaskets, Paint, Controls, Bushings, Oil processing, fans, LTC, LA's | HVB 944 Bay LA's and As Needed |
| CF2 | Gaskets, Paint, Controls, Bushings, Oil processing, fans, LTC, LA's | HVB 964 Bay LA's and As Needed |
| ETF5 | Paint, coolers, controls, LA, oil processed | Yard LA's and As Needed |
| ETF7 | Gaskets, paint, fans, controls, bushings, LA, oil processed | Yard LA's and As Needed |
| VKF2 | Gaskets, paint, fans, controls, bushings, oil processed | Yard LA's and As Needed |
| PWT5 | Gaskets, spot paint, controls, oil processed | As Needed |
| PES6 | Gaskets, paint, fans, controls, bushings, LA, oil processed | Yard LA's and As Needed |

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Potential New Transformers:

| |
|---|
| PES 5 |
| ETF 5 |
| ETF-C A3 |
| ETF-C E6 |
| ETF 7 |
| ETF 2 |
| PWT 3 |
| PWT 2 |
| ETF 6 |

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## 2. APPLICABLE DOCUMENTS

2.1. Government Documents:

2.1.1. AEDC Environmental Policy: https://media.defense.gov/2022/Nov/29/2003122404/-1/-1/1/ARNOLD_AFB_ENVIRONMENTAL_COMMITMENT_STATEMENT_15JUL22.PDF

2.1.2. AEDC Safety, Health, and Environmental (SHE) Standards latest version, provided in Appendix H.

| | | |
|---|---|---|
| A. | A6 with Supplement | User and Subcontractor Safety (Mandatory Documents/Submittals), |
| B. | A9 | Hazard Communication. |
| C. | A10 | Job Safety Analysis. |
| D. | B1 | Master Work Permit. |

| | | |
|---|---|---|
| E. | B2 | Lockout/Tagout (LOTO). |
| F. | B3 | Control of Hazardous Areas Using Safety Signs, Tags, and Barricades. |
| G. | B4 | High Voltage Electrical Work. |
| H. | B5 | Confined Spaces. |
| I. | B6 | Low Voltage Electrical Systems and Related Work Practices. |
| J. | C2 | Contaminated Equipment. |
| K. | D5 | Hoisting Devices. |
| L. | D6 | Aerial Work Platforms. |
| M. | D8 | Portable Power Tools. |
| N. | D9 | Fixed and Portable Ladders. |
| O. | D10 | Scaffolding. |
| P. | E7 | Asbestos. |
| Q. | E11 | Oil Pollution Prevention and POL Storage Tank Management. |
| R. | E17 | Oil and Hazardous Substance Spill Response. |
| S. | E18 | Hazardous Material Management Program. |
| T. | E19 | Lead and Heavy Metals. |

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|---|---|---|
| U. | F2 | Personal Protective Equipment. |
| V. | F5 | Hazardous Noise and Hearing Conservation (NRR Method for Estimating the Adequacy of Hearing Protector Attenuation). |
| W. | F6 | Fall Protection. |

2.1.3. <u>Code of Federal Regulations</u>:

| | | |
|---|---|---|
| A. | 29 CFR 1910.134 | Respiratory Protection. |
| B. | 29 CFR 1910.1200 | Hazard Communication. |
| C. | 29 CFR 1926.55 | Gases, Vapors, Fumes, Ducts, and Mists. |
| D. | 29 CFR 1926.57 | Ventilation. |
| E. | 29 CFR 1926.59 | Hazard Communication. |
| F. | 29 CFR 1926.62 | Lead Standard. |
| G. | 29 CFR 1926.1101 | Asbestos. |
| H. | 29 CFR 1926.1407 | Power Line Safety (up to 350 kV)-Assembly and Disassembly. |
| I. | 29 CFR 1926.1408 | Power line safety (up to 350 kV)-Equipment Operations. |
| J. | 40 CFR 61 National | Emission Standards for Hazardous Air Pollutants (NESHAP). |
| K. | 40 CFR 82 Subpart F | Protection of Stratospheric Ozone, Recycling and Emissions Reduction. |
| L. | 40 CFR 82.161 | Technician Certification. |
| M. | 40 CFR 112 | Oil Pollution Prevention. |
| N. | 40 CFR 260 | Hazardous Waste Management Systems: General. |
| O. | 40 CFR 261 | Identification and Listing of Hazardous Waste. |

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P. 40 CFR 262 — Standards Applicable to Generators of Hazardous Waste.

Q. 40 CFR 279 — Standard for the Management of Used Oil Authority.

R. 40 CFR 370 — EPA Hazardous Chemical Reporting and Community Right to Know Requirement.

S. 49 CFR 172 — Department of Transportation (DOT) Regulations for Use of Hazardous Materials Tables and for Communication.

T. 49 CFR 178 — DOT Specifications for Packaging.

2.1.4. Environmental Protection Agency:

A. EPA SW-846 — Proposed Sampling and Analytical Methodologies for Additions to Test Methods for Evaluating Solid Waste: Physical/Chemical Methods, 2021.

2.1.5. Public Law:

A. PL 101-637 — The Asbestos School Hazard Abatement Reauthorization Act (ASHARA), 1990.

2.1.6. Tennessee Department of Environment and Conservation Standard:

A. Chapter 1200-01-20 — Asbestos Accreditation Requirements, 2009.

B. Chapter 1200-3-11-02 — Hazardous Air Contaminants, Asbestos, 2018.

2.1.7. Specifications: The following CSI Specification sections are included in Appendix A as AEDC specific requirements and are listed as follows:

A. Appendix A.1 – 01 10 00 – Job Conditions at AEDC.

B. Appendix A.2 – 01 33 00 – Submittal Procedure.

C. Appendix A.3 – 01 74 19 – Construction Waste Management and Disposal.

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D. Appendix A.4 – 02 08 00 – Asbestos Removal.

E. Appendix A.5 – 02 08 50 – Lead and Heavy Metal Removal

F. Appendix A.6 – 09 90 00 – Painting and Coating.

2.1.8. Drawings:

A. Reference Drawings – Appendix E.

B. Nameplate Data – Appendix F.

C. Photographs – Appendix G.

2.2. <u>Non-Government Documents:</u>

2.2.1 <u>American Society of Safety Professionals:</u>

| | |
|---|---|
| A. ASSP Z9.2-18 | Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems. |

2.2.2 <u>American Society of Mechanical Engineers:</u>

| | |
|---|---|
| A. ASME A13.1-23 | Scheme for the Identification of Piping Systems. |
| B. ASME B16.5-20 | Pipe Flange and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard. |

2.2.3 <u>ASTM International:</u>

| | |
|---|---|
| A. ASTM D877-19 | Standard Test Method for Dielectric Breakdown Voltage of Insulating Liquids Using Dish Electrodes. |
| B. ASTM D923-23 | Standard Practices for Sampling Electrical Insulating Liquids. |
| A. ASTM D924-23 | Standard Test Method for Dissipation Factor (or Power Factor) and Relative Permittivity (Dielectric Constant) of Electrical Insulating Liquids. |
| B. ASTM D971-20 | Standard Test Method for Interfacial Tension of Oil Against Water by the Ring Method of Oil |

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| C. | ASTM D974-22 | Standard Test Method for Acid and Base Number by Color-Indicator Titration. |
| D. | ASTM D1275-24 | Standard Test Method for Corrosive Sulfur in Electrical Insulating Oils. |
| E. | ASTM D1298-23 | Standard Test Method for Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum and Liquid Petroleum Product by Hydrometer Method-API Designation. |
| F. | ASTM D1500-17 | Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale). |
| G. | ASTM D1524-22 | Standard Test Method for Visual Examination of Used Electrical Insulating Oils of Petroleum Origin in the Field. |
| H. | ASTM D1533-20 | Standard Test Method for Water in Insulating Liquids by Coulometric Karl Fischer Titration. |
| I. | ASTM D1816-19 | Standard Test Method for Dielectric Breakdown Voltage of Insulating Oils of Petroleum Origin Using VDE Electrodes. |
| J. | ASTM D2668-13 | Standard Test Method for 2,6-DitertiaryButyl Paracresol and 2,6-Ditertiary-Butyl Phenol in Electrical Insulating Oil by Infrared Absorption. |
| K. | ASTM D2945-09 | Standard Test Method for Gas Content of Insulating Oils. |
| L. | ASTM D3487-17 | Standard Specification for Mineral Insulating Oil Used in Electrical Apparatus. |
| M. | ASTM D7151-23 | Standard Test Method for Determination of Elements in Insulating Oil. |

2.2.4 <u>Institute of Electrical and Electronic Engineers:</u>

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A. IEEE C57.19.00-23 General Requirements and Test Procedure for Outdoor Power Apparatus Bushings.

B. IEEE C57.19.01-17 Performance Characteristics and Dimensions for Outdoor Apparatus Bushings.

C. IEEE C57.93-19 Guide for Installation and Maintenance of Liquid-Immersed Power Transformers

D. IEEE C57.106-15 Guide for Acceptance and Maintenance of Insulating Oil in Equipment.

E. IEEE 4-13 Standard Techniques for High-Voltage Testing.

F. IEEE 315-93 Graphic Symbols for Electrical and Electronics Diagrams.

2.2.5 International Electrotechnical Commission:

A. IEC 60068-2-1-07 Environmental Testing – Cold.

B. IEC 60068-2-2-07 Environmental Testing – Dry Heat.

2.2.6 National Electrical Manufacturers Association:

A. NEMA 250-20 Enclosures for Electrical Equipment (1000 Volts Maximum).

B. NEMA ICS 1-22 Industrial Control and Systems: General Requirements.

2.2.7 National Electrical Testing Association:

A. NETA MTS-23 Maintenance Testing Specifications.

2.2.8 The Society for Protective Coatings:

A. SSPC Guide 6-21 Guide for Containing Surface Preparation Debris Generated During Paint Removal Operations.

B. SSPC PA 1-16 Shop, Field, and Maintenance Painting of Steel.

C. SSPC PA 2-22 Measurement of Dry Coating Thickness with Magnetic Gages.

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|---|---|---|
| D. | SSPC SP1-16 | Solvent Cleaning. |
| E. | SSPC SP2-18 | Hand Tool Cleaning. |
| F. | SSPC SP3-18 | Power Tool Cleaning. |
| G. | SSPC SP6-07 | Commercial Blast Cleaning. |
| H. | SSPC SP12-02 | Surface Preparation and Clean of Metals by Waterjetting Prior to Recoating. |

2.2.9 Underwriters Laboratories, Inc.(UL):

| | | |
|---|---|---|
| A. | UL 50-20 | Enclosures for Electrical Equipment. |
| B. | UL 467-22 | Grounding and Bonding Equipment. |
| C. | UL 586-22 | High-Efficiency, Particulate, Air Filter Units. |

## 3. REQUIREMENTS

3.1. General:

3.1.1. The Contractor shall provide all labor, materials, and equipment required for disconnecting and reconnecting of all primary 161kV high voltage conductors and secondary medium voltage conductors required for the on-site field survey, bushing replacement, and transformer refurbish service. Contractor shall have capability to rewind transformer as necessary.

3.1.2. Provide a single point of contact or project manager to serve as on-site field service support for supervision of bus duct replacement, repair and refurbishment, transformer reconditioning, oil processing, startup, and commissioning of the refurbished transformer. Provide a Field Application Engineer, environmental coordinator, and safety professional.

3.1.3. Provide a list of personnel assigned to this project with telephone numbers and email addresses. Identify the team design lead and respective design discipline of each member of the team.

3.1.4. Perform all work in accordance with Appendix A.1, Job Conditions at AEDC.

3.1.5. All meetings shall be conducted, as necessary, as agreed to by the Government.

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3.1.6. A Microsoft Team's meeting between the Contractor and the Government Representative shall be held at least weekly for technical discussions and updates.

3.1.7. Contractor shall submit final construction drawings and final as-built drawings to the Government for approval.

3.1.8. Final design documentation (drawings) shall become the sole property of the Government.

3.1.9. The Contractor shall field verify existing site conditions.

3.1.10. Schedule and outages:

A. Submit plan to initially test the transformer within 3 days of notice to proceed. The Government expectation for initial electrical testing is defined in this SOW. The test plan shall comply with this SOW or deviations shall be approved by the Contracting Officer.

B. Submit a design plan and schedule within 10 days of notice to proceed.

C. Contractor shall submit request for outages 30 days in advance. The Government will provide outages for additional equipment upon request. This includes but is not limited to the following services:

1. All work or repairs that require a crane to lift equipment or personnel. Submit critical lift plan for review with request for outage in accordance with AEDC SHE Standard D5.
2. All work required in the 161kV substation control room.

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3.1.11. Safety:

A. All work shall comply with AEDC SHE Standards A6 with supplement, A9, A10, B1, B2, B3, B4, B5, B6, C2, D5, D6, D8, D9, D10, E7, E11 E17, E18, E19, F2, F5, and F6. Contractor shall determine if additional SHE Standards are required. Submit a Safety Plan in accordance with AEDC SHE Standard A6 prior to beginning onsite work be performed in accordance with AEDC SHE Standards.

B. All work stoppages due to unsafe conditions will be at the Contractor's expense and will not be grounds for outage extension.

C. All work or repairs that require a crane to lift equipment or personnel shall comply with the provisions of 29 CFR 1926.1407 and 1408, which states that cranes may be used within a minimum clearance of 15-feet from energized 161kV and shall comply with the requirements of a critical lift.

D. Work inside transformer tanks is considered to be a confined space and shall be in accordance with AEDC SHE Standard B5. Submit confined space procedure within 3 days of notice to proceed. Comply with the confined space procedure whenever personnel are inside the tank.

E. Prepare Job Safety Analysis per AEDC SHE Standard A10.

3.1.12. Do not order materials or start refurbishment work until receiving approval of the final submitted design plan, schedule, and drawings unless otherwise approved by the Government Representative.

3.1.13. Provide new components and devices, complete, with all necessary materials for installation. Provide materials, components, parts, and equipment to be incorporated into the work, which are manufactured in accordance with recognized and accepted standards such as IEEE, ASTM, and NEMA or similar. Where possible, use materials, components, parts, and equipment which have been tested and listed by an independent agency such as UL. All components, parts and equipment shall be designed for operation 24 hours per day, seven days per week for 25 years. Reject damaged and inferior materials, components, parts, and equipment. All materials used shall be new and unused. The Government reserves the right to inspect all material for discrepancies, damage, and inferiority once the material arrives on-site.

3.1.14. Allow a minimum of 14 calendar days for the Government to review submissions made by the contractor.

3.1.15. Reference drawings of the facilities are attached to the SOW in Appendix E. The contractor shall be responsible for verifying the accuracy of Government-furnished drawings.

3.1.16. Disconnect the transformer as required, including high voltage connections, auxiliary power, medium voltage power, control cables and wires, relay wiring, and conduits.

3.1.17. Upon completion of vacuum filling, re-connect the transformer, including high voltage connections, medium voltage connections, auxiliary power, control cables and wires, relay wiring, and conduits.

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3.1.18. For all repairs, perform using factory trained and qualified filed engineer and technician. Submit certification for qualifications to perform repairs. Repair shall include but not limited to:

A. Necessary internal repairs.

B. Necessary external repairs.

3.2. Survey:

3.2.1. The Contractor shall perform a detailed on-site field survey to review the current condition of the transformer(s) and verify the accuracy of the bushing data in Appendix E and F and all reference drawings in Appendix E as required to refurbish the transformer and replace wiring as per the specifications. As part of the on-site field survey, inspect and review existing wiring installation, document exact detail of the field wiring and hardware connected to the transformer control panels. During the on-site field survey, the Contractor shall verify the information on Appendix E and F and complete internal inspections required to gather dimensional data for the replacement high voltage, neutral, and low voltage bushings.

3.2.2. Provide tanker, pumping equipment, oil processing and filter equipment, and oil spill contingency to drain oil to prepare for initial inspection per Appendix D. Submit certification cleanliness of actual tanker to be used prior to pumping oil.

3.2.3. Provide dry air supply to maintain positive pressure during the oil removal process. Provide dry air throughout the internal inspection phase.

3.3. Internal Inspection:

3.3.1. Perform internal inspection using factory trained and qualified field engineer and technician. Submit certification for qualifications to perform inspection.

3.3.2. The internal inspection shall include but not be limited to the following: A.

Leads and lead structure(s).

B. De-energized tap changer contacts and operating mechanism.

C. Current Transformers (CT)s and associated mountings and wiring.

D. Tightness and integrity of all the coils.

E. Transformer core.

F. All low voltage bushings.

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G. Internal tank structure.

3.3.3. Document inspection and findings with photographs taken with Government provided camera.

3.3.4. Discuss findings with Government within 4 hours of completing the inspection.

3.3.5. Submit preliminary written report with detailed findings and anomalies observed within 24 hours of completion of inspection.

3.3.6. Submit recommendations for revisions to initial test plan within 3 days of completion of the internal inspection.

3.3.7. Perform an engineering evaluation to determine the need to replace the original tap changer with a new unit with appropriate ratings to meet or exceed the original equipment specifications. Submit evaluation with recommendation. Do not perform additional work unless approved by Government.

3.3.8. Test oil in accordance with Appendix D. Submit written results within three days of drawing the sample.

3.3.9. Perform electrical test of the transformer in accordance with the requirements in this SOW. Discuss findings with Government within 24 hours of completing the testing. Submit written electrical results with recommendations for all repairs outside the scope of this statement of work within three days of completion of the tests.

3.3.10. Evaluate and replace the control cables if approved by the Government.

3.4. <u>Baseline Testing</u>:

3.4.1. Perform baseline electrical testing on the transformer in accordance with NETA MTS prior to starting the repair. Where the manufacturer's requirements are more stringent than the standard NETA requirements, the manufacturer's requirements shall prevail. Submit a detailed report of the test results as part of the consolidated transformer test report. Inform the Government Representative following completion of testing of all test values that fall outside the recommended range. The electrical testing shall include but not be limited to the following tests:

A. External visual inspection of all transformer devices and general conditions. Note any nitrogen piping that appears to have been recently replaced and in good condition.

B. Insulation power factor.

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C. Insulation resistance.

D. DC winding resistance on all tap positions for all windings.

E. Winding excitation on all tap positions, verify proper test pattern.

F. Transformer turns ratio on all tap positions for all windings.

G. Leakage reactance.

H. Each current transformer ratio and polarity.

I. Sweep frequency response analysis.

3.4.2. Submit preliminary design drawings for repairs within 90-days of notice to proceed. Design review documentation shall include Contractors', manufacturers or fabricators' drawings and shall be submitted to the Contracting Officer.

3.4.3. Within 25-days of conclusion of the on-site field survey submit final shop drawings detailing demolition and refurbishment work, wiring diagrams, and schematics in accordance with IEEE 315, using drawings in Appendix E as a basis. Drawings shall include wiring diagrams, schematic diagrams, one-line and three-line diagrams, mounting details, outline, and complete dimensional information. The drawings shall also include all bushings, gauges, mechanical relief device, control cabinet power, controls, auxiliary, and alarm components. Provide drawings meeting the following requirements:

A. All drawings shall clearly indicate the location and arrangement of all materials and equipment.

B. All drawing submittals shall include three electronic copies on CD-ROM in AutoCAD 2020 format or later format and three legible and reproducible paper copies. Drawings shall be completely legible when the printed size is 11 by 17 inches.

C. Provide a project title block for all drawings provided. The title block shall include the drawing number and description for each drawing sheet.

D. Provide detailed schematic wiring and fabrication shop drawings for the custom designed transformer control cabinet enclosures (Refer to Appendix C for detailed requirements), the transformer temperature cooling system remote operator control panels and the nitrogen regulator enclosure which accurately and completely describe the control circuits, all wire designations, terminal points and labels, control devices and device contact points shown and labeled with proper ANSI device numbers.

E. Provide detailed wiring and connection diagram for all components that will be wired to terminals in the transformer control panel. This includes detailed transformer component wiring connections and wiring connections in the transformer control panel. Drawings shall include but not be limited to:
   1. Conduit installation drawings or specifications for connection of transformer components to the transformer control panel.
   2. Detailed installation drawings and specifications for mounting the transformer control cabinets.
   3. Detailed operational specifications of transformer components to be wired. For example, a high temperature switch the detail shall indicate "OPERATES ON TEMPERATURE RISE AT 100 DEGREES-F".
   4. Detailed installation drawings and specifications for mounting the remote operator control panel.
   5. Detailed conduit and wiring installation drawings and specifications for connection of the transformer control cabinets, the remote control panels, and the nitrogen regulator enclosures. All wiring and installation drawings shall include cable identification labels.
   6. Detailed fabrication drawings for the oil winding temperature indicator brackets.
   7. Detailed fabrication drawings for the oil flow indicator brackets.
   8. Detailed wiring and installation drawings required to replace the field wiring detailed in this SOW.

3.4.4. Provide detailed bill of materials (BOM) drawing sheet(s) listing each specific part with brand name, part number, quantity, detailed description, and reference drawing for sheet(s) where part is shown. This drawing shall have part numbers that can be referenced on a drawing sheet(s) with a part number symbol (number). The detailed BOM for hardware shall include the manufactures documented Mean Time Between Failure (MTBF), and Mean Time to Repair (MTTR) information for all components and assemblies.

3.4.5. Provide detailed three-phase wiring schematic for the replacement transformer cooling system controls.

3.4.6. Provide a detailed logic flow diagram for the transformer cooling system controls. The logic flow diagram shall be the basis for the logic programming of the Intelligent Electronic Device (IED) detailed in Appendix C.

3.4.7. Provide a drawing sheet for the replacement diagrammatic nameplate detailed within this SOW.

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3.4.8. Provide a drawing sheet for the replacement high voltage bushings, the low voltage neutral bushings, and the low voltage phase bushings.

3.5. Oil Processing:

3.5.1. Provide tanker, pumping equipment, oil processing and filter equipment, and oil spill contingency to drain oil for necessary on-site repairs per the procedures in Appendix D. Submit certification of cleanliness of actual tanker to be used prior to pumping oil if using tanker different than previous.

3.5.2. Provide dry air supply to maintain positive pressure during the oil removal process. Adjust oil flow rate to maintain a slight positive pressure on the tank at all times. Provide dry air throughout the on-site repair phase.

3.5.3. Provide new Type II oil for all oil filled cable compartments, tap changers, and flushing oil. New oil shall meet the requirements listed in ASTM D3487. Submit a certified test report to the Contracting Officer on the characteristics of the new oil prior to off-loading to verify that the oil meets the specifications. Provide tanker, pumping equipment, oil processing and filter equipment, and oil spill contingency in accordance with the procedures in Appendix D. Government storage is within one mile of the job location where the oil can be stored. If that storage method is chosen, deliver the oil to the Government storage vessel.

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3.6. Transformer Control Cabinet:

3.6.1. Design, fabricate and provide a new transformer control cabinet with enclosures and the transformer insulation oil cooling system motor controls in accordance with Appendix C. Exceptions include transformers: ETF #2, PWT #1. PWT #2 that have already had the control cabinet replaced.

3.6.2. Within 10-days following fabrication the Contractor shall provide inspection and testing of the transformer control cabinet enclosures and transformer temperature cooling system remote operator control panels. At least 14 days before scheduled inspection and testing submit plans and details for on-site preshipment factory testing and inspection. The purpose of this inspection is to verify that the design and fabrication of the cabinets meets all requirements. The Contractor submitted plans shall include but not be limited to a detailed agenda for functional testing and documentation review and list the proposed date and site of testing. Government reserves the right to witness testing.

3.6.3. Provide power inverter suitable to operate the entire load of the control cabinet controls. Normal source shall be 120VAC with capability to switch to 125VDC station battery upon loss of 120VAC. For example, the existing power inverters are Majorpower and Majorsine. Notify the Government of the required circuit ampacity with the initial design submittal. The Government will provide the location for the new circuit. Provide all new components include 2pole knife switch, fuses, wire, and install the new circuit as required. Work shall be performed by the contractor to install the power inverter and all necessary components.

3.7. Cooling System Refurbish Service:

3.7.1. Remove and refurbish transformer insulating oil cooling system assembly.

3.7.2. Cooling Fans: Disassemble and replace fans, fan motors, fan guards, fan brackets, and all fan wiring. Replacement bolts, mounting brackets, and all hardware shall be fabricated from non-corrosive material or plated to prevent corrosion. Relinquish the existing undamaged fan assemblies to the Government.

3.7.3. Steam clean to remove oil, dirt, and grime from radiators.

3.7.4. Perform an engineering evaluation on the need to replace the oil pumps and submit evaluation with recommendation.

3.7.5. Replace drain valves suitable for draining oil from the cooler pumps as required. Valves shall be for use with insulating oil and shall be capable of holding hot oil without leaking. Valves shall have National Pipe Threads. Provide plugs for valves.

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3.7.6. Clean all surfaces and sand as needed. Radiators shall be flow coated and receive 2 coats of epoxy primer and 1 coat of final coating in accordance with Appendix A.6.

3.7.7. Test all the cooling fans and pumps individually and all together. Perform an engineering evaluation on the recommended settings for forced oil initiation, forced air initiation, and alarm settings for the oil and winding temperature gauges. Verify that the fans and pumps turn on per each of the recommended settings via each gauge. Verify that the SCADA alarms are initiated for each recommended temperature level. Submit a certified test report on all testing after the testing is complete as part of final transformer test reports.

3.7.8. Replace existing insulating oil cooling system flow indicators. Remove the existing oil cooling system flow indicators and install new oil flow indicators. Each device shall include the following features and options:

A. One set of normally open and normally closed form C alarm contacts with interrupting rating of 0.5-amps non-inductive suitable for use on 125VDC. The Contractor shall install wiring connections from the alarm contacts to a terminal block for the IED in the transformer control panel. B. NEMA 4X housing to protect wiring connections.

3.8. <u>Tap Changer:</u>

3.8.1. Rebuild each load tap changer (LTC).

A. Provide and erect scaffolding as required.

B. Perform visual inspection of LTC and note all abnormalities.

C. Drain and excess oil from LTC.

D. Replace door gasket.

E. Wipe down the LTC active part with clean rags.

F. Check bypass contacts.

G. Check selector contacts.

H. Replace contacts, all replaceable components, and broken components.

I. Perform timing measurements (used to determine excessive backlash in drive train).

J. Full motor drive inspection.

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K. Perform a sequence check on the LTC.

L. Perform an operational check in the manual and electrical modes.

M. Provide new oil and fill LTC compartment thru filter press.

N. Perform check for leaks.

O. Provide a separate written report of inspection results and work.

P. Replace tap changer controllers with new modern controllers. Provide one spare controller of each model installed. Provide one complete set of spare parts that includes all replaceable components for each LTC.

3.8.2. If rebuild isn't possible, recommend replacement with the initial bid.

3.9. <u>Auxiliary Equipment:</u>

3.9.1. Replace the existing nameplate with a new stainless-steel nameplate permanently etched in black ink that shows all the new information, including weights, connections, gallons of oil, impedance, etc. The new nameplate, as a minimum, will have the same information as the original. Discuss with Government Representative on each transformer to determine if nameplate replacement is necessary.

3.9.2. Evaluate C-air and C-exhaust transformer's neutral current transformers if the neutral current transformer looks like it has been recently replaced, inform the Government representative so a decision can be made about replacement. If the units are not replaced, turn the new replacement units over to the Government. Otherwise, replace the two existing neutral current transformers. Neutral current transformers shall be outdoor type rated 15kV, 110kV BIL, 60 Hertz, 50/100 primary amp rating to 5-amp secondary ratio, for example our existing neutral current transformers are ABB type KOR-11, Style 7524A13G05 . Verify the existing ratio before disconnecting then connect new current transformers secondary wiring for the proper ratio.

3.9.3. Provide a dial-type liquid temperature indicating thermometer with its sensing element located in the path of the hottest oil. Locate the thermometer on the transformer tank approximately five feet above the base of the transformer. Provide thermometer dials with a range of 0 to 120 degrees C and three adjustable circuit-closing alarm contacts suitable for use with an ungrounded 125-volt dc circuit and with a resettable drag hand to capture the hottest temperature experienced. Analog 4-20 mA output proportional to temperature. Install wiring connections from the temperature indicating device analog outputs to a terminal block in the control cabinet then connect to the SEL-2414. Program the SEL-2414 to accept the analog input and display the oil temperature. In

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addition to the liquid temperature indicator, provide RTD sensors to detect top oil temperature and ambient temperature and connect the sensors to the SEL-2414. Program the SEL-2414 to accept the RTD inputs.

3.9.4. Replace the transformer main tank oil level indicator. The indicators shall be a magnetic-type oil gage equipped with low-level alarm contacts, NEMA form C with an interrupting rating of 0.5 amps (minimum) non-inductive suitable for use on 125VDC. The Contractor shall install wiring from the alarm contacts to a terminal block in the transformer control panel. The Contractor shall include details for the oil level indicator on the detailed BOM included in the design drawings. The oil level indication device shall include:

A. 6-inch dial face with lever (radial) or heavy-duty gear driven arm float arm, marked to show 25 degrees Celsius oil level and the minimum and maximum oil levels.

B. NEMA 4X housing to protect electronics.

C. Analog 4-20 mA output proportional to level. Install wiring connections from the temperature indicating device analog outputs to a terminal block in the control cabinet then connect to the SEL-2414. Program the SEL2414.

3.9.5. Replace the sudden-pressure relay mounted to detect a sudden rise in oil pressure in the tank. Provide a relay with at least one set of normally open and one set of normally closed contacts suitable for use on a 125VDC ungrounded circuit. The sudden-pressure relay shall be connected to the transformer oil space through a valve approved by the Government. The sudden-pressure relay shall be wired to a seal-in relay provided by the Contractor that shall be installed in the transformer control cabinet. Provide a seal-in relay with not less than one set of normally open and one set of normally closed alarm contacts and one set of normally open trip contacts with surge protection suitable for use on 125VDC. The seal-in relay contacts shall be rated to make 20-amps, carry 10-amps, and break 0.5-amps (minimum) resistive. Include details for the replacement sudden pressure relay, seal-in relay and the connection valve hardware on the detailed BOM included in the design drawings.

3.9.6. Replace the mechanical pressure relief device (MRD). Design the MRD to be self-sealing to protect the tank against an explosion due to arcing within the tank. Design the MRD to permit vacuum filling of the tank and to minimize oil discharge and to exclude air and water after it opens and shall be equipped with a visible flag to indicate operation. Equip the MRD with not less than one set of normally open and one set of normally closed alarm contacts with interrupting rating of 0.5 amps (minimum) non-inductive suitable for use on 125VDC. Install wiring connections from the pressure relief device to a terminal block in the control

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cabinet. Include details for the replacement device on the detailed BOM included in the design drawings.

3.9.7. Replace all winding temperature indication equipment devices. Include details for the replacement winding temperature indicator on the detailed bill of materials (BOM) included in the design drawings. All replacement winding temperature indicating equipment shall be mounted on a custom mounting bracket located directly above or beside the transformer control cabinet enclosure. The mounting height of winding temperature indicators shall be 84-inches maximum (unless approved by the Government). The mounting brackets shall be designed, fabricated and provided by the Contractor. The fabricated mounting brackets shall be painted in accordance with Appendix A.6. The winding temperature indication equipment device shall include:

A. 6-inch thermometer dials with a range of 0-160 degrees Celsius with a resettable drag hand to capture the hottest temperature experienced.

B. Four field adjustable switches with visible settings.

C. One set of normally open and normally closed form C alarm contacts with interrupting rating of 0.5-amps non-inductive suitable for use on 125VDC. The Contractor shall install wiring connections from the temperature indicating device alarm contacts to a terminal block in the transformer control cabinet.

D. Analog 4-20mA output proportional to temperature. The Contractor shall install wiring connections from the temperature indicating device analog outputs to a terminal block in the transformer control cabinet.

E. E. NEMA 4X housing to protect electronics.

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3.9.8. Replace the oil temperature indication equipment device. The Contractor shall include details for the replacement oil temperature indicator on the detailed BOM included in the design drawings. The replacement winding temperature indicating equipment shall be mounted on the custom mounting bracket located directly above or beside the transformer control cabinet enclosure. The mounting height of oil temperature indicators shall be 84-inches maximum (unless approved by the Government). The oil temperature indication equipment device shall include:

A. 6-inch thermometer dials with a range of 0-120 degrees Celsius with a resettable drag hand to capture the hottest temperature experienced.

B. Three field adjustable switches with visible settings.

C. One set of normally open and normally closed form C alarm contacts with interrupting rating of 0.5-amps non-inductive suitable for use on 125VDC. The Contractor shall install wiring connections from the temperature indicating device alarm contacts to a terminal block in the control cabinet.

D. Analog 4-20mA output proportional to temperature. The Contractor shall install wiring connections from the temperature indicating device analog outputs to a terminal block in the control cabinet.

E. NEMA 4X housing to protect electronics.

3.9.9. Replace all existing external conduit and wiring between all transformer protective devices and components and between the transformer protective devices and components and the transformer control cabinet (unless approved by the Government). Use rigid galvanized conduit and/or liquid-tight metallic flexible conduit for all replacement conduit. All connections to transformer protective devices and components with type NPT thread connections shall be made with liquid-tight metallic conduit and stainless-steel connectors. All wiring for transformer protective devices with NPT conduit connections shall be installed in conduit, unless approved by the Government. All other transformer protective devices shall be connected with outdoor rated type cable and connectors with water-tight seal and fabricated with corrosion resistant materials. All conduit connections to the transformer control panel enclosure shall be made with liquid-tight flexible metallic conduit and PVC coated connectors. All conduit bodies used for pulling conductors or connection conduits shall be cast-iron construction and painted in accordance with Appendix A.6.

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3.9.10. Replace all gaskets for all oil-to-air openings and valves on outside of the transformer and all gaskets for oil-to-air transformer secondary bushing connections that are leaking; do not replace gaskets that show no evidence of leaking.

3.9.11. Nitrile rubber shall be used for all gasket joints having a recessed gasket grove. Include details for each replacement gasket on the detailed BOM included in the design drawings.

3.9.12. Replace the following gaskets:

A. High voltage phase bushings.

B. Low voltage neutral bushings.

C. Manholes and hand-holes.

D. Pass through for current transformer wiring.

E. Pass through for temperature sensing elements.

F. Plug on stiffener on Northwest corner of main tank.

G. Drain plugs on both oil pumps.

H. Replace seals and repack leaking gate and globe valves.

I. All gauges replaced by this contract.

J. Low voltage cable housing and bushings.

3.9.13. Many of the existing fasteners for hardware noted above to be re-gasketed are rusted or damaged. Replace all fasteners including bolts, nuts and washers for all hardware to be re-gasketed. All replacement fasteners shall be exact replacement or full equivalent to existing fastener hardware. All replacement fastener hardware shall be fabricated from corrosion resistant material or plated with corrosion resistant material. The details for all replacement fasteners shall be included on the BOM drawing sheets.

3.9.14. Provide station class lightning arrestors mounted to the top of the transformer tank within three feet or as close as possible to the high-voltage terminals as determined to be required for each transformer between the Contractor and Government representative. The lightning arrestors shall have a Maximum Continuous Operating Voltage (MCOV) rating of 106kV and for nominal system voltage of 161kV. Existing lightning arrestors are ABB type Style Q144SC115C.

3.9.15. Replace the existing lightning arresters in the following substations below. Provide new mounting beams or modify the existing mounting as needed for proper height. Provide proper high voltage, ground connections, and mounting hardware. The lightning arrestors shall have a Maximum Continuous Operating Voltage (MCOV) rating of 106kV and for nominal system voltage of 161kV. Existing lightning arrestors are ABB type Style Q144SC115C.

A. ETF 1 Substation - one set of lightning arresters.

B. ETF 2 Substation– one set of lightning arresters.

C. ETF 7 Substation - one set of lightning arresters.

D. ETF- C Exhaust Substation– three sets of lightning arresters.

E. Main Substation – Bay 944 - one set of lightning arresters.

F. Main Substation – Bay 964 – one set of lighting arresters.

G. PES Substation – one set of lighting arresters.

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H. VKF Substation - one set of lighting arresters.

3.9.16. Provide all hardware, materials, and labor to replace the high voltage phase bushings, the low voltage neutral bushings, and the low voltage phase bushings if determined to need replacement by the Contractor. Discuss findings with Government Representative before proceeding with decision for bushing replacement. Provide suitable termination to match existing connections. The new bushings shall meet or exceed all electrical requirements of the existing bushings listed in Appendix E and F. The Basic Insulation Levels (BIL) for replacement bushings shall be 750kV BIL for the 161kV bushings and 110kV BIL

for the low voltage and neutral bushings. The electrical requirements of the new bushings shall be designed in accordance with IEEE C57.19.00 and IEEE C57.19.01 for application on power transformers. The Contractor shall include details for each replacement bushings on the detailed BOM included in the design drawings. The bushings shall be integrated with the CTs. The replacement specifications shall include but not be limited to:

A. Provide suitable termination to match existing connections for all bushings.

B. Subject bushings to power factor, partial discharge, C1 and C2 capacitance factory tests in accordance with IEEE C57.19.01. C. Subject bushings to hot collar test.

D. The new bushings shall be an exact mechanical fit as determined by the critical dimensions of the original bushings. Exception: standard bushings with mechanical dimensions for mounting flange connections less than the existing bushings may be used provided a factory designed and provided adapter plated is used to retrofit the new replacement bushing. The mechanical dimensions of the new bushings shall be in accordance with IEEE C57.19.01 except where this will preclude matching critical dimensions of the original bushing. Critical dimensions include:

1. Dimensions above the bushing flange.
2. Dimensions below the bushing flange.
3. Bushing flange to transformer interface.
4. Flange dimensions.
5. Top connector fit.
6. Bottom connector fit.
7. Minimum oil level.
8. Clearance in transformer opening.

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E. Drain the oil from all oil filled bushings that are to be replaced. Test each oil-filled bushing for printed circuit board (pcb) concentration prior to draining. Submit copies of pcb results with bushing common description and serial number. Handle and dispose of oil in accordance with 40 CFR 279. Transport old bushings after pcb results have been deemed acceptable to Government designated salvage yard.

F. If existing bushings have recently been replaced, salvage the existing undamaged bushings, and provide to the Government. Place the bushing into the shipping containers provided with the new bushings.

3.9.17. Install fiber optic cable from the transformer control cabinet enclosure to the existing Supervisory Control and Data Acquisition (SCADA) System panel located in the 161kV substation control room. Each cable shall only be terminated

in the transformer control panel at the fiber optic patch panel. The opposite end of the cable installed to the existing SCADA system panel shall be cut (with 240inches of spare length) and stored in the existing SCADA system panel and not be terminated. The cable for each transformer shall be uniquely labeled on each end with an oil resistant label, yellow with 0.25-inch black lettering, including a designation for each specific transformer refurbished for this contract. Example: "TRANSFORMER A FUTURE SCADA SYSTEM RTU CABLE". The cable end not terminated and installed to the existing SCADA system panel shall be covered with insulated cold shrink tubing to prevent damage and contamination. Each fiber cable shall be tight buffer cable with 6 single-mode fibers, indoor/outdoor color orange outer jacket, UL-listed type OFNR, riser rated and UV water and fungus resistant.

3.9.18. Evaluate the existing nitrogen cabinet, piping, and valves in place. Replace all damaged nitrogen piping and valves. Replace nitrogen piping with new stainless-steel piping and fittings in accordance with ASME B16.5. Replace the nitrogen gauge with new gauge that matches or exceeds the existing gauge features and replace the alarm wiring. All piping noted as being in good condition shall be left in place. Leak check the system after the installation is complete. If the existing nitrogen cabinet is determined to have been replaced recently, inform Government representative so a decision can be made about replacement.

3.10. <u>Painting:</u>

3.10.1. After completion of all repair work and component replacement the transformer shall be painted in accordance with Appendix A.6.

3.11. <u>Process and Test Oil:</u>

3.11.1. Fill transformer and process oil per Appendix D in preparation of final baseline electrical testing.

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3.11.2. Test oil per Appendix D. Submit written results within three days of drawing the sample.

3.12. Verification and Testing:

3.12.1. Perform final (baseline) electrical testing as per NETA MTS and IEEE 4 after the refurbishment of the transformer. Where the manufacturer's requirements are more stringent than the standard NETA requirements, the manufacturer's requirements shall prevail. The Contractor shall inform the Government Representative following completion of testing of all test values that fall outside the recommended range. All testing shall be performed by technicians certified by the National Electrical Testing Association (NETA) or National Institute for Certification in Engineering Technologies (NICET). All testing shall be accomplished using calibrated test equipment, which have recorded accuracy traceable to the National Institute of Standards Technologies (NIST). The Contractor shall provide a detailed report of the test results. Clean the new bushings prior to testing. The electrical testing shall include:

A. External visual inspection of all transformer devices and general conditions.

B. Insulation power factor.

C. Insulation resistance.

D. DC winding resistance on the present tap position for all windings. Test on all tap positions if tap changer was repaired or replaced.

E. Winding excitation on the present tap position, verify proper test pattern. Test on all tap positions if tap changer was repaired or replaced.

F. Transformer turns ratio on present tap position for all windings. Test on all tap positions if tap changer was repaired or replaced. G. High voltage bushing power factor and capacitance.

H. Neutral bushing hot collar.

I. Low voltage bushing hot collar.

J. Leakage reactance.

K. Control wiring insulation resistance.

L. Applied potential test of control wiring and auxiliary circuits for one minute.

M. Sweep frequency response analysis for new baseline.



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N. Oil testing for the concentration of Furanic compounds for new baseline.

O. Test the transformer tank for leaks when filled with oil at a test pressure of 5 pounds per square inch and maintain the test pressure for at least four hours with no leaks. Repair the leaks and document satisfactory test results.

P. Verify all Contractor supplied equipment and systems are operational and within industry and manufacturers tolerances and is installed in accordance with design specifications.

3.12.2. After the final electrical tests of the transformer. Discuss findings with the Government within 24 hours of completing the testing. Submit written electrical results within three days of completion of the tests.

3.12.3. Upon completion of the repair and after final testing is complete, torque all high voltage power, medium voltage power, low voltage power, and all control connections. Log the torque value and specific location of each bolt or fastener. Submit log within 24 hours of completing the work.

3.12.4. The Contractor shall provide a technician to oversee re-energization of the transformer and checkout of auxiliary components including a functional demonstration of all new transformer protective devices, gauges, coolers, and controls. The technician shall be on-site as required to complete this service and monitor the performance of the re-energized transformer for a period of 48-hours. The Contractor designated Field Application Engineer(s) shall also be on-site during this service to assist the technician and verify and document results. Plan and check-out task shall include:

A. Verify equipment has been installed as per specifications.

B. Verify and document proper performance of equipment.

C. Verify performance of the transformer refurbished insulating oil coolers.

D. Check for oil leaks and verify the quality of the gasket replacement.

E. Verify performance of the modified nitrogen piping and regulators.

F. Document changes to drawings and manuals to reflect all changes as a result of the checkout.

G. Following transformer re-energization checkout by the Contractor, a separate transformer performance and checkout will be managed and completed by the Government. The Contractor technician shall provide up to two 12-hour days of on-site support for the checkout managed by the Government Representative.

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3.13. Spare Parts:

3.13.1. Furnish a complete set of spare parts as recommended by the original equipment manufacturer. As a minimum this shall include not less than:

A. One of each model number high voltage bushing.

B. One of each model number low voltage neutral bushing.

C. One of each model number low voltage phase bushing.

D. Two fans, including fan blades, motors, guards, and mounting brackets.

E. One SEL-2414.

F. One of each model number SEL-2440.

G. One fiber optic switch.

H. One of each ampacity molded case circuit breaker utilized.

I. One of each size overload relay utilized.

J. One of each model oil flow switch.

K. Three lightning arresters.

3.14. Warranty:

3.14.1. Within 2 days of completing total equipment checkout and testing and prior to final acceptance of the work the Contractor shall in conjunction with the Government Representative convene a field service final walk through which shall include but not be limited to the following:

A. A listing of warranty issues known by the Contractor or Government Representative.

B. All suggested changes or revisions recommended by the Contractor or Government Representatives to help ensure system reliability.

C. At this walk through, all unresolved issues (punch list) will be addressed. Upon resolution of punch list items and submittal of final as-built drawings, the work shall be turned over to the Government.

3.14.2. Within 5 days after the final walk through, the Contractor shall submit a statement of warranty period for all Contractor provided hardware, quality of workmanship, and services included in this contract. All hardware and installation work shall be

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warranted for a minimum of 12 months from the date of acceptance testing, or have a standard manufacturer warranty, whichever is longest.

3.14.3. The Contractor and/or the IED hardware and software vendor shall provide warranty and technical support and assistance during the warranty period for the IED hardware and software and provide the Government all software upgrades for the software package provided in this contract.

3.15. Submittals:

3.15.1. Provide all submittals in accordance with Appendix A.2 – Submittal Procedure.

3.15.2. Within 30-days of the final walk through, submit a CD-ROM of all final documentation including record (as built) drawings, specifications, and cut sheets for all Contractor-provided hardware and components. Final design documentation shall become property of the Government without encumbrances or restrictions.

3.15.3. Within 30-days of test completion, submit three copies on paper and an electronic copy (Microsoft® Word® 2000 or later or Adobe® Acrobat® format) of an operations and maintenance manual for all Contractor provided equipment and components. Include all test results and engineering evaluation in one comprehensive document. The test results shall be reviewed and approved by a NETA certified technician or a licensed Professional Engineer. Each manual shall be provided in a heavy duty 3-ring loose leaf binder bound in 8.5 inch by 11-inch format with a detailed table of contents and dividers for each section of the manual. The manual shall have a cover sheet listing the transformer name and location, Contractor name, project name, Contractor job number, Government Project number, and project date. The manual shall include but not be limited to the following:

A. Factory user manuals and technical specifications for all Contractor provided hardware.

B. List of recommended critical spare parts not readily available off the shelf (with lead times greater than three weeks). The Contractor shall include brand name and model number as detailed on Contractor design drawings and name of the local supplier and/or distributor of the specific parts. C. List of recommend spare parts for all other items provided.

D. Recommended preventative maintenance requirements.

E. Repair instructions.

F. Safety considerations.

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G. Routing procedures and guide for troubleshooting, disassembly, repair, and reassembly instructions.

H. Updated outline drawings of the refurbished transformer.

I. A detailed certified final test report comparing the oil sampling test results obtained with minimum or maximum values recommended in NETA MTS and/or the manufacturer's instruction book for oil testing.

J. A detailed test report comparing the transformer test results obtained with minimum or maximum values recommended in NETA MTS and/or the manufacturer's instruction book for transformer testing prior to service. A detailed certified test report on the characteristics of the processed oil after the processing has been completed.

K. A detailed report of the operators log of all oil handling operations including dew points, vacuum limits and times and temperatures.

L. If new oil is required, a certified test report that new oil added to the transformer meets the requirements listed in ASTM D3487.

M. A detailed certified test report documenting final assembly and testing including test data and details on all required repairs for the transformer insulating oil cooling system refurbished.

N. A detailed test report comparing the transformer test results obtained with minimum or maximum values recommended in NETA MTS and/or the manufacturer's instruction book for transformer testing following completion of the refurbishment service.

O. A list of all testing equipment used by the transformer testing Contractor. Provide certification of calibration and the calibration dates for the test equipment.

3.15.4. Submit bushing factory test report with shipment of bushings. Furnish three copies.

3.15.5. Submit three copies of certifications and warranties.

3.15.6. Submit a complete set of standard software documentation and a complete set of documentation for all project specific software. Provide a set of logically coordinated documents, a comprehensive and detailed description of the software required for operation and maintenance of the IED hardware. Describe the overall function of each IED, source code, hardware configuration and database. The IED software documentation shall include a Certificate of License agreement listing the Government as the owner of the software license. The

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certification shall include the manufacture part number, serial number, and date of license agreement.

3.15.7. Factory Test Reports: Submit three copies of CT factory test report with the shipment of the CT's. Include typical excitation curves for each turn ratio and resistance of the secondary winding at 75 degrees C.

3.15.8. Submittal Table:

| No. | Paragraph Reference | Submittal Description | Due |
|---|---|---|---|
| 1 | Par 3.1.7 | Drawings | Before final inspection |
| 2 | Par 3.1.10.A | Initial test Plan | Within 3 days of notice to proceed |
| 3 | Par 3.1.10.B | Design Plan and Schedule | Within 10 days of notice to proceed |
| 4 | Par 3.1.10.C | Request for Outage | 30 days prior to outage |
| 5 | Par 3.1.10.C.1 | Critical Lift Plan | With request for outage |
| 6 | Para 3.1.11.A | Safety Plan | Within 10 days of notice to proceed |
| 7 | Para 3.1.11.D | Confined Space Procedure | Within 3 days of notice to proceed |
| 8 | Para 3.2.2 | Certification of Cleanliness | Prior to pumping oil |
| 9 | Para 3.3.1 | Certification for Qualifications | Before beginning on-site work |
| 10 | Para 3.3.5 | Preliminary Written Reports | Within 24 hours of completion of inspection |
| 11 | Para 3.3.6 | Revision to Initial Test Plan | Within 3 days of completion of the internal inspection |
| 12 | Para 3.3.7 | Engineering Evaluation | Within 3 days of completion of the internal inspection |
| 13 | Para 3.3.8 | Test Oil Results | Within three days of sampling |
| 14 | Para 3.3.9 | Electrical Results | Within three days of test completion |
| 15 | Para 3.4.1 | Detailed Test Results | With Final Electrical Results |
| 16 | Para 3.4.2 | Preliminary Drawings | Within 90-days of notice to proceed |
| 17 | Para 3.4.3 | Final Shop Drawings | Within 25-days of completion of on-site field survey |
| 18 | Para 3.5 | Certification of Cleanliness | Prior to pumping oil |
| 19 | Para 3.5.3 | Certified Test Results | With Final Documentation |
| 20 | Para 3.6.2 | Control Cabinet Test Plans | 14 days prior to inspection and testing |
| 21 | Para 3.7 | Certification for Qualifications | Before beginning on-site work |

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| 22 | Para 3.7.3 | Engineering Evaluation Recommendation | Within 3 days of completion of evaluation |
| 23 | Para 3.7.7 | Certified Test Results | With Final Documentation |
| 24 | Para 3.11.2 | Test Oil Reports | Within 3 days of sampling |
| 25 | Para 3.12.2 | Final Electrical Results | Within three days of test completion |
| 26 | Para 3.12.3 | Torque Log | Within 24 hours of completing work |
| 27 | Para 3.14.2 | Warranty | Within 5 days of final walk through |
| 28 | Para 3.15.2 | Final Documentation | Within 30-days of the final walk through |
| 29 | Para 3.15.3 | Operators and Maintenance Manual | Within 30 days of final test completion |
| 30 | Para 3.15.4 | Bushing Test Report | With shipment of bushings |
| 31 | Para 3.15.5 | Certifications and Warranties | With Final Documentation |
| 32 | Para 3.15.6 | Software Documentation | With Final Documentation |
| 33 | Para 3.15.7 | CT Factory Test Reports | With Final Documentation |
| 34 | Appendix A 1.3.C | Outage Request | 2 weeks prior to outage |
| 35 | Appendix A 1.3.L | Proposed scaffolding | Before assembly and use |
| 36 | Appendix A 1.3.M | As-built drawings | Before final inspection |
| 37 | Appendix A 1.4.B | Existing conditions report which could be construed as damage resulting from work | Prior to startup of on-site work |
| 38 | Appendix A 1.5.A | Safety and Health Plan | 10 days after task order award |
| 39 | Appendix A 1.11.A | Disallowed products locations | Before final inspection |
| 40 | Appendix A 1.12.A | Asbestos Locations | Before final inspection |
| 41 | Appendix A 1.13.A | Waste Identification (WIF) and Chemical Waste Data Sheet (CWDS) | Before final inspection |
| 42 | Appendix A 1.14.A.1.a | Hazardous Material Info | Before beginning on-site work |
| 43 | Appendix A.2 1.2.B | Summarized Submittal Listing with Deliver Dates | 14 days after task order award |
| 44 | Appendix A.3 1.4.A | Construction Waste Management Plan | Before beginning on-site work |
| 45 | Appendix A.4 1.3.A | Employee Certification | Before beginning on-site work |
| **No.** | **Paragraph Reference** | **Submittal Description** | **Due** |
| 46 | Appendix A.4 1.3.B | Training and Medical Certifications | Before beginning on-site work |
| 47 | Appendix A.4 1.3.C | Training and Hazard Communications Certification | Before beginning on-site work |
| 48 | Appendix A.4 1.3.D | Product List | Before beginning on-site work |
| 49 | Appendix A.4 1.3.E | Equipment List | Before beginning on-site work |
| 50 | Appendix A.4 1.3.F | Notice of Violation | Before beginning on-site work |
| 51 | Appendix A.4 1.3.G | Environmental, Health, and Safety Plan | Before beginning on-site work |
| 52 | Appendix A.4 1.3.H | Asbestos Removal Plan | Before beginning on-site work |
| 53 | Appendix A.4 1.3.I | Air Sampling Reports | Before beginning on-site work |
| 54 | Appendix A.4 1.3.J | Testing Certifications | Before beginning on-site work |

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| 55 | Appendix A.4 1.3.K | CPR Certification | Before beginning on-site work |
| 56 | Appendix A.4 1.3.L | Insurance Coverage Certification | Before beginning on-site work |
| 57 | Appendix A.4 1.3.M | Sampling List | Before beginning on-site work |
| 58 | Appendix A.4 1.3.N | Operator's Log and Shipper's Log | Before beginning on-site work |
| 59 | Appendix A.4 1.3.O | Daily Logs: Sign-In and Field Notes | Before beginning on-site work |
| 60 | Appendix A.5 1.4A | Employee Certification | Before beginning on-site work |
| 61 | Appendix A.5 1.4B | Training and Medical Certification | Before beginning on-site work |
| 62 | Appendix A.5 1.4C | Notice of Violation | Before beginning on-site work |
| 63 | Appendix A.5 1.4D | Environmental, Health, and Safety Plan | Before beginning on-site work |
| 64 | Appendix A.5 1.4E | Training and Hazard Communications Certification | Before beginning on-site work |
| 65 | Appendix A.5 1.4F | Product List | Before beginning on-site work |
| 66 | Appendix A.5 1.4G | Lead Abatement Plan | Before beginning on-site work |
| 67 | Appendix A.5 1.4H | Air & Substrate Sampling Reports | Before beginning on-site work |
| 68 | Appendix A.5 1.4I | Testing Laboratory Qualifications | Before beginning on-site work |
| 69 | Appendix A.5 1.4J | Air Monitoring Results | Before beginning on-site work |
| 70 | Appendix A.5 1.4K | Equipment List | Before beginning on-site work |
| 70 | Appendix A.5 1.4L | Rental Equipment List | Before beginning on-site work |
| 71 | Appendix A.5 1.4M | Shower Water Test Results | Before beginning on-site work |
| 72 | Appendix A.6 1.2 A | Paint - Product Data Sheets | Before beginning on-site work |
| 73 | Appendix A.6 1.2 B | Paint - Material Safety Data Sheets | Before beginning on-site work |
| 74 | Appendix A.6 1.2 C | Paint - Manufacturer's Certification | Before beginning on-site work |
| 75 | Appendix D 3.1.4 | Plans and procedures | Two weeks after award |
| 76 | Appendix D 3.1.6 | Transformer testing before oil processing | No less than 10 days after test |
| 77 | Appendix D 3.1.17 | Vapor pressure (dew point) measurement | Prior to final oil filling |
| 78 | Appendix D 3.1.18 | Electrical testing before oil procession | No less than 10 days after test |
| 79 | Appendix D 3.1.24 | Final oil analysis and electrical testing reports | Prior to release of transformer for re-energizing |
| 80 | Appendix D 3.1.25 | Test reports | Prior to re-energization |
| 81 | Appendix D 3.2.1 | Personnel information assigned to project | Before beginning on-site work |

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Table I. Required Submittal List – Completed.

APPENDIX A
CSI SPECIFICATION SECTIONS

- Appendix A.1 – 01 10 00 – Job Conditions at AEDC.
- Appendix A.2 – 01 33 00 - Submittal Procedure.
- Appendix A.3 – 01 74 19 – Construction Waste Management and Disposal.
- Appendix A.4 – 02 08 00 - Asbestos Removal.
- Appendix A.5 – 02 08 50 – Lead and Heavy Metal Removal.
- Appendix A.6 – 09 90 00 - Painting and Coating

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## PART 1 GENERAL

### 1.1 DEFINITIONS

A. Certain terms used in the contract documents are defined below. Definitions and explanations contained in this section are not complete but are general for the work to the extent that they are not stated more explicitly in another element of the contract documents.

1. Furnish. The term "furnish" means to supply and deliver to the project site, ready for unloading, unpacking, assembly, installation, and similar operations.
2. Install. The term "install" describes operations at the project site, including the actual unloading, unpacking, assembly, erection, placing, anchoring, applying, working to dimensions, finishing, curing, protecting, cleaning, and similar operations.
3. Provide. The term "provide" means "provide complete in place"; that is, "furnish and install."
4. Indicated. Where "as indicated" or words of similar import are used, it shall be understood that the reference is made to the drawings accompanying this contract unless stated otherwise.
5. Postconsumer material: a material or finished product that has served its intended use and has been discarded for disposal or recover, having completed its life as a consumer item. Postconsumer material is a part of the broader category of "recovered materials".
6. Recovered material: waste materials and by-products recovered or diverted from solid waste. The term does not include those materials and by-products generated from, and commonly reused within, an original manufacturing process.
7. Biobased product: a product determined by the U.S. Department of Agriculture (USDA) to be a commercial or industrial product (other than food or feed) that is composed, in whole or in significant part, of biological products, including renewable domestic agricultural materials (including plant, animal, and marine materials) or forestry materials.
8. USDA-designated item: a generic grouping of products that are or can be made with Biobased materials (1) that is listed by USDA in a procurement guideline; and (2) for which USDA has provided purchasing recommendations.

### 1.2 CODES AND STANDARDS

A. AEDC Environmental Policy: https://media.defense.gov/2022/Nov/29/2003122404/-1/-1/1/ARNOLD_AFB_ENVIRONMENTAL_COMMITMENT_STATEMENT_15JUL22.PDF

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A. AEDC Safety, Health, and Environmental (SHE) Standards latest version, provided in Appendix H.

| | | |
|---|---|---|
| 1. | A6 with Supplement | User and Subcontractor Safety (Mandatory Documents/Submittals), |
| 2. | A9 | Hazard Communication. |
| 3. | A10 | Job Safety Analysis. |
| 4. | B1 | Master Work Permit. |
| 5. | B2 | Lockout/Tagout (LOTO). |
| 6. | B3 | Control of Hazardous Areas Using Safety Signs, Tags, and Barricades. |
| 7. | B4 | High Voltage Electrical Work. |
| 8. | B5 | Confined Spaces. |
| 9. | B6 | Low Voltage Electrical Systems and Related Work Practices. |
| 10. | C2 | Contaminated Equipment. |
| 11. | D5 | Hoisting Devices. |
| 12. | D6 | Aerial Work Platforms. |
| 13. | D8 | Portable Power Tools. |
| 14. | D9 | Fixed and Portable Ladders. |
| 15. | D10 | Scaffolding. |
| 16. | E7 | Asbestos. |
| 17. | E11 | Oil Pollution Prevention and POL Storage Tank Management. |
| 18. | E17 | Oil and Hazardous Substance Spill Response. |
| 19. | E18 | Hazardous Material Management Program. |
| 20. | E19 | Lead and Heavy Metals. |
| 21. | F2 | Personal Protective Equipment. |
| 22. | F5 | Hazardous Noise and Hearing Conservation (NRR Method for Estimating the Adequacy of Hearing Protector Attenuation). |
| 23. | F6 | Fall Protection. |

B. Code of Federal Regulations, latest version:

| | | |
|---|---|---|
| 1. | 29 CFR 1926.1101 | Asbestos. |
| 2. | 40 CFR 261 | Identification and Listing of Hazardous Waste. |
| 3. | 40 CFR 370 | EPA Hazardous Chemical Reporting and Community Right to Know Requirement. |

## 1.3 JOB CONDITIONS

A. Plan and execute this project in a manner to minimize downtime. Schedule all work in advance with the Government Representative.

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B. Furnish new components and devices, complete, with all necessary materials for installation to meet this objective.

C. Do not interrupt existing utilities or commence power outages without signed authorization from the Government Representative. Submit each utility outage request for Government approval at least two weeks prior to each requested outage date. Outages may be granted depending upon the Government's schedule for facility use and may occur on weekends and/or holidays.

D. Keep work area clean and free from waste and debris. Dispose of waste as directed by the Government Representative.

E. Do not close or obstruct egress width to building or site exit without written permission from the Government Representative.

F. Do not disable or disrupt building fire or life safety systems without written permission from the Government Representative.

G. Conform to applicable procedures when hazardous or contaminated materials are discovered.

H. Cease operations immediately for unsafe conditions and notify the Government Representative. Do not resume operations until authorized by the Government Representative.

I. Provide temporary barricades and other forms of protection as required to protect Government personnel and the general public from injury due to the work. Provide temporary signage as needed or directed in accordance with AEDC SHE Standard B3.

J. Remove protections at the completion of work.

K. Do not close, block, or otherwise obstruct streets, walks, or other occupied or used facilities without written permission from the Contracting officer.

L. Submit sketches of proposed scaffolding, if utilized. All scaffolding shall be in accordance with AEDC SHE Standard D10 and shall be inspected, approved, and tagged by the Contractor's scaffold competent person after it has been erected and prior to use.

M. Submit record as-built drawings.

## 1.4 PROTECTION OF EXISTING IMPROVEMENTS:

A. Protection of the grounds, walkways and structures from deterioration due to excessive vehicular traffic, equipment disturbance, and material staging is critical to maintaining the quality appearance of AEDC. The Contractor shall protect against damage incurred by service vehicles, construction

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activities, maintenance and repair, and all other work that detracts from the overall appearance of the grounds and increases the costs of grounds maintenance.

B. Prior to commencement of work, inspect areas in which work will be performed. Submit a report with the Contracting Officer listing existing conditions which could be construed as damage resulting from the work. The Government Representative may also photograph or video tape areas or improvements susceptible to damage to document conditions prior to the start of work.

C. The activity of vehicles and equipment on the grounds shall be limited to only that which is essential and required for completing the work. The Government Representative will approve all access and haul routes required for the work prior to any on site operations to preclude damage to sensitive utilities or improvements which may not be readily apparent.

D. Minimize travel paths and staging areas to limit the impact and damage to work areas. Lay out the work site, material and equipment staging, and vehicle and equipment activity to minimize visibility and aesthetic impact of activities to surrounding areas.

E. Protect above and below grade utilities, plant life, lawns, and other features or improvements to remain.

F. Protect benchmarks, existing structures, sidewalks, curbs, and pavements from excavation equipment and vehicular traffic.

G. Use protective materials, such as sheeting, steel plating, rigger pads, to protect grounds and pavement from heavy equipment activity and damage.

H. Protect existing facilities from falling debris during construction such as roof materials, ductwork scraps, or other materials, by use of protective materials such as sheeting, steel plates, netting, or other materials.

I. Provide temporary weather protection during any interval between demolition and removal of existing construction on exterior surfaces and installation of new construction to ensure no water leakage or damage occurs to the structure, interior areas, or contents of the existing facility.

J. Implement a daily procedure for area clean up and organized parking of vehicles and equipment.

K. Provide temporary barricades, shoring, bracing, supports or other forms of protection as required to protect all existing improvements not scheduled to be demolished for both above and below ground improvements and to prevent movement, settlement, collapse or damage of any kind.



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L. Maintain personnel and vehicular access and existing utilities to all buildings and other structures during the course of the work. All changes or interruptions shall be coordinated and approved by the Government Representative well in advance. Temporary protections or services may be required as a condition of approval of any such change or interruption.

M. The Contractor shall be responsible for damage caused by the work and shall repair or replace same to the satisfaction of the Government. Obtain Government Representative’s approval of methods, materials, equipment or products for repair or replacement prior to use. All such repair or replacement shall be performed promptly at no additional cost to the Government. The Government will place special emphasis and requirements for repair or replacement of damage to underground utilities, lawns, grassed areas, existing structures, sidewalks, curbing, railings and pavements.

N. Remove all temporary barricades, shoring, bracing, supports or other forms of protection at completion of work. Restore all adjacent structures and surfaces to conditions existing prior to construction. Clean up all stockpiles and spoil areas and restore all surrounding areas to conditions existing prior to construction.

## 1.5 SAFETY REQUIREMENTS

A. All work shall be accomplished in compliance with AEDC Safety Standards A6 with supplement, A9, A10, B1, B2, B3, B4, B5, B6, C2, D5, D6, D8, D9, D10, E7, E11, E17, E18, F2, F5, and F6.

## 1.6 DISPOSAL OF CONTAMINATED EQUIPMENT

A. The Contractor shall clean equipment or material of contaminants prior to disposal at the AEDC Salvage Yard (with the exception of lead paint, unless the paint is flaking in which case the flaking paint shall be removed prior to transport to prevent contamination during transport). If cleaned equipment is to be turned-in to the Government for storage or excessing, a Special Material Turn-in Certification, shall be completed and submitted to the Government with the equipment or material prior to disposal. The Government Representative will assist the Contractor in filling out the form and in coordinating with Industrial Hygiene and Environmental Quality.

## 1.7 CONNECTIONS TO POTABLE WATER

A. Contractor shall not make any connections to an AEDC potable water system without specific approval from the Government Representative.

B. Notify Government Representative if connections to hose bibs are needed. Do not connect to any hose bibs without Government-supplied backflow preventer and specific approval from Government Representative.

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1.8 DELIVERY, STORAGE, AND HANDLING

A. Deliver all equipment properly packaged in factory containers and/or mounted on shipping skids.

B. Provide and store all equipment in a clean, dry space, and protect from dirt, fumes, moisture, construction debris, and traffic. Ensure that all equipment is prepared for anticipated storage conditions.

C. Where equipment is stored outdoors, ensure electrical components are stored above grade and enclosed with watertight wrapping. Provide 120V temporary power to space heaters or provide temporary heat if required to protect equipment.

D. Notify Government Representative a minimum of two weeks prior to delivery of large equipment being stored on-site. Contractor shall offload equipment at a location designated by the Government Representative.

E. All handling, loading, and offloading of all equipment shall be done by the Contractor.

F. Handle and store all equipment in accordance with manufacturer's instructions.

1.9 TOOL CONTROL: Coordinate tool control with the Government Representative.

1.10 ELIMINATION OF ODS CHEMICALS

A. The use of Class I ozone depleting substances (ODS) is prohibited.

1.11 DISALLOWED PRODUCTS

A. Do not use products or materials that contain lead, chromium, mercury, cadmium, silver, barium, selenium, beryllium, or arsenic on this project except as expressly authorized by the Contracting Officer. If no substitutes for products containing the listed materials are available, and the Contracting Officer approves the use of products containing the listed materials, highlight and detail their exact location on the drawings.

1.12 ASBESTOS PRODUCTS

A. Do not use products or materials that contain asbestos on this project except as expressly authorized by the Contracting Officer. If no substitutes for asbestos products are available, and the Contracting Officer approves the use of asbestos products, highlight and detail their exact location on the drawings and identify their location in the field following 29 CFR 1926.1101 guidelines.

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## 1.13 HAZARDOUS WASTE

A. Where hazardous waste (as identified in 40 CFR 261) is generated, follow the procedures in AEDC SHE Standard E18, Chemical and Petroleum Products Waste Management, for storing and turning in hazardous waste. These procedures include the requirement to complete the Waste Identification Form (WIF) and, if applicable, the Chemical Waste Data Sheet (CWDS), which will be furnished by the Government Representative.

## 1.14 HAZARDOUS MATERIAL IDENTIFICATION AND MATERIAL SAFETY DATA

A. The following procedures shall be followed to meet 40 CFR 370, EPA Hazardous Chemical Reporting and Community Right to Know Requirement.

1. Procedures:
    a. The Contractor shall furnish information on the hazardous materials he brings on AEDC property prior to beginning on-site work. Hazardous materials may include solvents, paints, adhesives, acids, or any other substance which could be included within the definitions in paragraph 2.
    b. The information required is:
        1) Company name, point of contact, and phone number.
        2) Brief statement indicating how the hazardous material will be used within the scope of the contract.
        3) List of all hazardous materials to be used (product name, manufacturer's name, and address).
        4) Amount of each material to be stored on site and where it will be stored.
        5) Where the product Safety Data Sheet (SDS) will be maintained.
    c. All unused product is the responsibility of the Contractor and shall be removed from AEDC property at the completion of the project.
    d. The Contractor shall coordinate with the Government Representative and the Hazardous Waste Operations Group (454-3628) if any hazardous waste is to be generated.
2. Definitions
    a. Hazard communication standard. A chemical right-to-know law under OSHA that requires chemical manufacturers and importers to assess the hazards of chemicals they make or import and to distribute this information to inform workers of the hazards associated with these chemicals. This written information is a Safety Data Sheet (SDS).
    b. Hazard classes. Hazardous materials that have been grouped into classes by the Department of Transportation (DOT). These classes include explosives, flammables, oxidizers and organic peroxides, compressed gases, corrosives, and poisons.

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c. Hazardous material. Any substance that may be harmful when used. Specific substances have been designated as hazardous under the Clean Water Act, the Resource and Conservation and Recovery Act, and the hazardous air pollutants under the Clean Air Act.

d. Hazardous waste. Any waste that may cause or significantly contribute to serious illness or death or that may pose a substantial threat to human health or the environment, if not properly managed. Hazardous wastes may be solids, liquids, semi-solids, or compressed gases.

1.15 PROPRIETARY AND BRAND NAMES: Proprietary and brand names used in this document are meant to describe physical, performance, and quality characteristics of items and materials, and are not meant to limit competition or exclude Contractors, Vendors, or Manufacturers from submitting like products to the Contracting Officer for approval.

END OF APPENDIX A.1

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APPENDIX A.2
SUBMITTAL PROCEDURE

## PART 1 GENERAL

1.1 DESCRIPTION OF REQUIREMENTS. This section specifies procedural requirements for non-administrative submittals, including but not limited to shop drawings, product data, manufacturer's certificate, design data, calculations, and verifications, manufacturer's instructions, manufacturer's field service reports, samples, operation and maintenance manuals, and other miscellaneous work-related submittals. These submittals are required to amplify, expand, and coordinate other information contained in the contract. This section describes the type and content of submittals that may be required for this work. The required submittals are included in the Statement of Work.

Non-work-related submittals are addressed elsewhere in the contract rather than in the specification and may include items such as: contract progress schedule, permits, payment applications, performance and payment bonds, insurance certificates, and progress reports.

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections of the Statement of Work.

1.2 SUBMITTAL PROCEDURE

A. Definition: For the purposes of this section, the term “Contractor” denotes the prime Contractor.

B. Listing: At the end of this section is a summarized listing of item submittals requiring approval for the work. The listing is included for the convenience of users of the contract documents. The listing may not be all inclusive and additional item submittals may be required. Within 14 calendar days after receiving contract award, Contractor shall submit a copy of the summarized submittal listing with calendar dates assigned to each item submittal indicating when submittal will be received by the Government. If contract requires a pre-construction conference, Contractor may submit completed submittal listing at the conference.

C. Risk: Do not proceed with the part of the work covered by an item submittal including purchasing, fabricating, and delivering until approval is received. Machinery, equipment, material, and articles that do not have the required approval shall be installed or used at the risk of subsequent rejection. Any fabrication or other work performed in advance of the receipt of accepted item submittals and approvals shall be entirely at the Contractor's risk and expense.

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D. Transmittal Timing: Coordinate the preparation and processing of item submittals with the performance of the work. Prepare and transmit each item submittal to the Contracting Officer sufficiently in advance of the performance of related work and other applicable activities. Transmit related item submittals for the same unit of work so that processing will not be delayed by the Government's need to review submittals concurrently for coordination. No delay damages or time extensions will be allowed for time lost in late submittals.

E. Content: Each submittal shall be complete and in sufficient detail to allow ready determination of compliance with contract requirements. All submittals which are generic and list more information than is specifically required shall be marked to identify required information. Complete a material approval submittal and attach as cover sheet for each submittal. Contractor may include multiple item numbers on a material approval submittal.

F. Language: All item submittals shall be written in English, including documents, notes on drawings, sketches, and/or samples, calculations, manuals, and all other instances of written text and communication.

G. Units: All item submittals shall be marked or show dimensions and values in the same units as specified in the contract documents.

H. Review Time: Allow sufficient time so that contract performance will not be delayed as a result of the time required to properly process submittals, including time for re-submittals, if necessary. Allow 14 calendar days for initial Government processing of each submittal. No extension of time will be authorized because of the Contractor's failure to transmit submittals to the Government sufficiently in advance of the work.

I. Contractor Certification: All submittals shall be carefully reviewed by an authorized representative of the Contractor prior to submission to the Government. Each submittal shall be dated, signed, and certified by the Contractor as being correct and in strict conformance with the contract documents. No consideration for review by the Government of any Contractor’s submittal will be made for any items which have not been so certified by the Contractor. All noncertified submittals will be returned to the Contractor without action taken by the Government, and any delays caused thereby shall be the total responsibility of the Contractor.

J. Deviations: Should any item submittals required by the contract documents show deviations from the contract requirement, the Contractor shall make specific mention of such deviations in the letter of transmittal, including stating cost effects, and product and system limitations which may adversely affect the work, in order that if acceptable, suitable action may be taken for proper adjustment of the contract; otherwise the Contractor will not be relieved of the responsibility for executing the work in accordance

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with the contract documents and the approved submittals. Contractor shall clearly mark the proposed variation in all documentation and specifically point out deviations from contract requirements in transmittal letters. Failure to point out deviations may result in the Government requiring rejection and removal of such work at no additional cost to the Government. Deviations from contract requirements require Government approval and will only be considered when advantageous to the Government. When proposing deviations, deliver written request to the Contracting Officer, with documentation of the nature and features of the deviation and why the deviation is desirable and beneficial to the Government. If lower cost is a benefit, include an estimate of the cost savings. In addition to documentation required for deviation, include the submittal information required for the item. Allow an additional 14 calendar days beyond normal submittal review period for consideration by the Government of submittals with deviations.

K. Approved Submittals: The part of the work covered by the approved item submittal may proceed provided it complies with the requirements of the contract documents. Final acceptance will depend upon that compliance. The term “Approved” shall only indicate that there is no exception taken to the submittal. Approval of the item submittal shall not be construed as a complete check and indicates only that the general method or other information appears to meet the contract requirements. Approval does not relieve the Contractor of the responsibility for any error which may exist. After item submittals have been approved by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment, or changes to any other information will be considered unless accompanied by an explanation of why a substitution or change is necessary.

L. Disapproved Submittals: Contractor shall correct disapproved item submittals and resubmit for approval within timeframe noted. If no date is given, Contractor shall submit corrected submittal within 14 calendar days. If the Contractor considers any correction or notation on the returned submittal to constitute a change to the Contract Drawings, Specification or Statement of Work the Contractor shall notify the Contracting Officer.

M. Responsibility: The Government’s review of Contractor submittals shall not relieve the Contractor of the entire responsibility for the correctness of details and dimensions and conformance to the specifications. The Contractor shall assume all responsibility and risk for any mistakes and/or costs due to any errors in submittals.

N. Inconsistencies: If a conflict or inconsistency arises between an approved item submittal and the contract documents, the contract documents shall govern.

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1.3 SPECIFIC SUBMITTAL DESCRIPTIONS AND REQUIREMENTS: Submittal requirements for individual units of work are specified in the Statement of Work (SOW). Except as otherwise indicated in the individual specification sections or SOW, comply with the following requirements for each type of submittal.

A. Shop Drawings: These are technical drawings and data specially prepared for this contract including fabrication and installation drawings, setting and seaming diagrams, and coordination drawings (for use on-site). Shop drawings include drawings, diagrams and schedules specifically prepared to illustrate some portion of the work, diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project, or drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated. Information required on shop drawings includes dimensions, identification of specific products and materials which are included in the work, compliance with specified standards, and notations of coordination requirements with other work. Provide special notations of dimensions that have been established by field measurements. Highlight, encircle, or otherwise indicate deviations from the contract documents on the shop drawings. Furnish one hard copy and one electronic copy in AutoCAD version 2020 or earlier drawing file format (.dwg). The only permissible text font is ROMANS. Custom fonts and shapes are prohibited. The minimum allowable font size shall be 0.125-inch. The maximum allowable font size shall be 0.25-inch.

B. Product Data: This data includes standard printed information on manufactured products that has not been specially prepared for this contract, including manufacturers' product specifications illustrating size, physical appearance and other characteristics of materials, installation instructions, standard color charts, catalog cuts, illustrations, schedules, standard wiring diagrams, standard product operating and maintenance manuals, and samples of warranty language when the contract requires extended product warranties. General information required specifically as product data includes manufacturers' standard printed recommendations for application and use, compliance with recognized standards of trade associations and testing agencies, the application of their labels and seals (if any), special notation of dimensions which have been verified by way of field measurement, and special coordination requirements for interfacing the material, product, or system with other work. Furnish three hard copies and one electronic copy.

C. Samples: These are physical examples of work, including, swatches showing color, texture, and pattern, color-range sets, and units of work to be used for independent inspection and testing. Samples include fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged, color samples from the manufacturer’s standard line (or custom color samples if specified)

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to be used in selecting or approving colors for the project, and field samples and mock-ups constructed on the project site to establish standards by which the ensuring work can be judged. This includes assemblies or portions of assemblies which are to be incorporated into the project and those which will be removed at conclusion of the work. Submit samples for the Contracting Officer's visual review of general kind, color, pattern, and texture for a final check of the coordination of these characteristics with other related elements of the work and for quality control comparison of these characteristics between the final sample submittal and the actual work as it is delivered and installed. Submit one each.

D. Design Data: Design data includes design calculations, mix designs, analyses or other data pertaining to a part of work. Refer to individual sections of the Specification or Statement of Work for required quantities, formats, and signatures or certifications required. Furnish three hard copies and one electronic copy.

E. Test Plans and Reports: Test plans shall include planned testing, including a description of the test, equipment and supplies needed, and step-by-step notation of test activities and tasks. Test reports include reports signed by an authorized official of a testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accordance with the specified requirements. Testing shall have been within three years of date of contract award for the project. Reports also include findings of tests required to be performed by the Contractor on an actual portion of the work or prototype prepared for the project before shipment to the job site, findings of tests made at the job site or on a sample taken from the job site, investigation reports, daily logs and checklists, and final acceptance test and operational test procedures. Furnish three hard copies and one electronic copy of each such report required.

F. Certificates: These include statements printed on the manufacturer’s letterhead and signed by the responsible officials of the manufacturer of a product, system or material attesting that the product, system or material meets specification requirements. Certificates include documentation required of the Contractor, or of a manufacturer, supplier, installer or subcontractor through the Contractor, to further demonstrate the quality of orderly progressions of a portion of the work by documenting procedures, acceptability of methods or personnel qualifications. Examples include confined space entry permits, and text of posted operating instructions. Certificates shall be dated after award of the project contract and clearly name the project. Furnish three hard copies and one electronic copy of each certificate required.

G. Manufacturer’s Instructions: Preprinted material describing installation of a product, system or material, including special notices and Material Safety Data sheets concerning impedances, hazards and safety precautions. Furnish three hard copies and one electronic copy.

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H. Manufacturer's Field Reports: Documentation of the testing and verification actions taken by a manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with the manufacturer's standards or instructions. The documentation shall be signed by an authorized official of a testing laboratory or agency and shall state the test results, and indicate whether the material, product, or system has passed or failed the test. Furnish three hard copies and one electronic copy.

I. Operation and Maintenance Data: Data furnished by the manufacturer, or the system provider, to the equipment operating and maintenance personnel, including manufacturer's help and product line documentation necessary to maintain and install the equipment, and needed by operating and maintenance personnel for the safe and efficient operation, maintenance, and repair of the item. The data is intended to be incorporated in an operations and maintenance manual or control system. Furnish two hard copies and one electronic copy.

J. Final Design Drawings and Documentation: Final design documentation including drawings, specifications, estimates, and other required deliverables. Electronic documentation submitted shall include every associated component file of drawings and other documentation, including raster images, jpeg, tiff, Excel, and any related items required to view, modify, or manipulate the electronic files. Furnish a CD-ROM of all final design documentation, and drawings in AutoCAD 2020 version or earlier drawing file format (.dwg), to the Contracting Officer for approval prior to applying for final payment. The drawings shall be fully editable using AutoCAD software. In addition, drawings shall be delivered to clearly show the location and arrangement of all materials and equipment. The only permissible text font is ROMANS. Custom fonts and shape files are prohibited. The minimum allowable font size shall be 0.125-inch. The maximum allowable font size shall be 0.25-inch.

K. Record (As-Built) Drawings: Record drawings showing final configuration of work accomplished. Show all changes, additions, and deviations from the original contract drawings and documentation. If no changes occur, furnish certification to that effect. Drawings shall be redlined electronic drawings, unless waived by the Government, and shall accurately show as-built conditions during the progress of the job. Furnish drawings on a CD-ROM, in AutoCAD 2020 or earlier version, including every associated component file of the drawings, including raster images, jpeg, tiff, Excel, and any related items required to view, modify, or manipulate the electronic files. Submit to the Contracting Officer for approval prior to applying for final payment.

L. Miscellaneous Submittals: These are work-related, non-administrative submittals that do not fit in the previous categories, including the following:

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1. Maintenance agreements. Furnish one hard copy and one electronic copy.
2. Survey data and reports. Furnish one hard copy and one electronic copy.
3. Project photographs. Furnish both hard copies and digital files.
4. Keys and other security protection devices.
5. Maintenance tools, spare parts, and overrun or maintenance stock. Refer to individual sections of the specification for required quantities of spare parts, extra and overrun stock, maintenance tools and devices, keys, and similar physical units to be submitted.
6. Qualification certificates. Furnish one hard copy and one electronic copy.
7. Employee training certificates and documentation showing successful completion of training.
8. Documentation of percentage of recovered material content used during contract.
9. Contractor safety and work plans and schedules.
10. Warranties.

END OF APPENDIX A.2

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APPENDIX A.3
CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

## PART 1 GENERAL

### 1.1 SUMMARY

A. This section is applicable to construction/demolition debris, excavated materials, reusable and salvageable items, and recyclable and non-recyclable wastes. Hazardous or special wastes are identified in separate related sections.

B. Section includes:
1. Construction waste management plan.
2. Construction waste recycling, management, and disposal instructions.

C. Related Sections:
1. Section 01 10 00 - Summary.
2. Section 02 08 00 - Asbestos Removal.
3. Section 02 08 50 - Lead and Heavy Metal Removal.

### 1.2 CODES AND STANDARDS

A. AEDC Safety, Health, and Environmental (SHE) Standards latest version, available upon request from the Government.

| | | |
|---|---|---|
| 1. | E18 | Managing Wastes Containing Chemical or Petroleum Products. |

B. Code of Federal Regulations, latest version:

| | | |
|---|---|---|
| 1. | 40 CFR 82 Subpart F | Protection of Stratospheric Ozone, Recycling and Emissions Reduction. |
| 2. | 40 CFR 82.161 | Technician Certification. |
| 3. | 40 CFR 261 | Identification and Listing of Hazardous Wastes. |

### 1.3 DEFINITIONS

A. Clean: Not contaminated with oils, solvents, caulk, or other chemicals or materials that result in the waste being considered hazardous or special waste.

B. Construction/Demolition Debris: Building materials, debris, and rubble resulting from construction, remodeling, repair, and demolition operations.

C. Construction Waste Management Plan: Required project submittal documenting the project activities and procedures to be used to minimize landfill disposal of construction and demolition waste and to optimize recycling opportunities within the project.

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D. Hazardous Waste: A used or discarded material that can damage the environment and be harmful to health. Hazardous waste possesses at least one of four characteristics (ignitibility, corrosivity, reactivity, or toxicity as defined by applicable State and Federal regulations) or appears on federal or state official lists of hazardous wastes.

E. Hydrovac Method: An excavation method using a high-pressure water jet to loosen soil, after which the soil is removed from the excavation using a vacuum truck.

F. Industrial Area: Area at Arnold AFB occupied by industrial operations, including test cells and test cell support structures, water treatment plant, and sewage treatment plant.

G. Non-Recyclable Waste: Any non-hazardous product or material (not including construction/ demolition debris) unable to be reused, returned, recycled, or salvaged.

H. Recyclable: The ability of a product or material to be recovered at the end of its life cycle and remanufactured into a new product for reuse.

I. Recycle: To remove a waste material from the project site for reuse.

J. Recycling: The process of sorting, cleaning, treating and reconstituting solid waste and other discarded materials for the purpose of reuse. Recycling does not include burning or incinerating waste.

K. Reusable: Able to be used again, especially after salvaging or special treatment or processing.

L. Salvageable: Discarded or damaged equipment or material that can be saved for further use.

M. Source Separation: The act of keeping different types of waste materials separate beginning from the first time they become waste.

N. Special Waste: Waste which is not regulated hazardous waste, which has physical or chemical characteristics, or both, that are different from construction/demolition debris, recyclable, and non-recyclable wastes, and which potentially requires special handling. Some examples include oil-contaminated soil, oily sorbents, and water treatment plant sludge.

O. Waste: Extra material or material that has reached the end of its useful life in its intended use. Waste includes salvageable, returnable, recyclable, and reusable material.

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1.4 SUBMITTALS

A. Construction Waste Management Plan describing methods to be used for waste recycling and minimization, procedures for implementation and compliance monitoring to include the following:
   1. Construction waste materials anticipated for recycling and reuse.
   2. Method of waste stream segregation prior to disposal or reuse.
   3. Method of excavated soils management.
   4. Identification of disposal or recycling facility destination for each waste stream (construction/demolition landfill, Recycling Operations Center, Arnold AFB Salvage Yard, or off-site.)
   5. Method of temporary site storage of wastes.
   6. Method of transporting waste to recycling facility or to landfill.

**PART 2 PRODUCTS – Not Used**

**PART 3 EXECUTION**

3.1 CONSTRUCTION WASTE MANAGEMENT PLAN

A. Develop and implement Construction Waste Management Plan with approval by the Government.

B. To the maximum extent possible, divert/dispose of construction demolition, and land-clearing debris from landfill.

C. Direct recyclable material to Arnold AFB recycling.

D. Dispose of hazardous wastes in accordance with AEDC SHE Standard E18.

E. Implement Construction Waste Management Plan at start of construction.

F. Distribute approved Construction Waste Management Plan to subcontractors and others affected by plan requirements.

G. Review Construction Waste Management Plan with Government and all onsite Contractor personnel at preconstruction meeting and progress meetings.

H. Oversee Construction Waste Management Plan implementation and instruct construction personnel for plan compliance.

3.2 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

A. Use source separation method suitable to sorting and processing of wastes selected for recycling. Separate recyclable materials from construction/demolition debris and non-recyclable waste prior to transportation to Arnold AFB recycling facilities as described herein.

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B. Store construction waste materials to prevent environmental pollution, fire hazards, hazards to persons and property, and contamination of stored materials.

C. Cover construction waste materials subject to disintegration, evaporation, settling, or runoff to prevent polluting air, water, and soil.

D. Separate recyclable, reusable and salvageable materials from construction/demolition debris and non-recyclable waste and sort into separate bins, containers, or lay-down areas, prior to transportation for recycling. The Arnold AFB Recycling Operations Center (ROC) is located in Building 1426 near the Arnold AFB Salvage Yard. Coordinate with ROC personnel (454-6068) to coordinate transfer of recyclable materials. Recyclable materials shall be further sorted for reuse as follow:

1. Cardboard: Cardboard shall be stockpiled on site prior to transportation to the Arnold AFB Recycling Operations Center or shall be placed in any of the cardboard receptacles located throughout the facility.
2. Aluminum Cans: Aluminum cans shall be stockpiled on site prior to transportation to the Arnold AFB Recycling Operations Center or shall be placed in aluminum can recycling bins located inside many Arnold AFB buildings.
3. Recyclable Plastics: Recyclable plastics shall be stockpiled on site prior to transportation to the Arnold AFB Recycling Operations Center. Plastic bottles (such as soft drink bottles) shall be placed in plastic bottle recycling bins located inside many Arnold AFB buildings.
4. Scrap Metals including non-PCB light fixture ballasts: Unless otherwise directed by the Government Representative, scrap metals shall be stockpiled on site prior to transportation to the Arnold AFB Salvage Yard or shall be placed in scrap metal bins located at some Arnold AFB buildings.
5. Equipment: Equipment left after demolition or remodeling shall be taken to the Arnold AFB Salvage Yard for disposal. The Contractor shall clean equipment prior to disposal at the Arnold AFB Salvage Yard (with the exception of lead paint, unless the paint is flaking in which case the flaking paint shall be removed prior to transport). If cleaned equipment is to be turned in to the Government for storage or excessing, a Special Material Turn-in Certification form shall be completed and submitted to the Government with the equipment. The Government Representative will assist the Contractor in filling out the form and in coordinating with Industrial Hygiene and Environmental Quality.
6. Salvageable/Reusable items: Identify items to be salvaged and/or reused. Protect items of salvageable value and remove as work progresses. Transport salvaged items from site to Government designated location. If salvageable/reusable items are to be turned in to the Government for storage or excessing, a Special Material

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Turn-in Certification form shall be completed and submitted to the Government with the items. The Government Representative will assist the Contractor in filling out the form and in coordinating with Industrial Hygiene and Environmental Quality.

7. Paper: Paper shall be disposed of in Cintas recycling bins located in various Arnold AFB buildings. Security-sensitive papers shall be placed into lockable bins, while other paper (magazines, newspapers, etc.) shall be disposed of in yellow open top bins.
8. Other Miscellaneous Recyclable Waste: Other miscellaneous recyclable waste shall be disposed as directed by the Government.
9. Used abrasives: If abrasives are verified as non-hazardous by the Government, used abrasives and lead-containing paint residue shall be placed directly in a roll-off box for disposal at the Arnold AFB construction/demolition landfill. Even if verified as non-hazardous, the State now requires a special waste permit for abrasives containing paint residue prior to disposal in the landfill. Coordinate with the Government representative to obtain the necessary permit prior to disposal. The Contractor shall be responsible for testing the used abrasive by an accredited laboratory and shall submit results to the Government. Based on schedule and project funding, the analysis may be conducted by the Arnold AFB on-site laboratory. Used abrasives shall be tested for TCLP Metals.

E. Dispose of construction/demolition debris separately at the Arnold AFB landfill. Food wastes are to be considered non-recyclable wastes and shall not be disposed of in the construction/demolition landfill. Coordinate with NAS Environmental Quality (454-4284) to obtain a disposal permit prior to transfer of materials and to coordinate schedule for disposal. The Arnold AFB landfill is located on 6th Street near the airfield and may be subject to limited operating hours. Construction/demolition debris shall be further sorted for disposal as follows:
   1. Concrete: Concrete shall be stockpiled at the landfill for later reuse.
   2. Asphalt: Asphalt shall be stockpiled at the landfill for later reuse.
   3. Milled Wood: Milled wood shall be placed into a large bin at the landfill for eventual recycling.
   4. Other Miscellaneous Debris: Other miscellaneous debris shall be placed into the Arnold AFB landfill for disposal.

F. Dispose of non-recyclable waste in any of the brown dumpsters located throughout the facility. Examples of non-recyclable waste include food scraps and food containers, empty glass containers, Styrofoam, empty and dry hazardous material containers (without lids).

G. Dispose of trees, shrubs, and other debris resulting from clearing/grubbing at the Arnold AFB landfill in a location as directed by the Government.

H. Management of excavated soils shall be as follows:

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1. Clean soil or granular fill material from outside the Industrial Area may be taken to the soil stockpile area located just inside Arnold AFB Gate 2 (Red Hill).

2. Soil excavated from the Industrial Area of Arnold AFB may be reused in the original excavation without the requirement of laboratory analysis.

3. Soil excavated from the industrial area of Arnold AFB that cannot be reused in the original excavation shall require sampling and analysis to confirm that it is clean and free from contaminants (such as oil or solvents) prior to disposal at Arnold AFB. The Contractor shall be responsible for testing the soil by an accredited laboratory and shall submit results to the Government. Based on schedule and project funding, the analysis may be conducted by the Arnold AFB on-site laboratory. The analysis shall include results showing quantities of TCLP Volatile Organics (VOCs), TCLP Metals, and Extractable Petroleum Hydrocarbons (EPH) at a minimum. Additional analyses may be required depending on the location of the excavation and known or suspected presence of additional contaminants (consult with the Government). Analytical results shall be submitted to the Government to verify the soil is clean (non-contaminated). Verified clean soil excavated using the Hydrovac Method shall be sufficiently dried so that it contains no free liquids prior to disposal at the C&D Landfill. Verified clean soil excavated using the Hydrovac Method may be taken directly to the soil stockpile area (Red Hill) without the drying requirement. Coordinate with the Government.

4. Soil Sampling for Laboratory Analysis: If possible, collect soil samples to be submitted for laboratory analysis prior to excavation. If this is not practical, soil may be sampled from the stockpile following excavation (note: having analytical results prior to excavation reduces double-handling of excavated materials). To collect and analyze soil, follow these guidelines:

   a. Determine the volume of soil to be excavated.

   b. For each 10 cubic yards of soil, collect one sample to be used for compositing. Samples should be collected at least one foot below the ground surface, preferably about the median depth of the proposed excavation.

   c. Submit composited samples for laboratory analysis for TCLP Metals, TCLP VOCs, and EPH, and any other contaminant as directed by the Government at the frequency indicated below. For example, if the estimated volume of soil to be excavated is 60 cubic yards, then collect six samples to be composited to create one lab sample. For larger volumes of material, consult the Contract Monitor for instructions.

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| Volume of Material to be Excavated (cubic yards) | Number of Samples to Submit for Laboratory Analysis |
|---|---|
| 0-60 | 1 |
| 60-240 | 2 |
| 240-480 | 3 |
| 480-720 | 4 |

d. Soil Stockpile Management: If it is not possible to collect soil samples prior to excavation, then it shall be necessary to create stockpiles while the soil is being dewatered from the Hydrovac Method and/or waiting for analytical results. If it is necessary to create a stockpile for dewatering purposes, follow these requirements:

1) Find a level area in an out of the way location (such as an equipment lay down yard). Consult with the contract monitor to verify the acceptability of the location.
2) Lay down overlapping plastic sheeting to create a water barrier.
3) Create berms around the periphery of the plastic sheeting by overlapping the edges on top of straw bales or other solid objects.
4) Place the excavated soil in the bermed area, being careful not to overtop the berms.
5) Allow the material to dry and obtain analytical results prior to taking the material for final disposal at the Red Hill area or at the construction/demolition (C&D) Landfill.

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I. Capture and recycle refrigerants from all appliances in accordance with 40 CFR 82, Subpart F. The refrigerant shall be removed by a certified technician as defined in 40 CFR 82.161 (or an apprentice directly under the supervision of a certified technician). Since the types of units being salvaged could be high- or low-pressure systems, the technician shall possess a “universal” certification. Provide proof of certification to the Government. The reclamation/recovery of all refrigerants shall occur prior to removal of equipment from Government property. All reclaimed/recovered refrigerant shall be turned over to the Government for final disposition. Refrigerant containing equipment may include but is not limited to: A/C units, refrigerators, freezers, water coolers, and dehumidifiers.

J. Hazardous Waste: Where hazardous waste (as identified or listed by 40 CFR 261) is generated, follow the procedures given in AEDC SHE Standard E18 for storing and turning in hazardous waste for proper disposal. These procedures include the requirement for completion of a Waste Identification Form and a Chemical Waste Data Sheet, which will be furnished by the Government Representative. Return the completed forms to the Government Representative and coordinate with the Arnold AFB

Hazardous Waste Operations Group (931-454-3628 or 931-454-7383) prior to transporting the waste to the designated Base accumulation point or Base permitted storage facility as directed.

3.3 CLEANUP

A. Clean up and dispose of all waste continuously throughout construction and thoroughly at job completion.

END OF APPENDIX A.3

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## PART 1 GENERAL

### 1.1 SUMMARY

A. Section includes removal and disposal of asbestos materials.

### 1.2 CODES AND STANDARDS

A. AEDC Safety, Health, and Environmental (SHE) Standards latest version, available upon request from the Government.
   1. A9 — Hazard Communications.
   2. B1 — Master Work Permit.
   3. B5 — Confined Spaces.
   4. E7 — Asbestos.
   5. E18 — Managing Waste Containing Chemical or Petroleum Products.

B. American Society of Mechanical Engineers:
   1. ASME A13.1-23 — Scheme for the Identification of Piping Systems.

C. American Society of Safety Professionals:
   1. ASSP Z9.2-18 — Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems.

D. Code of Federal Regulations, latest version:
   1. 29 CFR 1910.134 — Respiratory Protection.
   2. 29 CFR 1910.1200 — Hazard Communication.
   3. 29 CFR 1926.55 — Gases, Vapors, Fumes, Dusts, and Mists.
   4. 29 CFR 1926.59 — Hazard Communication.
   5. 29 CFR 1926.1101 — Asbestos.
   6. 40 CFR 61 — National Emission Standards for Hazardous Air Pollutants (NESHAP).
   7. 40 CFR 260 — Hazardous Waste Management Systems: General.
   8. 40 CFR 261 — Identification and Listing of Hazardous Waste.

E. Public Law:
   1. PL 101-637 — The Asbestos School Hazard Abatement Reauthorization Act (ASHARA), 1990.

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F. Tennessee Department of Environment and Conservation Standard:

| | | |
|---|---|---|
| 1. | Chapter 1200-01-20 | Asbestos Accreditation Requirements, 2009. |
| 2. | Chapter 1200-3-11-02 | Hazardous Air Contaminants, Asbestos, 2018. |

## 1.3 SUBMITTALS

A. Evidence satisfactory to the Contracting Officer that the firm removing asbestos has at least one designated employee on site during all abatement activity who has received certification by completing an Asbestos Abatement Contractor Training Course approved or sponsored by the U. S Environmental Protection Agency (EPA) and who will be responsible for the work whenever any phase of the project is in progress. The course shall be a 5-day supervisory course. A 1-day supervisory refresher course also is acceptable if documents showing completion of the initial 5-day course are submitted and the refresher course or courses have been completed within the required time frame to maintain currency in EPA certifications. This training shall have been completed within 12 months prior to the bid opening date. Certification shall remain current throughout the course of the job and shall comply with 40 CFR 61 and PL 101-637. In addition, submit for each individual a copy of their current asbestos accreditation/license issued by the State of Tennessee for all asbestos disciples that the individual shall perform on this contract. The accreditation shall remain current for the length of the current contract. Submit the most current accreditation/license and provide an updated copy as new accreditations are issued by the State.

B. Written certification that all employees involved in the asbestos removal have received training and medical examinations as required by 29 CFR 1926.1101 and 29 CFR 1910.134, respectively. Certification includes respirator fit test records and a copy of respirator operating procedures and program, as specified in 29 CFR 1910.134. Include with the certification's dates of the most recent training, medical examinations, and a physician's statement indicating that workers are physically able to perform asbestos work and use the required respiratory and general body protection. Provide this information for all personnel including management and any air monitoring personnel on the job site before their first entry onto the job site. Keep information current during all phases of the job.

C. A copy of the hazard communications program and certification that all employees have been trained concerning the hazard communications standards and the written program in accordance with 29 CFR 1910.1200 and/or 29 CFR 1926.59. All personnel involved in asbestos related activities as defined in TDEC Chapter 1200-01-20 shall possess and maintain current asbestos accreditations during the course of the work. New accreditation is required annually per regulations.

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D. A list of products to be used and a Safety Data Sheet (SDS) for each. Products include, but are not limited to, aerosol sprays of any kind, wetting and cleaning agents, fuels, solvents, and paints. SDS's will be kept in a notebook and will be indexed for easy reference. This SDS notebook shall remain available to all employees on the job site at all times.

E. A list of all equipment to be used and manufacturers' literature showing that the equipment and materials meet all EPA, Occupational Safety and Health Administration (OSHA), and ANSI standards for use in asbestos abatement activities. Do not use materials that are or will create hazardous waste as described in 40 CFR 261.

F. Any citation or notice of violation from any Government agency issued as a result of work performed under this contract or any contract conducted in the last three years. If none have been received, submit a letter certifying that none have been received.

G. Environmental, health, and safety plan that addresses all environmental, health, and safety aspects of the job. Submit this plan within 30 calendar days after award of the contract and before any field work begins. Include methods to ensure safety including a lockout/tagout plan; job safety analysis; toolbox safety meeting minutes; accident reports and investigations; lead-testing data/certification; fall protection systems; shop drawings; procedures for disposing of waste, scrap, and excess materials; and procedures for work involving transportation or disposal of hazardous waste. The plan shall address all other environmental, health, and safety concerns associated with the job, including a hazardous waste management plan in accordance with 40 CFR 260, a fire safety plan, and procedures for addressing other emergencies within the work area and in compliance with 29 CFR 1926.55.

H. Asbestos removal plan including the precautions to be taken in this work. Do not proceed without the Contracting Officer's written approval of the plan. The plan shall include location of control areas and change rooms; layout of change rooms; location of hot and cold running water shower facilities; types of air machines used; kinds of interface of trades involved in the construction; schedule for sequencing of asbestos-related work; plan for asbestos disposal; type, manufacturer, and name of wetting agent and asbestos sealer to be used; air monitoring; and a detailed description of the pollution control method to be used. The plan shall also state the method proposed to handle oversized asbestos material (too large for bagging). Include dates of proposed work commencement and completion (by phases, if more than one phase is required or proposed).

I. Air sampling reports are to include the results of daily area and personal air and excursion limit sampling along with negative pressure differential documentation.

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J. Testing Certifications.
1. Evidence that all air sampling is analyzed by a laboratory in full compliance with the OSHA Reference Method and which participates in an inter-laboratory quality assurance program or is accredited by the American Society of Safety Professionals (ASSP).
2. Evidence that all persons analyzing samples have successfully completed the required National Institute for Occupational Safety and Health (NIOSH) and EPA-approved courses and been certified proficient by successfully participating in a NIOSH Proficiency Analytical Testing (PAT) (air) or National Institute of Standards and Technology (NIST) program within the last year.
3. Should the Contractor choose to collect and analyze bulk samples, submit evidence that the laboratory analyzing asbestos bulk samples is a NIST-accredited laboratory. (Bulk samples may be collected only with the permission of the Contracting Officer and shall be returned to AEDC for disposal.) Samples shall only be collected by State of Tennessee accredited inspectors.

K. Certification that persons monitoring work in confined spaces have successfully completed a course in cardiopulmonary resuscitation (CPR) by the American Red Cross or the American Heart Association.

L. Evidence that the firm removing asbestos has suitable insurance to cover any asbestos liabilities.

M. A list of the sampling numbers required by paragraph 3.1B.

N. AEDC Asbestos Landfill Operator's Log and Shipper’s Log. This form will be provided by the Government Representative.

O. Daily logs.
1. Sign-in logs will be submitted at least monthly and when the job is completed. Sign-in logs will include the following information for all persons entering the controlled area:
   a. First and last name (must be legible).
   b. Company and organization.
   c. Social security number.
   d. Date and time of arrival and departure.
   e. Reason of visit.
2. Field notes will also be submitted at least monthly and when the job is completed.

1.4 QUALITY ASSURANCE

A. Demolish, remove, and dispose of asbestos materials as indicated on the drawings and specified herein. For the purposes of this specification, full-gross removal containment is required for any removal activity that takes two people over four hours to complete or that is larger in size than allowed

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in paragraph (g)(5)(vi) of 29 CFR 1926.1101 (mini-enclosure). See 1.4.C below for additional information. Mini-enclosures shall be constructed of similar materials as required for full-gross removal per Part 3 of this specification. This applies only to removal activity and not to enclosure construction or final area clean-up.

B. Use glove bag techniques as described in 29 CFR 1926.1101 and paragraph 3.7C, for small sections. If samples taken during initial glove bag work exceed 0.01 fiber per cubic centimeter (f/cc), stop the job and remove all remaining asbestos using full-gross removal containment. Gross removal methods utilizing full decontamination units as described in 29 CFR 1926.110 shall be the method of removal.

C. On small sections of pipe, valves, or other small areas of abatement where the glove bag is not suitable and full-gross removal containment is not required, mini-enclosures as specified in 29 CFR 1926.1101 may be used. Mini-enclosures shall be equipped with high-efficiency particulate air (HEPA) filtered exhaust ventilation. Ventilation is accomplished through the use of negative air machine. A HEPA vacuum does not meet this requirement.

D. Assume unidentified insulating material to be asbestos.

**PART 2 PRODUCTS**

2.1 AIR RETURN FILTERS

A. 1-inch-thick disposable random fiber.

2.2 ENCAPSULANTS

A. Dryable to clear appearance and paintable with standard latex paints that are designed to encapsulate and seal asbestos containing material.

**PART 3 . EXECUTION**

3.1 CONTROL OF WORK: Control work locations where the generation of asbestos dust could expose persons not properly protected.

A. Use safety ropes and barricades and post with 20-inch by 14-inch signs bearing the following warning:

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**DANGER**

**ASBESTOS**

**MAY CAUSE CANCER**
**CAUSES DAMAGE TO LUNGS**
**AUTHORIZED PERSONNEL ONLY**
**WEAR RESPIRATORY PROTECTION**
**AND PROTECTIVE CLOTHING**
**IN THIS AREA**

B. Record all sampling numbers identified on asbestos material to be removed.

C. Ensure isolation of the work area or construction of the enclosure complies with 29 CFR 1926.1101. In addition, a continuous layer of true 6-mil polyethylene or other impermeable material shall first be applied to the floor and extend 12 inches up the wall. Use a layer of true 4-mil polyethylene for the first wall and ceiling and extend 12 inches onto the floor. Place the second layer of true 6-mil polyethylene on the floor and extend 18 inches up the wall and follow by the second layer of true 4-mil polyethylene on the walls and ceiling and extend 18 inches onto the floor. Polyethylene used for floors and walls shall be installed in such a manner to prevent ballooning of the polyethylene from the walls or floors. Drop and splatter sheets shall also be used in all containments in addition to the use of the two layers of polyethylene on the walls and the three layers on the floors of the containment. In special situations, such as enclosures constructed outside, you may elect to construct the entire containment out of true 6-mil polyethylene or other impermeable material also supported by plywood or other rigid material. Coordinate such construction with the Government Representative. Enclosures constructed to contain asbestos work such as around equipment, utility systems, windows, doors, or other industrial systems shall be constructed of lumber (such as two by fours) or other rigid material supported on centers no greater than 48 inches apart and in a manner that shall maintain the integrity of the containment and prevent containment failure and release of asbestos fibers outside the work area. The use of wire, cable and other non-rigid systems shall not be used as containment framework or to otherwise hold polyethylene unless also supported by lumber or other rigid material on no greater than 48-inch centers. Support of polyethylene by wire, cable or other non-rigid materials by themselves shall not be allowed. Polyethylene sheeting shall be attached to the lumber or other rigid material and held in place through the use of spray glue, wooden screen molding nailed or screwed in place, duct tape and/or other mechanical methods.

D. Construct hygiene facilities for decontamination of workers and equipment in the same way except use true 6-mil polyethylene for walls, ceiling, and floor. Hygiene facilities for decontamination of workers and equipment must be contiguous with the work area in all instances. Construct doors so that

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flaps completely isolate the rooms in the event of air exhaust ventilation failure and allow easy access for personnel and equipment. The clean room shall be large enough to accommodate at least three workers. Prevent direct viewing into the shower, clean room, or dirty room by other personnel by constructing the walls and ceilings of these areas of black polyethylene.

E. All negative pressure enclosures shall be smoke tested after initial setup and at the beginning of each work shift.

F. Ensure that the enclosure walls, floor, and ceiling do not billow or pull out more than 6 inches from the walls or floors to allow for effective cleaning and easy movement of personnel and equipment while inside.

G. Repair damaged barriers and correct defects as soon as they are discovered. Visually inspect containment barriers at the beginning of each work period. The Government Representative may use smoke tube methods to test barrier effectiveness.

H. Do not commence work until signs are posted, required isolation barriers are erected, and the Contracting Officer or the Government Representative has authorized the work to begin. In addition, all equipment such as ladders, scaffolds, HEPA vacuums, air machines, trucks, and other tools and equipment are subject to visual inspection and bulk sampling to ensure that no asbestos debris or contamination is brought onto AEDC from the Contractor's previous jobs. Wrap in two layers of true 6-mil polyethylene sheeting or properly bag any items that have visible debris, label ASBESTOS, and remove from AEDC. Encapsulation of items is not sufficient justification to use contaminated equipment. Items that do not pass visual inspection shall not be cleaned at AEDC.

I. Turn off all building ventilation air systems during preparation and until the area has passed final visual inspection and final air sampling by the Government Representative. Remove all heating, ventilation, and air conditioning system filters before commencing asbestos removal and treat them as asbestos contaminated. Seal all air supply and return openings with true 6-mil polyethylene. Replace filters with new ones following the approved inspection.

J. Clean the work areas of all visible asbestos debris prior to placing polyethylene sheets or beginning asbestos containment work. Establish critical barriers before beginning clean-up work.

K. Completely decontaminate all ladders, vacuum cleaners, air machines, and other equipment used during abatement activities prior to removal from the abatement area. Cover all such equipment with true 4-mil polyethylene sheeting and duct tape prior to abatement activity. Seal all openings to air machines with true 6-mil polyethylene prior to their removal from an abatement area and any time they are not in operation. Wrap vacuum

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cleaner hoses with polyethylene. Seal all open ends of vacuum hoses or intake openings of vacuums with duct tape when not in operation to contain the asbestos. Seal ladder rungs, steps, and sides in polyethylene before use in an abatement area to ensure their complete decontamination following abatement. Clean and completely decontaminate all pump-up sprayers, tools, and equipment following abatement. If they cannot be decontaminated, dispose of them as asbestos material in the AEDC asbestos landfill, which is located approximately 2 miles west of the North Hap Arnold Drive and Sixth Street intersection.

3.2 RESPIRATORY PROTECTION REQUIREMENTS

A. Establish a respiratory protection program as required by CFR 1910.134 and 29 CFR 1926.1101. The Government will strictly enforce the OSHA "no facial hair/respiratory policy" for all personnel who wear respirators at any time during the job.

B. Ensure workers are clean shaven daily immediately preceding their work shift and before wearing respiratory protection.

C. Provide spectacle inserts to personnel wearing full-face respirators who normally wear spectacles, otherwise they will not be allowed in the containment area. Do not allow contact lenses to be worn in asbestos areas.

3.3 PROTECTIVE EQUIPMENT

A. Use protective equipment that meets all Government standards for use in asbestos abatement. Use coveralls having headcovers and booties attached.

3.4 LOCAL EXHAUST SYSTEM

A. Provide a local exhaust system in the asbestos control area in accordance with ASSP Z9.2. Equip exhaust with absolute HEPA filters. When possible, HEPA-filtered air shall be exhausted to the outside of buildings. Local exhaust equipment shall be sufficient to maintain a minimum pressure differential of minus 0.02 inches of water column relative to adjacent unsealed areas and provide a minimum of four complete air changes per hour. Provide continuous 24-hour-per-day monitoring of the pressure differential with an automatic recording instrument. In no case shall the building exhaust system be used as the local exhaust system for the asbestos control area. Filters on vacuums and exhaust equipment shall conform to ASSP Z9.2. Change HEPA filters at least every 700 hours for 12-inch HEPA filters or more often as required to ensure proper filtration of air. Change pre-filters as soon as visible accumulations occur on the filters and change intermediate filters at least once per shift. If filter loading occurs (i.e., visible accumulations on prefilter), change more often. Conduct air monitoring during asbestos removal to ensure filter integrity and asbestos levels outside the enclosure remain at or below 0.01 f/cc. Provide and install

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a back-up HEPA air exhaust ventilation system to be used in the event of primary system failure. Do not use a system with a remote filter housing inside gross removal areas.

3.5 COMMUNICATION DEVICES

A. Do not use any two-way communication devices unless pre-approved by the AEDC Security Forces.

3.6 CONFINED SPACES

A. Ensure entry into confined spaces is consistent with AEDC SHE Standard B5. Before entering a confined space, make oxygen and Lower Explosive Limit (LEL) measurements using a NIOSH-approved $O_2$/LEL metering device. While persons are working in a confined space, designate a stand-by person to remain outside who has been trained within the last 12 months in cardiopulmonary resuscitation (CPR) by the American Red Cross or American Heart Association.

3.7 ASBESTOS REMOVAL

A. General: Comply with the rules of Tennessee Department of Environment and Conservation, Chapter 1200-3-11-02, and 40 CFR 61. The Government will notify the Tennessee Air Pollution Control Board (ref 1200-3-11-.02 [2][d]1[ii] and 2). The removal of asbestos insulation from existing piping or other surfaces shall be subject, but not limited, to the following:

1. Require personnel who work with asbestos to use disposable coveralls; disposable head, neck, and shoe coverings; non-porous gloves; eye goggles; and a NIOSH-approved respirator.
2. Saturate all asbestos materials with wetting agent and ensure the material stays damp during removal. Do not allow asbestos insulation to drop to the floor or ground. Place asbestos in a proper container and lower to the floor or ground as appropriate.
3. Protect all existing machinery, equipment, floors, and walls from contamination by asbestos waste.
4. Control the dispersal of asbestos particles through methods such as isolation and wetting of material before removal. Keep a HEPA-filter vacuum on the job site at all times for use in clean-up of asbestos debris and during glove bag removal.
5. Do not wear protective clothing off the job site or take home for laundering. Provide a decontamination locker room and a clean locker room for personnel required to wear whole-body special clothing. Keep street clothes in the clean locker. While still wearing respirators in the decontamination room, vacuum and remove asbestos-contaminated disposable protective clothing and seal in impermeable bags or containers for disposal. Locate a shower between the decontamination and clean locker rooms and require all

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employees to shower before changing into street clothes. Filter shower water to 1 micron prior to disposal in a sanitary sewer.

6. Do not smoke, eat, drink, chew tobacco or gum, or apply cosmetics at the job site. Ensure that workers are fully decontaminated prior to conducting any of these activities.

B. Small Section Removal: Use glove bags to remove asbestos material in small sections with two-person teams specifically trained in glove bag procedures. The Government will enforce strict observance of the glove bag techniques described in 29 CFR 1926.1101 and as specified herein.

1. Place polyethylene under the work area.
2. Wear full body protection (e.g., coveralls, booties, headcovers, and gloves) in addition to respiratory protection.
3. Turn off ventilation systems located in the area during asbestos removal.
4. Clean up and seal ventilation openings in the area.
5. Establish critical barriers by sealing doors and windows or wall penetrations as necessary.
6. Do not allow asbestos insulation to drop to the floor or ground.

C. Glove Bag Requirements: In areas where insulation has caused contamination under the pipeline, pre-clean the work area of all contamination before applying polyethylene worksheet or conducting any repair or glove bag activity.

1. Use true 6-mil-thick transparent polyethylene glove bags.
2. Ensure that the diameter of the pipe insulation does not exceed one-third of the bag's working length.
3. Secure the glove bag with an air-tight seal of duct tape. Place duct tape around the pipe insulation first to form a smooth seal.
4. Ensure the glove bag is sealed by conducting a smoke test. A smoke test is conducted by inserting a smoke tube used in ventilation system analysis through the bag. If smoke leaves the bag, the seal is inadequate; and work will be discontinued until an adequate seal is achieved. Retest the glove bag after each failure.
5. Wrap any damaged pipe insulation, adjacent to the work area or capable of creating asbestos fallout as a result of glove bagging, in at least true 6-mil-thick plastic sheeting and seal tight with duct tape, or seal and repair with insulation mastic. In areas where insulation to be removed has deteriorated and the temporary repair may create a potential fiber release, use HEPA local exhaust ventilation during repair or removal.
6. Saturate all asbestos-containing materials within the glove bag with amended water prior to stripping. When using pump-up sprayers for wetting agents and encapsulants, place the spray wand through the side of the bag and seal holes prior to the start of asbestos removal.
7. Saturate the pipe with amended water after the insulation has been stripped and scrub with a brush to remove all visible asbestos material.



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8. Seal any piping insulation ends created by the repair with an EPA-approved encapsulant and an insulation mastic.
9. Use a HEPA vacuum to collapse the glove bag. Seal the vacuum in the glove bag prior to asbestos removal and run continuously during shifting of glove bag.
10. Use negative pressure enclosure with HEPA local exhaust ventilation in areas where removal of badly deteriorated insulation is to occur, regardless of the amount of asbestos to be removed.
11. Double-bag the glove bag and all other asbestos-containing waste for disposal.

D. Raised Floor Removal Requirements:

1. The concrete floor beneath the raised floor is coated with the original asbestos containing mastic or asbestos floor tile or both from the original construction of the building. At the time when the raised floor was installed, asbestos floor tile and mastic was not removed or may not have been treated in accordance with current practices. Therefore, most (not all) of the tile was removed but very little mastic removed. The mastic is in various states of dryness that can easily be scrapped from the floor if significant activity on the mastic itself is conducted. Because removal or repair of the raised floor will disturb the mastic or some pieces of floor tile still in the area, during removal and re-installation of the floor the following shall be conducted:
2. Removal of raised floor shall be performed by asbestos certified workers in accordance with the asbestos specification 02 08 00 and the following:
    a. The area shall be isolated from other activities during floor removal and installation. If entire control rooms cannot be isolated in total, then work shall be phased using small containments framed with two by fours and covered with six mil polyethylene ventilated with HEPA filtered negative air machines. Critical barriers shall also be established from the bottom of the raised floor to the concrete floor beneath.
    b. Equipment remaining in the work area shall be protected from contamination.
    c. The floor and the floor area under the raised floor shall be HEPA vacuumed to remove any flaking floor mastic, and tile debris or pedestal mastic. All mastic shall be removed. In areas where existing ground wiring or computer, etc. wiring will not permit the removal of the mastic, then only vacuum the floor to remove the mastic, loose tile, etc. without damaging the wiring to remain.
    d. While the cleaning and removal work is being conducted, all ventilation to the area shall be turned off.
3. Pedestal glue or mastic is present under most pedestals. Removal of this material will damage the original asbestos mastic underneath. Dispose of entire pedestal as asbestos and remove all mastic remaining from the floor beneath.

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E. Floor Tile and Mastic Removal Requirements:

1. All mastic shall be treated and presumed to contain asbestos unless clearly stated otherwise in contract documents.
2. Post warning signs at the entrance to the renovation area.
3. Shut off the building HVAC system during tile removal and subsequent clean-up. Seal ductwork openings with polyethylene sheeting as required in paragraph 3.1I. Establish critical barriers to the work area by sealing doors, windows, and any wall penetration.
4. Conduct no other construction work in the renovation area while removing asbestos tile or during the clean-up of removed tile.
5. Pre-clean the floor tile to be removed of all visible construction debris and dust. This pre-cleaning is not considered asbestos removal unless damaged floor tile is removed during the process.
6. Seal all holes, floor penetrations, and utility tracks found in the floor to prevent asbestos contamination by tile, mastic, or contaminated water.
7. Cover the walls of the floor tile removal area with one layer of true 4-mil polyethylene to a height of 4 feet. Do not damage wall coverings during this process.
8. Remove floor tiles using methods that will not create any visible dust. The preferred method is to wet the floor, cover it with polyethylene, and let the floor remain wet overnight or longer. The water under the plastic should loosen the glue allowing easy dust-free removal of the tile. Use a hand sprayer to mist tiles as they are being removed to further reduce dust. Remove glue that might remain on the floor using a method that does not create dust. If a sanding method is used, it shall be done under wet conditions. Use a HEPA-filtered vacuum to remove any residue left on the floor by wet methods.
9. Dispose of floor tiles, residue, mops, and rags as asbestos-containing materials in the asbestos landfill. Ensure any solvents used for mastic removal do not result in a residue that is considered to be a hazardous waste as defined by the EPA. Do not place solvents in the AEDC asbestos landfill in a free-liquid state. Dispose of all asbestos-contaminated or asbestos-containing materials in two true 6-mil polyethylene bags especially designed for asbestos disposal. If the floor is mopped with water, treat mop water as asbestos-contaminated, solidify, and place in metal drums. Rinse water used to clean asbestos mastic from surfaces shall not be disposed of in the sanitary sewer system. Do not overload bags to the point that they might rupture. Pack asbestos waste containing sharp ends in a manner that will prevent any puncture of bags. Bagged asbestos floor tile or linoleum shall be disposed of in air-tight metal drums. When using metal drums, ensure they meet the requirements of paragraph 3.10D. Transport waste to the asbestos landfill in a covered truck.
10. When the odor of solvents used for mastic removal is detected in areas adjacent to removal area, ventilate the removal area using a HEPA-filtered air exhaust system until such odors are no longer

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detected. While this is occurring, stop all solvent use and general removal until odors are no longer detected in adjacent areas.

11. Monitor area, personnel, and clearance air in accordance with OSHA and EPA guidelines (see paragraph 3.8) during floor tile removal and document asbestos fiber concentrations.
12. The Government Representative will visually inspect the area prior to job completion. Remove all tile and glue residue from floors prior to inspection. The floor shall be considered clean when mastic and floor tile on all surfaces of the floor have been removed. Clean cracks and holes, 1/16-inch or larger in width or diameter, of all mastic that may be removed using solvents, common utility knife, brushes, and HEPA vacuum without damaging the floor by chipping. Cracks and holes smaller than 1/16-inch shall be considered clean if their surfaces have been cleaned of mastics using solvents, brushes, and HEPA vacuum and shall generally be treated the same as non-cracked surfaces.
13. Ensure compliance for Class II asbestos as stated in 29 CFR 1926.1101.

F. Carpet tack strips:

1. When tack strips are present under carpet to be removed remove the carpet carefully assuring that no asbestos tile is removed with the carpet including small flakes or chips from damage caused by installing the tack strips during the original installation.
2. Remove any asbestos from the carpet using HEPA vacuum.
3. To remove the tack strips, carefully pry up the tack strips one at a time and HEPA vacuum the floor area removing any asbestos debris created during the process. Repeat this process until all the tack strips are removed.
4. Tack strips and contaminated carpet shall be disposed of as asbestos waste.
5. During this process only asbestos workers are allowed in the work area. However, if the work is carefully done, furniture may remain in the area provided that it is protected from asbestos contamination.
6. Follow other sections of the specifications such as barriers, signage and air sampling as required.

G. Base molding and glue or mastic:

1. Follow other requirements of this specification as appropriate such as the use of barriers, signage and air sampling in addition to the following:
   a. Protect the floor area with polyethylene sheeting prior to the removal of the base board and mastic.
   b. Carefully remove the base molding and dispose of the material as asbestos waste. If scraping of the molding is required to remove it from the wall, use HEPA vacuum and wet methods during the removal and cleanup process.
   c. If asbestos glue or asbestos mastic is present, scrape the material from the walls using HEPA vacuum methods and or chemicals to remove all material from the wall. Minor stain may remain as long as all solid material has been removed.
   d. If chemicals are used during this process, HEPA ventilation is required to eliminate chemicals from the building. Where

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necessary to control odors, establish critical barriers and ventilate the area.

e. During this process only asbestos workers are allowed in the work area.

f. When floor tile and floor mastic is also to be removed, the above may be modified to allow the proper sequencing of floor tile and base molding removal.

H. Window Glazing Requirements: Remove windows containing only asbestos glazing compounds by doing the following in addition to the requirements of other applicable sections of this specification.

1. Establish critical barriers to the general work area by sealing doors, wall penetrations, and ductwork vents as appropriate.
2. Post the asbestos warning signs.
3. Using duct tape and polyethylene, seal all exposed asbestos materials to prevent fiber release during window removal.
4. Pre-clean the windows, seals, floor, and ground as necessary to remove asbestos contamination located in these areas. Outside, clean the ground a distance of approximately 3 feet from the building and 2 feet to the left and right of each set of windows to remove ground contamination. Remove visible debris only. If it is more efficient to remove debris and soil or gravel to ensure complete decontamination of the area, remove soil or gravel at a sufficient depth to ensure complete asbestos removal (usually to a depth of one inch unless specified elsewhere in the specification or on a drawing). Replace gravel or soil back to grade with clean non-asbestos soil or gravel as appropriate.
5. Remove the windows with care. Wrap each in two layers of 6-mil polyethylene, label as asbestos, and dispose of in the asbestos landfill as described in paragraph 3.10.
6. Protect all existing finishes, furniture, and fixtures.
7. Coordinate the window removal with the window replacement.
8. Ensure compliance for Class II asbestos as stated in 29 CFR 1926.1101.

I. Window Caulking Requirements: Remove windows containing asbestos caulking compounds by doing the following in addition to the requirements of other applicable sections of this specification.

1. Establish critical barriers to the general work area by sealing doors, wall penetrations, and ductwork vents as appropriate.
2. Post the asbestos warning signs.
3. Pre-clean the windows, sills, floors, and ground as necessary to remove asbestos contamination located in these areas. Outside, clean the ground a distance of approximately 3 feet from the building and 2 feet to the left and right of each set of windows to remove contamination. Remove visible debris only. If it is more efficient to remove debris and soil or gravel to ensure complete decontamination of the area, remove soil or gravel at a sufficient depth to ensure complete asbestos removal (usually to a depth of

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one inch unless specified elsewhere in the specification or on a drawing). Replace gravel or soil back to grade with clean non-asbestos soil or gravel as appropriate.

4. Construct mini enclosures on the inside and outside of the windows and exhaust to the outside using HEPA filtration machines.
5. If asbestos glazing compounds are present, seal all exposed asbestos materials to prevent fiber release. Remove caulking compounds using wet methods and HEPA-filtered vacuums. Clean windows of all caulking materials or dispose of the entire window in the asbestos landfill as described in paragraph 3.10.
6. Protect all existing finishes, furniture, and fixtures.
7. Coordinate the window removal with the window replacement.
8. Ensure compliance for Class II asbestos as stated in 29 CFR 1926.1101.

J. Ventilation Ductwork Removal Requirements: Remove ventilation ductwork, which is insulated with a non-asbestos material held in place by asbestos-containing glue or mastic, by doing the following in addition to the requirements of other applicable sections of this specification and 29 CFR 1926.1101.

1. Establish critical barriers to the general work area by sealing doors, windows, wall penetrations, and ductwork vents as appropriate.
2. Post the asbestos warning signs.
3. Place a true 6-mil polyethylene worksheet under the work area.
4. Carefully wet the insulation mastic and the wrap the ductwork with one layer of true 6-mil polyethylene.
5. When cutting a duct in sections, strip insulation and mastic from the area of the cut using wet methods and seal the edges of the insulation and mastic before cutting. Take care to prevent mastic fallout from the duct due to vibration caused by cutting.
6. After removing the duct from the ceiling area, wrap the duct with a second layer of true 6-mil polyethylene and label for disposal.
7. Clean any residue mastic remaining on the floor, roof deck, duct hangers, or ceiling frames.
8. Ensure compliance for Class II asbestos as stated in 29 CFR 1926.1101.

3.8 AIR MONITORING: Monitor airborne concentrations of asbestos fibers in accordance with 29 CFR 1926.1101 and as specified below:

A. Monitoring During Asbestos Work: Provide personnel and area monitoring and establish an 8-hour time-weighted average and 30-minute excursion level concentration during the first exposure to airborne asbestos to document exposure levels and determine respiratory protection requirements. Thereafter, when the same type of work is being performed, provide area monitoring once per work shift inside the asbestos control area, outside the entrance to the asbestos control area, and at the exhaust opening of the local exhaust system. Due to other areas of the building

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being occupied during asbestos removal, collect samples from all sides of the work area to verify air quality outside the containment. This includes sampling on the second floor above the asbestos work area. Sampling shall be done each shift. If monitoring outside the asbestos control area shows airborne concentrations above 0.01 f/cc, stop all work, notify the Government Representative immediately, and correct the condition causing the increase. Provide results of sampling to the Contracting Officer as soon as possible following collection of the sample. A primary calibration standard is the standard of choice. However, a secondary standard may be used if a calibration curve for that standard is on-site in the field with the secondary standard and the curve compared to a primary standard within 3 months of the sample collection date. Conduct air sampling following the current OSHA Sampling Reference Method, which includes field calibration of sample pumps immediately before and after air sampling.

B. Monitoring After Final Clean-Up: Provide area monitoring of asbestos fibers and establish a quality level of less than 0.01 f/cc after final clean-up but before removing the enclosure of the asbestos control area. If any of the final samplings indicates a higher value, take appropriate action to re-clean the area and repeat the monitoring. Provide sample results to the Contracting Officer prior to removal of any enclosures or barriers.

C. Provide the results of all air samples as soon as possible following collection and analysis. Include the location of their collection (for example, area [where], personnel [who]), sample number, start and stop times, dates of collection, duration of sampling, flow rate in liters per minute, sample volume, total fiber count, detection limit of the analysis and airborne fiber concentration in fibers per cubic centimeter of air, name of the laboratory, and name of the person analyzing the samples. Make field notes used at the job site during sample collection available at any time to the Government Representative upon request.

D. Should you analyze bulk samples, use a laboratory accredited by NIST. Persons collecting samples shall be accredited by State of Tennessee to inspect and collect bulk samples. Accreditation shall be current at time of work. Air samples shall be analyzed by a laboratory which is in full compliance with the OSHA Reference Method and participates in the required inter-laboratory quality assurance program. Any persons analyzing these samples shall have attended the required NIOSH- and EPA-approved courses and shall have been determined proficient by successful participation in a NIOSH-PAT (air) or NIST program within the last year. This proficiency shall remain current throughout the course of the project. Return bulk samples to AEDC for disposal.

## 3.9 VENTILATION

A. In a non-gross removal area, ventilate the local areas with a HEPA-filtered air exhaust ventilation system during clean-up of areas greater than 400



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square feet or when gross asbestos contamination presents such a hazard to warrant the use of a HEPA local exhaust system to control the hazard. See paragraph 3.1K for clean-up in a gross removal area.

3.10 DISPOSAL

A. Seal asbestos-contaminated material, disposable coveralls, disposable protective equipment, polyethylene, wood, and all other material used for enclosures and scrap in a clear true 6-mil sealed impermeable bag. HEPA vacuum all trapped air from each disposal bag before sealing and place in another true 6-mil labeled, sealed impermeable bag. Bags filled or rebagged at the asbestos landfill do not have to be HEPA vacuumed (see subparagraph C). Label the outer bag with the following warning:

| |
|---|
| **DANGER** |
| **CONTAINS ASBESTOS FIBERS** |
| **MAY CAUSE CANCER** |
| **CAUSES DAMAGE TO LUNGS** |
| **DO NOT BREATHE DUST** |
| **AVOID CREATING DUST** |

B. Capture water and fluids used in wet method control and cleaning, and place in labeled, sealed impermeable containers. Label the containers with the same warning required in paragraph 3.10A. Filter water through 1-micron filters before allowing water to pass to a sanitary sewer. Do not place free liquids in the AEDC asbestos landfill.

C. Dispose of all asbestos waste in the AEDC asbestos landfill. Coordinate times of delivery with the Government Representative (normally from 7 am to 1 pm, Monday through Friday). When landfill conditions preclude adequate covering of asbestos, disposal will not be permitted. This determination shall be made by the Government Representative. These conditions will include, but are not limited to, excessive moisture in the landfill caused by the weather. When this condition occurs, the asbestos will be stored at AEDC at the Contractor's expense until the Government grants disposal authority. Do not dispose of asbestos material in any area other than the asbestos landfill. Remove and dispose of all asbestos dust particles and waste generated during each work period at the end of each work period. Place bagged waste not taken to the landfill at the end of the shift in secure areas, such as a locked panel truck, prepared for disposal as indicated in paragraph 3.11.

D. Place asbestos materials, which contain sharp edges or are too heavy to be placed in true 6-mil polyethylene bags, in clean, new, or reconditioned,

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practically air-tight metal drums. Reconditioned drums are drums which have been emptied as specified in 40 CFR 261 and repainted inside and out.

E. Do not place any hazardous waste, as defined in 40 CFR 261, in any AEDC landfill. Where hazardous waste is generated or removed, follow the procedures given in AEDC SHE Standard E18. Coordinate with the Government Representative in completing Waste Identification Form (WIF) and the Chemical Waste Data Sheet (CWDS).

F. All users of the asbestos landfill are required to obtain an AEDC Disposal Permit. (Refer to AEDC SHE Standard E7 for permitting procedure and permit requirements.)

G. Complete AEDC Asbestos Landfill Operators Log and Shipper's Log, for each load of asbestos waste. This log will be provided by the Government Representative.

H. Only properly containerized, labeled, and adequately wet asbestos accompanied by a completed AEDC Asbestos Landfill Operator's Log and Shipper's Log and an AEDC Disposal Permit shall be transported to or disposed of in the asbestos landfill.

3.11 TRANSPORTATION

A. Transport properly bagged and identified asbestos waste in a metal panel truck, trailer, or dumpster (roll-off) which is prepared as follows:
   1. Bed lined with three layers of true 6-mil polyethylene which overlap walls by at least 12 inches; line the walls with two layers of true 4-mil polyethylene; and line the doors to the enclosed bed lined with two layers of true 4-mil polyethylene.
   2. Seal the truck, trailer or dumpster (roll-off) to prevent any water or contamination leakage.
   3. Equip the doors to the lined enclosure with a security lock.

B. The truck, trailer, or dumpster (roll-off) will be inspected by the Government Representative before asbestos is loaded and after disposal.

When transporting asbestos on the open highway, follow current Department of Transportation regulations.

3.12 SEALING

A. Reseal any asbestos material that is not in the job scope for removal but is exposed as part of this work. Seal with a Government-approved bridging encapsulant and insulation mastic to contain and prevent future damage of the asbestos. If outside, ensure material used shall withstand weathering.



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## 3.13 SAFETY

A. Ensure the safe passage of persons around the area of demolition. Conduct operations to prevent injury to personnel and damage to existing equipment and structures. Minimize the generation and spread of dust and flying particles.

## 3.14 HAZARD COMMUNICATIONS

A. Maintain and implement a written Hazard Communications Program as required by 29 CFR 1910.1200 and AEDC SHE Standard A9. Ensure all employees and anyone else involved with the abatement job are familiar with the program and its location. Ensure all other requirements outlined in AEDC SHE Standard A9 are met.

## 3.15 UTILITIES

A. Do not interrupt existing utilities or commence power outages without written permission from the Contracting Officer or the Government Representative. Obtain an approved, AF 103 form in accordance with AEDC SHE Standard B1. Do not remove asbestos from active steam or electrical lines. Wait for appropriate utility outages. Provide backflow prevention devices as required to prevent cross-contamination of water supplies.

## 3.16 GENERAL CLEAN-UP

A. Remove dust, dirt, and debris caused by demolition operations from adjacent structures and improvements. Return adjacent areas to their condition prior to the start of the work.

## 3.17 LABELING

A. Stencil new and replaced insulation with the word “NON-ASBESTOS,” in accordance with ASME A13.1, at the edges of replaced sections. Indicate the direction of replacement with arrows using a 1-inch stripe to indicate the boundaries. Place the word “NON-ASBESTOS” at intervals not exceeding 25 feet using a highly visible paint.

Place labels identifying piping systems (e.g., 30 lb. steam, raw water, heated potable water) as appropriate for newly insulated piping systems.

## 3.18 DEBRIS DISPOSAL

A. Transport debris, rubbish, waste, and other non-asbestos materials resulting from demolition from the site to the construction landfill which is located approximately 2 miles west of the Avenue E and Sixth Street intersection. Do not place edibles or garbage in the construction landfill; use existing dumpster boxes.

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B. Dispose of all material contaminated by asbestos in the asbestos landfill as described in paragraph 3.10.

## 3.19 VISUAL INSPECTIONS

A. Visually inspect the work area after pre-cleaning and before placing any polyethylene sheeting except for critical barriers that shall be in place before pre-cleaning may begin. Re-clean and inspect any area where cleaning has not been adequately done before placing polyethylene sheeting. Inspect polyethylene enclosures for adequacy prior to removing any asbestos. Do not start abatement procedures prior to release by a Government industrial hygienist who will visually inspect the area for cleanliness and enclosure adequacy.

B. Assist in the visual inspection of all areas (enclosure areas cleaned, disposal bags, drums, trucks, and equipment used in asbestos removal) as requested by the Government Representative. Include the opening of drums and bags or any other inspection activity.

## 3.20 ASBESTOS ABATEMENT COMPLETION

A. Do not remove protective barricades or enclosures until the Government Representative concurs in writing. The Government may conduct independent, aggressive air monitoring at the conclusion of the removal operation to determine air quality. A reading of not more than 0.01 f/cc of air is required before barricades and enclosures shall be removed. The Government Representative will visually inspect the affected surfaces for residual asbestos material and accumulated dust, and the Contractor shall re-clean all areas showing dust or residual asbestos materials. If re-cleaning is required, monitor the airborne asbestos concentrations after re-cleaning. Remove the decontamination facility from the area following the final visual inspection and upon concurrence by the Government Representative. Encapsulate interior of polyethylene walls, ceiling, floor, pipe surfaces, and other surfaces where asbestos has been removed following visual inspection. Keep the area sealed, barriers intact, and HEPA-filtered air exhaust ventilation in operation until the results of final air samples are received. The Government Representative will visually inspect the general work area following enclosure or barrier removal to ensure the work area has been adequately cleaned and to ensure that no damage has been done to buildings or equipment.

END OF APPENDIX A.4

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APPENDIX A.5
LEAD AND HEAVY METAL REMOVAL

**PART 1 GENERAL**

1.1 SUMMARY

A. Section includes removal and disposal of lead-containing materials and other heavy metals (barium, cadmium, silver, mercury, and chromium). Both containment and non-containment methods of paint removal are included in this section.

1.2 CODES AND STANDARDS

A. AEDC Safety, Health, and Environmental (SHE) Standards latest version, available upon request from the Government.

| | | |
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| 1. | A6 | User and Subcontractor Safety. |
| 2. | A9 | Hazard Communication. |
| 3. | B1 | Master Work Permit. |
| 4. | E17 | Oil and Hazardous Substances Spill Response. |
| 5. | E18 | Managing Waste Containing Chemical or Petroleum Products. |
| 6. | E19 | Lead and Heavy Metals. |

B. American Society of Safety Professionals:

| | | |
|---|---|---|
| 1. | ASSP Z9.2-18 | Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems. |

C. Code of Federal Regulations, latest version:

| | | |
|---|---|---|
| 1. | 29 CFR 1910.134 | Respiratory Protection. |
| 2. | 29 CFR 1910.1200 | Hazard Communication. |
| 3. | 29 CFR 1926.55 | Gases, Vapors, Fumes, Ducts, and Mists. |
| 4. | 29 CFR 1926.57 | Ventilation. |
| 5. | 29 CFR 1926.62 | Lead Standard. |
| 6. | 40 CFR 260 | Hazardous Waste Management Systems: General. |
| 7. | 40 CFR 261 | Identification and Listing of Hazardous Waste. |
| 8. | 40 CFR 262 | Generators of Hazardous Waste. |
| 9. | 49 CFR 172 | Department of Transportation (DOT) Regulations for Use of Hazardous Materials Tables and for Communication. |
| 10. | 49 CFR 178 | DOT Specifications for Packaging. |

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D. Environmental Protection Agency:

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| 1. | EPA SW-846 | Proposed Sampling and Analytical Methodologies for Additions to Test Methods for Evaluating Solid Waste: Physical/Chemical Methods, 2021. |

E. The Society for Protective Coatings:

| | | |
|---|---|---|
| 1. | SSPC Guide 6-21 | Guide for Containing Surface Preparation Debris Generated During Paint Removal Operations. |

F. Underwriters Laboratories, Inc.:

| | | |
|---|---|---|
| 1. | UL 586-22 | High-Efficiency, Particulate, Air Filter Units. |

## 1.3 DEFINITIONS

A. Action Level: Employee exposure, without regard to use of respirators, to an airborne concentration of lead of 30 µg/m3 averaged over an 8-hour period. As used in this section, "30 µg/m3" refers to the action level.

B. Area Monitoring: Sampling of lead concentrations within the lead-control area and inside the physical boundaries which is representative of the airborne lead concentrations which may reach the breathing zone of personnel potentially exposed to lead.

C. Change Rooms and Shower Facilities: Rooms within the designated physical boundary around the lead-control area equipped with separate storage facilities for clean protective work clothing and equipment and for street clothes which prevent cross-contamination.

D. Clearance Level: Prior to the moving or removal of enclosures used for lead abatement, air samples will be taken by the Government Representative to ensure that airborne levels of lead are at or below 3 µg/m3. In addition, a detailed visual inspection will be conducted by the Government Representative for all surfaces and equipment in the containment or control area. Surfaces include any portion of the containment including walls, ceilings, and floors, scaffolds, and any equipment or objects that are present in the containment or that have been used in the containment. The inspection will be conducted by wiping a clean cloth across all surfaces and inspecting the cloth for evidence of any dust. If any dust is found on the cloth, the Contractor shall re-clean the entire containment until a detailed inspection is passed. All dust will be assumed to be lead- or heavy-metal contaminated. When enclosures are not required, inspection of the work area will be conducted to ensure adequate decontamination of the area. This method will be used before moving or removing containments or enclosures. Before containments are removed from AEDC, wipe and/or microvac samples will be collected from representative surfaces to

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determine if the containments have been cleaned to a level of 500 µg /ft2 or less. If any one sample exceeds 500 µg /ft2, then the entire containment shall be re-cleaned.

E. Decontamination Room: Room designated for removal of contaminated personal protective equipment (PPE).

F. Designated Lead-Abatement Supervisor: A person who has attended any 3- to 5-day lead-abatement course taught in the United States. The person shall be knowledgeable of Occupational Safety and Health Administration (OSHA), Environmental Protection Agency (EPA), and other Government regulations.

G. Eight-Hour Time Weighted Average (TWA): Airborne concentration of lead averaged over an 8-hour workday to which an employee is exposed.

H. Grit blasting: Remove paint with recyclable steel grit or recyclable steel grit embedded in a synthetic open-cell polymer sponge.

I. High-Efficiency Particulate Air (HEPA) Filter Equipment: HEPA-filtered vacuuming equipment with a UL 586 filter system capable of collecting and retaining lead-contaminated dust. A high-efficiency particulate filter means it is 99.97 percent efficient against 0.3-micron-size particles. This equipment may be containment exhaust systems or hand held paint removal equipment such as peeners, needle-guns, grinders, or sanders.

J. Lead: Metallic lead, inorganic lead compounds, and organic lead soaps.

K. Lead-Control Area: An enclosed area or structure with full containment to prevent the spread of lead dust, paint chips, or debris of lead-containing paint-removal operations. The lead-control area is isolated by physical boundaries to prevent unauthorized entry of personnel.

L. Lead Permissible Exposure Limit (PEL): Fifty µg/m3 as an 8-hour TWA as determined by 29 CFR 1926.62. If an employee is exposed for more than 8 hours in a work day, the PEL shall be determined by the following formula: PEL (µg lead/m3) = 400/No. hrs. worked per day

M. Microvac: Alternate sampling method for surfaces that are not conducive to wipe sampling. Sampling is conducted using a 37mm air sampling cassette with 0.8 micron filters at a flow rate of approximately 4 liters per minute. Samples are vacuumed from a 6 in2 area unless conditions require a smaller or larger sample area. Results will be reported in µg /ft2.

N. µg/m3: Micrograms per cubic meter of air (refers only to lead in this document).

O. µg /ft2: Micrograms per square foot of surface (refers only to lead in this document).

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P. Personal Monitoring: Sampling of lead concentrations within the breathing zone of an employee to determine the 8-hour TWA concentration in accordance with 29 CFR 1926.62. Samples shall be representative of the employee's work tasks. Breathing zone shall be considered an area within a hemisphere, forward of the shoulders, with a radius of 6 to 9 inches and the center at the nose or mouth of an employee.

Q. Physical Boundary: Area physically roped or partitioned around an enclosed lead-control area or area where HEPA filtered hand or power tools are used or chemical paint removal is being conducted. The barriers are placed to limit unauthorized entry of personnel. As used in this section, "inside boundary" shall mean the same as "outside lead-control area but within the roped-off area." In areas where enclosures are not used, this is the area where lead abatement work is being conducted.

## 1.4 SUBMITTALS

A. Evidence satisfactory to the Contracting Officer that the firm performing lead abatement has at least one designated employee on site during all abatement activity who has attended a lead-abatement course taught in the United States and who is knowledgeable in all aspects of lead abatement. Show this by course certification and description of past lead-abatement experience which includes a list of previous clients and a resume. Have physically on each individual job site at least one such designated supervisor directly responsible for the work whenever any phase of the project is in progress. Multiple enclosures being worked at the same time shall require individual lead-abatement supervisors responsible for each enclosure.

B. Written certification that all employees involved in lead abatement have received training and medical examinations as required by 29 CFR 1926.62. Certification includes respirator fit-test and training records and a copy of the respiratory protection program. Include with the certifications, dates of the most recent training, medical examinations, and a physician's statement indicating that workers are physically able to perform lead-abatement work and use the required respiratory and general body protection. Provide this information for all personnel including management and any air-monitoring personnel on the job site before their first entry within the job site. Training shall be accomplished prior to the time of initial job assignment. Keep the job information current for all employees during all phases of the job.

C. Any citation or notice of violation from any Government agency issued as a result of work performed under this contract or any contract engaged in during the last 3 years. Submit a brief explanation of any cited incident. If none have been received, submit certification to that effect.

D. Environmental, health, and safety plan that addresses all environmental, health, and safety aspects of the job. Submit this plan within 30 calendar

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days after award of the contract and before any field work begins. The plan shall include the following information:

1. Identification of hazardous waste associated with the work.
2. Estimated quantities of wastes and/or hazardous wastes to be generated and disposed of.
3. Names and qualifications (experience and training) of personnel who will be working on site with hazardous waste.
4. List of the waste-handling equipment to be used in performing the work, to include cleaning, volume-reduction, and transport equipment.
5. Spill prevention, containment, and clean-up contingency measures to be implemented. Reference AEDC SHE Standard E17.
6. Work plan and schedule for waste containment, removal, and disposal. Waste shall be cleaned up and containerized daily.
7. Methods to control fugitive air emissions.
8. Methods to control employee exposure to lead during removal.
9. Methods to ensure safety including a lockout/tagout plan; job safety analysis; tool box safety meeting minutes; accident reports and investigations; lead-testing data/certification; fall protection systems; shop drawings; procedures for disposing waste, scrap, and excess materials; and procedures for work involving transportation or disposal of hazardous waste. The plan shall address all other environmental, health, and safety concerns associated with the job. The plan shall also include fire safety plan and procedures for addressing other work area emergencies in compliance with 29 CFR 1926.55, and a hazardous waste management plan in accordance with 40 CFR 260 and with applicable requirements of federal and local hazardous waste regulations.

E. A copy of the hazard communications program and certification that all employees have been trained concerning the hazard communications standards and the written program in accordance with 29 CFR1910.1200 and AEDC SHE Standard A9.

F. A list of products to be used and a Safety Data Sheet (SDS) for each. Products include, but are not limited to, aerosol sprays of any kind, wetting and cleaning agents, fuels, solvents, paints, etc. SDS's shall be kept in a notebook and indexed for easy reference. This SDS notebook shall remain available to all employees on the job site at all times.

G. A detailed job-specific lead abatement plan of the work procedures to be used in the removal of lead paint. The plan shall include a sketch showing the locations, size, and details of lead-control areas and the location and details of decontamination rooms, change rooms, shower facilities, and mechanical ventilation systems. Include eating, drinking, smoking, and restroom procedures; interface of trades; sequencing of lead-related work; collected wastewater (to include shower water) and paint debris disposal plan; air sampling plan; respirators; protective equipment; and a detailed

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description of the method of containment of the operation to ensure that airborne lead concentrations of 3 µg/m3 are not exceeded outside the lead-control area. Include air sampling, training, strategy, sampling methodology, frequency, duration of sampling, and qualifications of air-monitoring personnel in the air-sampling portion of the plan. The plan shall also include measures to minimize the amount of waste generated. This may include segregation of hazardous and non-hazardous waste to reduce volume, recycling of packaging materials, inventory controls to reduce left over chemicals, etc. The plan shall also include methods to protect spent abrasive media from the elements to prevent rainwater infiltration. Obtain approval of the plan prior to the start of paint-removal work.

H. Air and substrate sampling reports.

I. Testing laboratory qualifications including the name, address, and telephone number of the testing laboratory selected to perform the monitoring, testing, and reporting of airborne and substrate concentrations of lead. The laboratory shall be accredited by the American Society of Safety Professionals (ASSP). Provide ASSP documentation along with date of accreditation/re-accreditation. Samples collected to determine if materials are hazardous waste shall be analyzed by a laboratory qualified to conduct such analysis following Environmental Protection Agency Document SW-846. Provide split samples of any materials or media to the Government as requested for Government analysis.

J. Air-monitoring results submitted within 24 hours following the monitoring, signed by the person performing the air monitoring, the employee who analyzed the sample, and the designated site superintendent responsible for the lead-abatement operation. See paragraph 3.2.B.3 for additional information.

K. A list of all equipment, including water, air filters, and respirators to be used, and manufacturer's literature showing that the equipment and material meet all EPA, OSHA, and ANSI standards for use in lead-abatement activities. Include certification that vacuum- and air-filtration devices are filtered with HEPA filters. Include operating instruction for paint-removal equipment.

L. Equipment rental notifications (see paragraph 1.5C).

M. Shower water sample test results.

1.5 EQUIPMENT

A. Respiratory Protection Requirements: Establish a respiratory protection program as required by 29 CFR 1910.134 and 29 CFR 1926.62. The Government will strictly enforce the OSHA "no facial hair/respiratory policy" for all personnel who wear respirators at any time during the job.

1. Ensure workers are clean shaven daily immediately preceding their work shift and before wearing respiratory protection.

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2. Provide spectacle inserts to personnel wearing full-face respirators who normally wear spectacles; otherwise, spectacles shall not be worn in lead-abatement areas.

B. Special Protective Clothing: Furnish personnel who will be exposed to lead-contaminated dust with appropriate disposable protective whole-body clothing, head coverings, gloves, and foot coverings. Use coveralls having head covers and booties attached. Furnish appropriate disposable plastic or rubber gloves to protect hands. Reduce the level of protection only after obtaining concurrence from the Government Representative.

C. Rental Equipment Notification: If rental equipment is to be used during lead-containing paint handling and disposal, notify the rental agency in writing concerning the intended use of the equipment. Furnish a copy of the written notification to the Contracting Officer (see paragraph 1.4L).

D. Vacuum and Negative Air Machine Filters: UL 586-labeled HEPA filters.

E. Decontamination: Completely decontaminate all ladders, vacuum cleaners, air machines, and other equipment used during abatement activities prior to removal from the abatement area. If they cannot be decontaminated, then dispose of them as hazardous waste.

F. Condition: Clean all equipment used at non-AEDC job sites prior to arrival at AEDC. Any contaminated equipment identified during inspection of incoming vehicles shall be removed from AEDC until cleaned. Any such equipment shall not be cleaned at AEDC. Do not remove any equipment used at AEDC that has not been decontaminated and inspected by the Government Representative. All equipment and other articles are subject to inspection by Government Representatives upon arrival or exit from AEDC. Contaminated equipment identified on out-going vehicles will be impounded by the Government until the Contractor conducts adequate decontamination procedures.

## PART 2 PRODUCTS

2.1 ABRASIVE MATERIALS (If applicable)

A. Abrasive blasting materials shall meet the requirements as specified in the paint schedule under "Surface Preparation."

B. Limits on the Composition of Abrasive Materials: The soluble metal content and the total metal content shall not exceed values which would cause a material to be classified as a hazardous waste as defined in 40 CFR 261.

C. Unless approved by the Government, the use of abrasive media or additives to abrasive media that solidify after use is prohibited.

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## PART 3 EXECUTION

3.1 PROTECTION

A. Notification: Notify the Government Representative 30 days prior to the start of any paint-removal work.

B. Lead-Control Area Requirements:

1. Establish a lead-control area by completely enclosing the area or structure where lead-containing paint-removal operations are to be performed or isolate using barrier ropes and signs if containment is not required.
2. When enclosures are not required, place polyethylene sheeting on the ground or floor of the work area and out from the building a distance of at least twenty feet. Cover non-moveable objects with protective covering such as polyethylene. Close and tightly lock doors and windows when working near doors or windows. Protect brick and walls from contamination and remove prior dust and debris by HEPA vacuum and wet wiping. If doors and windows will not tightly close, seal with polyethylene from the inside of the building. If storm windows must be removed to allow repainting of windows, wet wipe and HEPA vacuum the entire storm window (both sides) to remove any lead contamination.
3. When building occupants are allowed to remain in the premises provide a safe, lead free access to and from the building during the work and at the end of the day. Provide adequate security to the work area and equipment to prevent any hazard to the area occupants.
4. Contain removal operations by the use of a negative-pressure full-containment system. Also see paragraph 3.1.I.
5. Enclosures used to control lead emissions shall consist of the lead-abatement work area, and a decontamination unit for personnel, consisting of a dirty equipment room, a shower equipped with hot and cold running water, and a clean change room for workers. A separate decontamination chamber shall be constructed for equipment decontamination and the safe passage of hazardous wastes from the work area to the outside. Removal of contaminated dust-collecting filters from the recycling abrasive blasting and vacuuming machines shall be accomplished in a manner to prevent the contaminated dust from entering the environment. All personnel assigned to changing filters and cleaning the machinery shall be fully clothed with approved protective clothing and equipment. The clean room shall be equipped with lockers where clean respirators and street clothes are stored. No contaminated articles shall enter the clean room. Contaminated articles shall remain in the work area until cleaned or disposed of as hazardous waste. Contaminated articles to include HEPA filters and PPE shall be segregated from waste abrasive media. The decontamination units shall be constructed contiguous to the work area (enclosure), and the shower shall be

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constructed in a manner that requires the worker to pass from the dirty room through the shower stall into the clean room.

6. Filter shower water through a 1-micron filter or other filter system that will result in equivalent water filtration. Collect water and sample to determine if lead levels in the water are below 100 ppb if tested by a qualified laboratory or 50 ppb if tested by an approved field kit. If levels are below these concentrations, then the water may be discharged into the sanitary sewer. All water shall be collected and sampled before discharge using either a field measuring kit as described below or the results from a qualified laboratory. Sample results from the qualified laboratory shall be submitted to the Contracting Officer for approval prior to discharging the water. The Government Representative may collect and test duplicate samples to ensure the integrity of the qualified laboratory performing the analysis. Field analysis conducted using portable test kits will be approved by the Government industrial hygienist prior to use. Colorimetric test kits such as CHEMetrics, Inc., Cat. No. K-8350, are such kits. Any water tested using field kits that indicates lead levels above 50 ppb shall be re-filtered and re-tested until field measurements are below 50 ppb or the water has been found to be less than 100 ppb using laboratory testing from a qualified laboratory. Sample results from the qualified laboratory shall be submitted to the Contracting Officer for approval prior to discharging the water. If field analysis is used, a Government Representative will be present during all testing and field analysis. The holding tank used for the collection of contaminated water will be locked and unlocked by the Government Representative to prevent the release of contaminated water to the environment before adequate filtering. The Contractor shall provide the means of locking the tank; however, the Government Representative will provide the lock.

7. Enclosures used for lead abatement shall be constructed of materials strong enough to withstand environmental elements (i.e., wind, rain, and snow) when outside. The containment shall comply with a Class 1 containment system as described in SSPC Guide 6 (CON). The containment shall be made of impermeable walls with rigid or flexible framing, fully sealed joints, airlock entryways, and HEPA-filtered negative air achieved by forced air flow (verified by instrument monitoring). Air flow in the containment shall be maintained at a pressure differential of minus 0.02 inch of water. Air flow in the containment shall be maintained at a minimum of 100 ft/min for the cross draft ventilation and at least 60 ft/min for the down draft ventilation. Construct hygiene facilities for decontamination of workers and equipment similar to the main containment. Construct doors so that flaps completely isolate the enclosure in the event of air exhaust failure and allow easy access for personnel and equipment. The clean room shall be large enough to accommodate at least three workers at any one time. Prevent direct viewing into the shower, clean room, or dirty room by other personnel by constructing

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the walls and ceiling of these areas of black polyethylene or similar material. Provide detailed specifications, drawings, and load calculations of containment structure for 100 percent containment of lead emissions, grit, and dust. If the containment is to be used for abrasive blasting, blast shields shall be used to protect the outside walls of the containment from damage by blast media.

C. Protection of Existing Work to Remain: Perform paint removal work without damage or contamination of adjacent areas. Where existing work is damaged or contaminated, restore work to its original condition or better.

D. Boundary Requirements: Provide physical boundaries around the lead-control area by roping off the area. Ensure that airborne concentrations of lead will not exceed 3 μg/m3 outside the lead-control area or enclosure.

E. Change Room and Shower Facilities: Provide clean change rooms and shower facilities within the physical boundary around the designated lead-control area in accordance with requirements of 29 CFR 1926.62 and as outlined in paragraph 3.1.

F. Mechanical Ventilation System:
   1. Use adequate ventilation to control personnel exposure to lead in accordance with 29 CFR 1926.57.
   2. Local exhaust system: Provide a local exhaust system in the lead-abatement area (enclosure) in accordance with ASSP Z9.2. Equip exhaust with absolute HEPA filters. HEPA-filtered air shall be exhausted to the outside of buildings when work is conducted inside buildings. Local exhaust equipment shall be sufficient to maintain a minimum pressure differential of minus 0.02 inches of water column relative to adjacent unsealed areas. Provide continuous 24-hour-per-day monitoring of the pressure differential with an automatic recording instrument. Filters on vacuums and exhaust equipment shall conform to ASSP Z9.2. Change pre-filters and HEPA filters often enough to ensure that lead concentrations at the exhaust are at or below 3 μg/m3. Provide and install a back-up HEPA air-exhaust ventilation system to be used in the event of primary system failure. Do not use a system with a remote filter housing inside the lead-removal area.

G. Personnel Protection: Personnel shall wear and use protective clothing and equipment as specified herein. Eating, smoking, and/or drinking are not permitted in the lead-control area. No one shall be permitted in the lead-control area unless they have received appropriate training and protective equipment.

H. Warning Signs: Provide warning signs at approaches to lead-control areas. Locate signs at such a distance that personnel may read the sign and take

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the necessary precautions before entering the area. Signs shall comply with the requirements of 29 CFR 1926.62.

I. During building renovations where abrasive blasting is not used and paint must be removed by other means, such as HEPA-shrouded mechanical removal equipment, critical barriers and polyethylene enclosures may be used. The requirement for showers and HEPA negative pressure exhaust shall be dependent on air concentrations. If air concentrations are below the action level for lead, then showers shall not be required. Hand and face washing facilities shall be required. Submit methods of removal and control as required in paragraph 1.4.D. If work is done outside, then air concentrations within the work area shall be within acceptable limits as indicated in paragraph 3.1.D. above. Submit methods of removal and control as required in paragraph 1.4.D.

3.2 WORK PROCEDURES

A. Perform removal of lead-containing paint in accordance with the approved lead- abatement plan. Use procedures and equipment required to limit occupational and environmental exposure to lead when lead-containing paint is removed in accordance with 29 CFR 1926.62 and 40 CFR 262, except as specified herein. (Dispose of removed paint chips and associated waste in compliance with federal and local requirements.) The hazardous waste shall be properly drummed and labeled as required by 49 CFR 172 prior to being moved by the Contractor to an accumulation point, which is within one mile of the job site (see paragraph 3.9F).

1. Personnel exiting procedures: Whenever personnel exit the lead-controlled area, they shall perform the following procedures and shall not leave the work place wearing any clothing or equipment worn during the work day:
    a. Vacuum themselves.
    b. Remove protective clothing in the decontamination room, and place them in an approved impermeable 6-mil polyethylene disposal bag.
    c. Shower.
    d. Change to clean clothes prior to leaving the physical boundary designated around the lead-contaminated job site.

B. Air Monitoring: Monitor airborne concentrations of lead in accordance with 29 CFR 1926.62 and as specified below:

1. Monitoring during lead-abatement work: Provide personnel and area monitoring and establish an 8-hour TWA during the first exposure to airborne lead to document exposure levels and determine respiratory protection requirements. Provide continuous area monitoring during each work shift inside the lead-control area, outside the entrance to the lead-control area, and at the exhaust opening of the local exhaust system. If monitoring outside the lead-control area shows airborne concentrations above 3.0 µg/m3, stop all work, notify the

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Government Representative immediately, and correct the condition causing the increase. Conduct air sampling following OSHA and NIOSH guidelines which includes field calibration of sample pumps immediately before and after air sampling.

2. Collect personal air-monitoring samples on employees who are anticipated to have the greatest risk of exposure. In addition, take air-monitoring samples on at least 25 percent of the work crew or a minimum of two employees, whichever is greater, during each work shift.
3. Submit the results of all air samples taken in support of the contract within 10 days following their collection. Include the location of their collection (for example, area where, personnel who, sample number, start and stop times, dates of collection, duration of sampling, flow rate in liters per minute, sampling volume, total lead concentration in µg/m3, detection limit of the analysis, TWA of the representative employee's exposure, name of the laboratory, and name of the person collecting the sample and analyzing the samples). This information shall be submitted in a formal report to the Contracting Officer. Within 24 hours of sample collection, make the sampling available for the Government Representative's review. These may be laboratory reports or rough draft field data. Make field notes used at the job site during sample collection available at any time to the Government Representative. Notify the Contracting Officer immediately of exposure to lead at or in excess of 3 µg/m3 outside the lead-control area.
4. Monitoring after final clean-up: Provide area monitoring of lead concentrations and establish an air quality level of 3 µg/m3 or less after final clean-up. Before moving or removing the enclosure from the lead-abatement control area, the Government Representative will conduct a visual inspection of the area to determine its cleanliness. Once the visual inspection has been passed, the Government Representative will collect clearance air samples to determine lead air concentrations. If the air samples indicate levels above 3 µg/m3, then the Contractor shall re-clean the enclosure, and the visual inspection and clearance air sampling shall be repeated. This shall continue until an inspection is passed and a clearance sample is obtained. The Contractor shall assist the Government Representative to ensure adequate inspection of all surfaces of the enclosure and work areas. See paragraph 1.3.D.

C. Once the visual inspection and air samples meet necessary requirements, remove the enclosure.

3.3 LEAD-CONTAINING PAINT REMOVAL

A. Comply with the applicable procedure in Annex B, AEDC SHE Standard E19 and the following: Manual or power sanding/grinding of interior and exterior surfaces is not permitted unless accomplished in enclosure or done

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so using proper barriers, signs, HEPA vacuum attachments on equipment, and wet methods. Also see paragraph 3.1.B. and I. Remove paint within the areas as required to allow cutting or painting as identified under scope of lead abatement in section 1.1B. on the drawings and in the paint schedule in order to completely expose the substrate. Take whatever precautions are necessary to minimize damage to the underlying substrate if painting.

1. Mechanical paint removal and blast cleaning: Perform mechanical paint removal and blast cleaning in lead-control areas using negative-pressure full-containment with HEPA-filtered exhaust. Collect paint residue and spent grit (used abrasive) from blasting operations for disposal in accordance with CFR and local requirements.
2. Abrasive blasting and vacuum filtering system: The system used to collect residue paint and grit blast shall be contained in a HEPA-filtered exhausted enclosure to ensure that the emptying of residue, the maintenance of systems, and/or the replacement of filters are done in an enclosed restricted area that shall prevent the contamination of the outside work area. This enclosure area shall be constructed in accordance with the requirements for the main enclosure and will be inspected and cleared by the Government Representative prior to its removal as indicated in paragraph 3.10.

B. Do not conduct paint removal if wind speeds at the job site are greater than 20-miles per hour unless paint removal is being accomplished by chemical methods. In addition, work must stop and cleanup of all debris must occur before any precipitation begins.

C. Do not leave debris on polyethylene or other parts of the work area overnight even if the work is not complete. Clean up all debris and contaminated polyethylene at the end of each shift.

## 3.4 CLEARANCE PRIOR TO PRIMER APPLICATION

A. Before primer application, a detailed visual inspection will be conducted by the Government Representative for unprepared surfaces and visible dust. Any visible dust will be assumed to be lead contaminated. See paragraph 1.3.D. The work area including enclosure floors, walls, and ceiling shall be cleaned. If visible dust or unprepared surfaces are identified, the work area shall be re-cleaned and the inspection will be repeated. The outer enclosure shall remain intact and HEPA-filtered exhaust shall be maintained until final clearance air and inspection is conducted before enclosure removal as indicated in paragraph 3.10. Any personnel entering the work area are required to wear protective coveralls, head cover, gloves, and other necessary equipment including respirator until final clearance sampling of 3 µg/m3 is obtained.

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## 3.5 SAFETY

A. Ensure the safe passage of persons around the area of work. Comply with AEDC SHE Standard A6. Conduct operations to prevent injury to personnel and damage to existing equipment and structures.

## 3.6 UTILITIES

A. Do not interrupt existing utilities or commence power outages without written permission from the Government Representative. Obtain an approved, AF 103 form, in accordance with AEDC SHE Standard B1, from the Government Representative prior to interrupting utilities. Do not remove lead from active steam, electrical lines, or high-pressure lines. Wait for appropriate utility outages. Provide back flow prevention devices as required to prevent cross-contamination of water supplies.

## 3.7 COMMUNICATION DEVICES

A. Do not use any two-way communication devices unless pre-approved by the AEDC Security Forces.

## 3.8 WORK CLEARANCES

A. Obtain work permits as required by AEDC SHE Standard B1. Perform hazard analysis to ensure all possible health hazards (e.g., toxic gases) have been evaluated and properly controlled. Before entering into a work space, make oxygen and Lower Explosive Limit (LEL) measurements using an NIOSH-approved 02/LEL metering device. While persons are working, designate a stand-by person, who has been trained within the last 12 months in cardiopulmonary resuscitation (CPR) by the American Red Cross or American Heart Association, to remain outside.

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## 3.9 CLEAN-UP AND DISPOSAL

A. Clean-Up: Maintain surfaces of the lead-control area free of accumulations of paint chips and dust. Restrict the spread of dust and debris; keep waste from being distributed over the work area. Do not dry sweep or use compressed air to clean up the area. At the end of each shift and when the paint-removal operation has been completed, clean the area of visible lead paint contamination by vacuuming with a HEPA-filtered vacuum cleaner. Prevent ground contaminants by enclosing the work area as specified in paragraph 3.1.B. Pre-clean the ground or floor of visible paint chips and obvious visible lead contamination prior to enclosure construction to remove visible contamination already present in the work area.

B. Visual Inspections: Visually inspect the work area after pre-cleaning and before placing any polyethylene sheeting. Re-clean and inspect any area where cleaning has not been adequately done before placing polyethylene sheeting. Inspect enclosures for adequacy prior to removing lead. Do not start abatement procedures prior to release by a Government industrial hygienist who will visually inspect the area for cleanliness and enclosure adequacy.

C. Inspection Assistance: The designated lead-abatement supervisor shall assist in the visual inspection of all areas (enclosure areas cleaned, drums, trucks, and equipment used in lead abatement) as requested by the Government Representative. This includes any inspection activity required.

D. Testing of Lead-Containing Paint Residue and used Abrasive: Where indicated or when directed by the Government, collect samples and deliver to Government Representative lead-containing paint residue and used abrasive in accordance with 40 CFR 261 and AEDC SHE Standard E18.

E. Non-Hazardous Debris Disposal: Transport debris, rubbish, demolition waste, and other non-hazardous wastes that are clearly construction and demolition debris resulting from work from the site to the construction landfill which is located approximately 2 miles west of the intersection of North Hap Arnold Drive and 6th Street. Do not place edibles or garbage in the construction landfill; use existing dumpster boxes. Dispose of all material contaminated by lead as hazardous waste in compliance with AEDC SHE Standard E18.

F. If project debris is verified as non-hazardous waste by the Government, lead-containing paint residue and used abrasive shall be placed directly in a roll-off box for disposal at the AEDC construction/demolition landfill. The Government will furnish roll-off boxes and transportation to the landfill.

G. Hazardous Waste Disposal:
1. Where hazardous waste (as identified or listed by 40 CFR 261) is generated, follow the procedures given in AEDC SHE Standard E18 for storing and turning in hazardous waste for proper disposal. These procedures include the requirement for completion of Forms Waste Identification Form (WIF) and the Chemical Waste Data Sheet (CWDS), which will be furnished by the Government Representative. Return the completed forms to the Government Representative and coordinate with the AEDC Hazardous Waste Operations Group (454-3628 or 454-3521) prior to transporting the drums to the designated Base accumulation point or Base permitted storage facility as directed by the Hazardous Waste Operations Group.
2. Collect lead-contaminated waste, scrap, debris, bags, containers, equipment, and clothing, which may produce airborne concentrations of lead particles.
3. Store removed paint, lead-contaminated clothing and equipment, and lead-contaminated dust and cleaning debris into U. S. Department of Transportation 49 CFR 178-approved 55-gallon drums which shall be provided by the Contractor. Provide Government approved closed head drums for liquid waste and open

tops for solid waste. Liquid wastes must not be stored or transported in open head drums. Government will test the contents to determine the hazardous characteristics, and provide the test results. The Contractor shall label and move the waste to a designated accumulation point in accordance with 40 CFR 262 and 49 CFR 172. The Government will be responsible for the further transportation, storage and disposal of the waste.

4. Do not place any hazardous waste, as defined in 40 CFR 261, in any AEDC landfill.

## 3.10 LEAD ABATEMENT COMPLETION

A. Samples and Tests: Do not remove protective barricades or enclosures until concurrence is received in writing from the Government Representative. The Government Representative will visually inspect the surfaces of both the enclosure and abated substrate for visible dust contamination, and the Contractor shall re-clean all areas as required. Also see paragraph 1.3.D. Wipe and/or microvac samples will be collected to determine that the lead surface contamination does not exceed 500 µg/ft2 of surface. If any wipe and/or microvac samples do not meet this criterion, re-clean the entire work area. If re-cleaning is required, monitor airborne lead concentrations during and after re-cleaning. Once the visual inspection has been made and wipe and/or microvac samples indicate clean surface levels, clearance air monitoring will be accomplished. If airborne lead concentrations exceed 3 $\mu g/m^3$, re-clean the area. Clearance monitoring will be repeated by the Government Representative as necessary. HEPA-filtered air systems shall be operated continually until adequate clearance levels are met. In addition to air and wipe samples, soil, gravel, and water samples will be taken in the work area to determine that lead contamination in the area is no greater than 3.7 parts per million above pre-construction levels in soil and gravel or 100 parts per billion in water regardless of the pre-construction levels. Sample results below these limits are required before enclosures or barricades are removed. Shower water shall be sampled prior to disposal to ensure that the 100-parts-per-billion level is met. See paragraph 3.1.B.4. Analysis of air and wipe samples collected and tested by the Government Representative may take 1-1/2 to 3 work days, with bulk and water analysis taking up to 5 work days. Keep the area sealed, barriers intact, and HEPA-filtered air exhaust ventilation in operation until the results of final air samples are received.

B. Work Area Inspection: The Government Representative will visually inspect the general work area following enclosure removal to ensure the work area has been adequately cleaned and to ensure that no damage has been done to buildings, structures, or equipment.

END OF APPENDIX A.5

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APPENDIX A.6
PAINTING AND COATING

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## PART 1 GENERAL

### 1.1 SUMMARY

A. Provide all materials, labor, paint, and equipment for cleaning, surface preparation, priming, and painting of exterior surfaces of the bus duct and the transformer and associated surfaces, in accordance with this Appendix. Varying degrees of lead may be present in the existing paint on the transformer. Lead abatement does constitute a portion of this project, and varying degrees of lead may be present in any resulting debris. Only debris from rusted areas that is wire brushed off (or similarly removed) needs to be fully contained. All other areas to be washed and rinsed need to be tested at intervals to verify no lead containing wash/rinse water is being discharged. The non-lead containing water does not have to be contained.

### 1.2 SUBMITTALS

A. Paint manufacturers' product data sheets.

B. Manufacturers' material safety data sheets for each paint product.

C. Paint manufacturers' certifications that materials meet or exceed requirements outlined in Part 2.

### 1.3 CODES AND STANDARDS.

A. <u>The Society for Protective Coatings:</u>

| | | |
|---|---|---|
| 1. | SSPC Guide 6-21 | Guide for Containing Surface Preparation Debris Generated During Paint Removal Operations. |
| 2. | SSPC PA1-16 | Shop, Field, and Maintenance Painting of Steel. |
| 3. | SSPC PA2-22 | Measurement of Dry Coating Thickness with Magnetic Gages. |
| 4. | SSPC SP1-16 | Solvent Cleaning. |
| 5. | SSPC SP2-18 | Hand Tool Cleaning. |
| 6. | SSPC SP3-18 | Power Tool Cleaning. |
| 7. | SSPC SP6-07 | Commercial Blast Cleaning. |
| 8. | SSPC SP12-02 | Surface Preparation and Clean of Metals by Waterjetting Prior to Recoating. |

### 1.4 DEFINITION

A. The term "paint," as used herein, means all coating system materials, including primers, emulsions, acrylics, and other applied products whether used as a prime, intermediate, or top coat.

## PART 2 PRODUCTS

### 2.1 PAINT

A. Materials specified are intended to establish standards of quality and generic types. All coatings shall be asbestos and heavy-metal free, and VOC compliant

1. Cleaning Agent: (for all surfaces):
   a. A biodegradable surface conditioner that helps remove soluble salts and other contaminates.
   b. Application: Apply liberally; by pump up sprayer, conventional or airless paint sprayer, brush or roller, hand sponge or rags. Do not thin.
2. Primer #1 (for all surfaces):
   a. Epoxy: Rust penetrating primer for use over marginally prepared surfaces when abrasive cleaning is not possible.
   b. Tightly adhered existing coatings are to remain.
      Coats: One full coat.
      DFT: 1.5 - 2.0 mils.
      Color: Transparent (add tinting).
      Application: Brush, Roller or Spray.
3. Primer #2: (for all surfaces).
4. All weather flat finish. Coating shall be low VOC, impact and abrasion resistant, and can be applied in normal moderate environmental conditions.
   a. Coats: One full coat.
      DFT: 3.5 – 4.0 mils.
      Color: Off white.
      Application: Brush, Roller or Spray.
5. Coating A: (for all surfaces):
   a. Coating shall be low VOC, impact and abrasion resistant, and can be applied in normal moderate environmental conditions.
      Coats: Two full coats.
      DFT: 3.5 – 4.0 mils each coat.
      Color: Galvano or Platinum.
      Application: Brush, Roller or Spray.
   b. Use a “flow-coat”, application instead of a brush, roller or spray application for all inaccessible areas such as between the radiator fins.

## PART 3 EXECUTION

3.1 All work performed on-site shall comply with the Statement of Work.

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3.2 Transformer cleaning and surface preparation before painting

A. The Contractor shall clean and inspect the transformer surface of all steel and iron to ensure that all weld splatter, weld slag, grease, oil, and other foreign materials have been removed. The Contractor shall also inspect the transformer for any abnormal conditions including severe corrosion, cracks, loose nuts or bolts, unconnected wiring. The Contractor shall report any abnormal condition to the Government Representative prior to the start of paint preparation for repair.

B. Cover and protect all new transformer components and electrical panels prior to start of the transformer cleaning and painting process.

C. Cover and protect all areas around each transformer included but not limited to power equipment and structures.

D. Remove all paint that is peeling, flaking, cracking, blistering, or lifting using power tool cleaning methods in accordance with SSPC SP3.

E. Clean, sand, and feather edge all bare and rusted metal spots before paint coating is applied. All cleaning shall be in accordance with SSPC SP12 and SSPC SP6.

F. If paint stripping is required, use a hot caustic stripping solution using the flow coat method. After stripping the area, stripped shall be rinsed with water, then with a 5-percent phosphoric acid solution, then with clean water until the surface is neutralized. The stripping and cleaning solutions shall be approved by the Government before usage.

3.3 Painting Requirements

A. All work shall be accomplished in compliance with SSPC Guide 6, SSPC PA1, SSPC PA2, SSPC SP1, SSPC SP2, and SSPC SP3.

B. Do not paint moving parts, rubber, stainless steel, aluminum, gauges, galvanized or similar surfaces. Verify all surfaces that shall not be painted are removed or covered to prevent being painted or affected by overspray.

3.4 The Contractor shall be aware of potential bird droppings in area and on surfaces to be repainted. Take necessary precautions and measures to protect personnel from possible contamination.

3.6 Refer to Appendix A.5 - Lead and Heavy Metal Removal.

3.7 Signage:

A. Prior to commencing painting, the Contractor shall obtain Government Representative Approval.

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B. The Contractor shall provide all signs, stands, and barricades necessary to post warnings along roads adjacent to the work area and at the entrances to all parking lots, through streets, and roads with a 500-foot perimeter around the work area. Signs shall be at least 2 by 2 feet and weighted for stability.

C. The signs shall provide the following information: NOTICE: SPRAY PAINTING IN PROGRESS, PASS AT OWN RISK". The signs shall be maintained by the Contractor.

D. The signs shall have a white background. The word "NOTICE" shall be 4-inch letters, white on blue, balance of words shall be 3-inch letters, black on white.

END OF APPENDIX A.6

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APPENDIX B
OIL TEST REPORTS
- PWT3

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END OF APPENDIX B

16332-24-001

SDMyers ACTS 4:12 *INTELLIGENT TRANSFORMER MANAGEMENT®*

**Date Printed** 1/9/2024 **TC# 9021**

| | | | | | |
|---|---|---|---|---|---|
| **Customer** | 9169120 National Aerospace Solutions, LLC | **City** | Arnold AFB, TN | **Location** | |
| **Sub-Name** | XFMR #3 MAIN TANK | **Unit No.** | PWT #3 | **Other** | MAIN TANK |

## NAMEPLATE DATA

| | | | |
|---|---|---|---|
| **Manufacturer** | ABB | **Equipment Type** | TRANSFORMER |
| **Manufacture Date** | 1/1/1992 | **Transformer Class** | |
| **Serial No.** | MNL5845-01 | **Impedance %** | |
| **KVA Rating** | 37,500 | **Phase/Cycle** | |
| **High Voltage** | 161,000 | **Liquid Type** | OIL |
| **Low Voltage** | 6,900 | **Gallons** | 6,845 |
| **Weight** | | **Other Access** | 1/4 INCH PLUG |

## ADDITIONAL EQUIPMENT

| | | | |
|---|---|---|---|
| **Radiators** | | **Conservator Tank** | |
| **Fans** | | **LTC Compartment** | |
| **Water Cooled** | | **Bushing Location** | |
| **Oil Pumps** | | **Breather** | |
| **Top FPV (inch)** | | **Hose Length (feet)** | |
| **Bottom FPV (inch)** | | **Service Online** | |
| **InsulationType** | | **Power Available** | |

## VISUAL INSPECTION

| DATE | LEVEL | SAMPLE TEMP | TOP TEMP | P/V | PAINT | LEAKS |
|---|---|---|---|---|---|---|
| 05/02/20 | | 19 | 25 | | | |
| 04/30/21 | | 25 | 55 | | | NONE |
| 04/29/22 | | 33 | 59 | | | |
| 04/14/23 | | 21 | | | | |

## FIELD SERVICE

| DATE | SERVICE |
|---|---|

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**Additional Information**

**Reason Not Tested**

## LIQUID SCREEN TEST DATA

| DATE | SERVICE | ACID | IFT | DIEL 877 | DIEL 1816 | GAP | COLOR | | SP. GRAV. | VISUAL | SEDIMENT |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 04/01/17 | | 0.020 AC | 37.0 AC | 40 AC | 38 UN | 2.0mm | 1.00 | AC | 0.880 AC | CLEAR AC | NONE AC |
| 04/17/17 | | 0.030 AC | 38.2 AC | 51 AC | 26 UN | 2.0mm | 1.00 | AC | 0.890 AC | CLEAR AC | NONE AC |
| 04/09/18 | | 0.030 AC | 37.6 AC | 38 AC | 30 UN | 2.0mm | 1.00 | AC | 0.880 AC | CLEAR AC | NONE AC |
| 04/17/19 | | 0.030 AC | 36.4 AC | 42 AC | 56 AC | 2.0mm | 1.00 | AC | 0.880 AC | CLEAR AC | NONE AC |
| 05/02/20 | | 0.030 AC | 39.9 AC | 36 AC | 38 UN | 2.0mm | 1.00 | AC | 0.880 AC | CLEAR AC | NONE AC |
| 04/30/21 | | 0.020 AC | 37.5 AC | 39 AC | 40 UN | 2.0mm | 1.00 | AC | 0.885 AC | CLEAR AC | NONE AC |
| 04/29/22 | | 0.020 AC | 38.9 AC | 41 AC | 35 UN | 2.0mm | 1.00 | AC | 0.882 AC | CLEAR AC | NONE AC |
| 04/14/23 | | 0.030 AC | 36.0 AC | 43 AC | 25 QU | 1.0mm | 1.50 | AC | 0.880 AC | CLEAR AC | NONE AC |

## INHIBITOR CONTENT

| DATE | PCT. BY WEIGHT |
|---|---|
| 09/08/11 | 0.11% QU |

NOTE - STUDIES SHOW THAT A LEVEL OF 0.3% INHIBITOR IS OPTIMUM FOR PRESERVATION OF IN-SERVICE TRANSFORMER OILS. OILS WITH A LEVEL BELOW 0.08% ARE CONSIDERED TO BE UNINHIBITED.

## LIQUID POWER FACTOR

| DATE | 25 C | 100 C |
|---|---|---|
| 04/17/17 | 0.012 AC | 0.257 AC |
| 04/09/18 | 0.038 AC | 0.405 AC |
| 04/17/19 | 0.013 AC | 0.500 AC |
| 04/29/22 | 0.013 AC | 0.607 AC |
| 04/14/23 | 0.003 AC | 0.247 AC |

KEY TO ABBREVIATIONS: AC - ACCEPTABLE QU - QUESTIONABLE UN - UNACCEPTABLE RS - RESAMPLE

**NOTE:** * After a result indicates that the test or service was performed by an outside source.

SDMyers ACTS 4:12 INTELLIGENT TRANSFORMER MANAGEMENT®

 **Date Printed** 1/9/2024 **TC# 9021**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Customer** | 9169120 National Aerospace Solutions, LLC | **S/N** | MNL5845-01 | | | | |
| **Sub-Name** | XFMR #3 MAIN TANK | **Mfg.** | ABB | **Gallons** | 6,845 | **High Volt.** | 161,000 |
| **Location** | | **Unit No.** | PWT #3 | **KVA** | 37,500 | **Low Volt.** | 6,900 |

## KARL FISCHER TESTING MOISTURE CONTENT EXPRESSED IN PPM

| DATE | AVG. TEMP | PPM | PCT. SATURATION | | MOISTURE BY DRY WEIGHT PCT. |
|---|---|---|---|---|---|
| 04/09/18 | 26 | 6 | 8.5 | QU | 0.96 |
| 04/17/19 | 23 | 7 | 10.4 | QU | 1.22 |
| 05/02/20 | 24 | 9 | 13.1 | QU | 1.53 |
| 04/30/21 | 30 | 5 | 6.0 | AC | 0.65 |
| 04/29/22 | 38 | 3 | 2.2 | AC | 0.21 |
| 04/14/23 | 26 | 4 | 5.7 | AC | 0.64 |

**RECOMMENDATION** RETEST 1 YEAR
The moisture content continues to be acceptable based on the equipment and liquid type. Continued normal monitoring is indicated.

## FURAN ANALYSIS EXPRESSED IN PPB

| DATE | 5H2F | 2FOL | 2FAL | 2ACF | 5M2F | TOTAL |
|---|---|---|---|---|---|---|
| 04/09/18 | ND | ND | 2 | ND | ND | 2 |
| 04/17/19 | ND | ND | 2 | ND | ND | 2 |
| 05/02/20 | ND | ND | 12 | ND | ND | 12 |
| 04/30/21 | ND | ND | 2 | ND | ND | 2 |
| 04/29/22 | ND | ND | 5 | ND | ND | 5 |
| 04/14/23 | ND | ND | 4 | ND | ND | 4 |

**RECOMMENDATION** RETEST 1 YEAR
NO DIAGNOSTIC CHANGES ARE NOTED IN FURAN LEVELS SINCE THE PREVIOUS ANALYSIS. THE CELLULOSIC INSULATION APPEARS TO BE IN GOOD CONDITION.

**CALCULATED DP** 800 **EST. LIFE REMAINING** 100%

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## GAS-IN-OIL ANALYSIS GAS CHROMATOGRAPHY EXPRESSED IN PPM

| DATE | HYDROGEN | OXYGEN | NITROGEN | METHANE | CARBON MONOXIDE | CARBON DIOXIDE | ETHANE | ETHYLENE | ACETYLENE | TOTAL COMBUST. | TOTAL GAS |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 04/01/17 | 3 | 6,622 | 81,933 | 11 | 13 | 831 | 10 | 6 | ND | 43 | 89,429 |
| 04/17/17 | 18 | 9,333 | 62,621 | 9 | 11 | 748 | 10 | 11 | ND | 59 | 72,761 |
| 04/09/18 | ND | 6,325 | 61,828 | 9 | 13 | 765 | 10 | 10 | ND | 42 | 68,960 |
| 04/17/19 | 1 | 7,683 | 58,564 | 5 | 17 | 750 | 6 | 7 | ND | 36 | 67,033 |
| 05/02/20 | ND | 5,412 | 88,415 | 5 | 87 | 1,164 | 7 | 7 | ND | 106 | 95,097 |
| 04/30/21 | ND | 1,542 | 62,512 | 4 | 60 | 1,199 | 6 | 6 | ND | 76 | 65,329 |
| 04/29/22 | 4 | 1,155 | 52,118 | 4 | 43 | 1,023 | 5 | 4 | ND | 60 | 54,356 |
| 04/14/23 | ND | 1,081 | 85,312 | 5 | 63 | 1,179 | 5 | 5 | ND | 78 | 87,650 |

**RECOMMENDATION** RETEST 1 YEAR
B-THE ANALYSIS OF THIS SAMPLE SHOWS NO SIGNIFICANT INCREASE IN THE COMBUSTIBLE GAS VOLUME. THIS INDICATES NORMAL OPERATION.

## ICP METALS-IN-OIL EXPRESSED IN PPM

| DATE | ALUMINUM | IRON | COPPER |
|---|---|---|---|
| 09/08/11 | <0.1 | <0.03 | <0.01 |

**RECOMMENDATION** RETEST 1 YEAR
THERE ARE NO DIAGNOSTIC LEVELS OF METALS IN THIS SAMPLE. THESE DATA CAN SERVE AS A BASELINE FOR FUTURE ANALYSES.

## PCB CONTENT EXPRESSED IN PPM

| DATE | 1242 | 1254 | 1260 | OTHER | TOTAL |
|---|---|---|---|---|---|
| 09/08/11 | ND | ND | ND | ND | ND |

**COLOR LABEL:** Green **CLASS:** NON-PCB

**Results in mg/kg**
**ND means None Detected**
**(<2 mg/kg per ASTM D4059)**

**NOTE:** * After a result indicates that the test or service was performed by an outside source.

**Transformer Control Cabinet:**

1. Transformer Control Cabinet Enclosure Replacement: Provide labor and materials to design, fabricate and provide new transformer control cabinet enclosure and the transformer insulation oil cooling system motor controls. The control cabinet shall be custom designed for the application. All equipment within the control cabinet shall be uniquely and clearly labeled for ease of identification. The control cabinet shall be fabricated to strictly comply with UL 50 and NEMA 250. The control cabinet shall be pre-fabricated and tested by the Contractor before shipment to Arnold.

2. The control cabinet enclosure manufacturer shall have a minimum of 10 years of experience in manufacturing this type of equipment.

3. The control cabinet enclosure shall be rugged weatherproof NEMA Type 3R construction, manufactured from 12-gauge stainless steel or galvannealed steel, and fabricated in full compliance with NEMA 250 and UL 50 standards. The control cabinet enclosure shall be painted with white epoxy polyester powder coated paint over the inside phosphatized surfaces and shall be painted ANSI-70 gray epoxy polyester powder coated finish over phosphatized surfaces on the exterior surface.

4. Provide a rain cap to prevent entry of water.

5. Provide a concealed continuous hinged full length door with a three point latching mechanism with corrosion resistant latching hardware and a stainless steel door handle kit. The continuous hinge for each door shall be provided with a stainless steel pin. A door stop shall be provided to prevent uncontrolled travel and securely hold the door in the open position.

6. Provide memory retaining oil resistant urethane gaskets.

7. Provide a tin plated copper ground bus-bar mounted inside the cabinet.

8. The ground bar shall be bonded to the enclosure door and the existing ground grid.

9. Provide a removable 10-gauge stainless or galvannealed steel sub-panel for mounting of all hardware to be installed in the cabinet enclosure. The sub-panel dimensions shall be 3 inches less than corresponding cabinet enclosure dimensions. Each sub-panel shall be painted with white epoxy polyester powder coated paint over phosphatized surfaces.

10. The control cabinet shall have 2 inches by 4 inches (minimum) hinged slotted Panduit type wire-way mounted on the sub-panel panel for installation of internal wire and cables. The wire-way shall be installed to provide a minimum 1.25-inch space between the edge of the wire-way and installed hardware. The wire-way shall be mounted within the dimensions of the sub-panel without overlapping.

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11. All control cabinet control wiring shall be 14 gauge (minimum unless otherwise noted) extra flexible copper conductors with moisture-resistant, oil-resistant and heat-resistant insulation suitable for conductor temperatures not exceeding 90 degrees C normal operation. All control wiring shall be stranded single-conductor coated annealed copper, insulated for not less than
600-volt service. All wires and cables shall be neatly trained and marked on each end with permanent oil-resistant pre-printed wire markers. Also all cables in each cabinet shall be labeled with an oil-resistant label over the outer jacket. No hand lettering will be allowed. All labels shall be visible after wiring termination. Stick-on cable supports shall not be used. All terminations shall be insulated ring-tongue lugs.

12. Provide terminal blocks rated for 600-volt alternating current (VAC) and provide fifteen percent spare terminals on each block or group of blocks. All terminal blocks shall be easily accessible and shall be installed so that maintenance personnel have access without removal of barrier plates, covers or wiring. All wiring leaving the enclosure shall be terminated on the terminal blocks and not from devices in the enclosure. All terminal blocks and jumpers shall be permanently and uniquely labeled in conformance with NEMA ICS 1. Provide separate terminal blocks for power and control wiring. All control wiring leaving the cabinet shall be No. 14 American Wire Gage (AWG) (minimum). All 480V three-phase power wiring shall be No. 12-AWG or larger as required by the NEC. Current transformer secondary wiring shall be type SIS of not less than 10-AWG. Connect each current transformer secondary lead to a short circuiting terminal block that is conveniently located in the control cabinet. Current transformer connections to terminal strips shall be made using ring-tongue terminals.

13. The control cabinet shall have a 240VAC-rated thermostatically controlled anti-condensation strip heater of adequate rating to prevent condensation when the outside temperature reaches 32 degrees Fahrenheit (F) with the heater operating at 50-percent of rated voltage. The strip heaters shall be operated on 120VAC power. The strip heater shall be provided with a metal perforated cover designed by the manufacturer to prevent accidental contact.

14. The control cabinet shall have two 120VAC-powered light kits with LED lamps and door activated switch for illumination of the control panel whenever the door is opened.

15. The control cabinet shall be provided with a duplex 125VAC, 3-wire, 15A-rated, GFCI grounding type, DIN-rail mountable receptacle mounted in the cabinet on a DIN- rail. A 120VAC-5A 1-phase circuit breaker shall be provided to isolate the 120VAC power for the receptacle.

16. The control cabinet shall use DIN-rail mountable circuit breakers for isolation of all 120VAC power circuits and over-current protection. At a minimum, individual circuit breakers shall be provided for light kits, receptacles and control power circuits. All electronic hardware installed in cabinets shall be DIN-rail mountable.

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17. All leads from the transformer devices and accessories to the control cabinet shall be enclosed in rigid galvanized conduit and connected to terminal blocks in the control cabinet, with exception that short lengths of liquid-type metallic flexible conduit (36-inches or less) are permitted between individual device terminals and nearby connection boxes. All fittings shall be watertight. For necessary connections, provide water-tight cast-metal boxes and outlet fittings with threaded outlets and gasketed covers. Do not use running threads on conduit. At a minimum the control cabinet shall be provided with the terminals listed below:

    A. Leads from each secondary tap of each bushing type current transformer (CT). Provide 600VAC-rated shorting terminal blocks for each CT.

    B. Leads from alarm contacts on the pressure-relief device.

    C. Leads from alarm contacts and 4-20 mA output on the magnetic-type oil gauge.

    D. Leads from alarm contacts and 4-20 mA output on the dial-type indicating thermometer.

    E. Leads from all winding temperature equipment.

    F. Leads from the fault pressure relay contacts.

    G. Leads from loss of voltage relay contacts.

    H. Leads from loss of oil flow alarm contacts.

    I. Leads from loss of fan power relay contacts.

    J. Power supply conductors for lighting, receptacle, and heater circuits.

    K. Three-Phase power supply conductors for pump and fan motors, separated from other circuits.

18. The control cabinet shall include an oil cooling motor control and monitoring system custom designed for each transformer oil cooling system. The oil cooling control system shall be fully equal and provide all control and interlock features as the existing oil cooling control system for the transformer. The monitoring and oil cooling control system shall provide fully automatic control with equipment designed to start and stop the fan motors and pump motors as the transformer winding temperature requires. The control system circuits shall be designed so that it will not be necessary to shut down the entire forced-oil/air equipment control system to isolate individual cooling banks for repairs. The Contractor shall provide all materials and labor to install and test the oil cooling control system. The oil cooling system controls shall be designed to operate on the same supply and control voltage as the existing transformer oil cooling system. The cabinet shall include but not be limited to the following features and hardware:

    A. Provide a main 3-pole molded case circuit breaker of proper current rating for the system three-phase supply power. Provide additional 3- pole molded-case circuit breakers as required for branch circuits. The branch circuit breakers shall not be rated less than 20A. Provide all individual 3-pole branch circuit breakers with auxiliary contacts wired to the transformer

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monitoring system to indicate the individual breaker is in the closed position. Include details for each main 3-pole molded case circuit breaker on the detailed BOM included in the design drawings.

B. Provide a 3-pole molded case motor circuit protector (MCP) of proper current rating for each oil cooler fan motor and pump motor.
The MCP shall include:

i. UL 489 Instantaneous Short Circuit Current Ratings: 65kA at 480V/3-Phase.

ii. HP and 3-phase current rating: sized for the motor application.

iii. Provided with 1-N.O. and 1-N.C. factory installed axillary contacts. All 3-pole MCP's shall be provided with factory installed auxiliary contacts wired to the transformer monitoring system to indicate the MCP is in the closed position. The Contractor shall include details for each MCP on the detailed BOM included in the design drawings. All 3-pole MCP's shall be provided with factory installed auxiliary contacts wired to the transformer monitoring system to indicate the MCP is in the closed position. The Contractor shall include details for each MCP on the detailed BOM included in the design drawings.

C. All three-phase branch circuit devices shall be connected from a three-phase power distribution block. No double conductor connections on three-phase branch circuit devices will be permitted. This includes both three-phase motor circuit protectors and three-phase molded case circuit breakers.

D. Provide individual DIN-rail mountable motor contactors with integral overload relays for each cooling fan and pump motor. Fan motors may be combined on the same branch MCP if they are in the same cooling zone. All motor contactors shall be designed with IEC 60947-4-1 and UL 508, should have an operating coil voltage of 120VAC(60Hz) and the amerage rating shell be sized for the motor application, and shall contain one spare normally-open (minimum) auxiliary contact and one spare normally-closed (minimum) auxiliary contact. All motor contactors shall be provided with surge suppressors designed to be mounted on the motor contactor assembly. The surge suppressors shall be designed for use with the contactor and provided by the contactor manufacturer. The integral overload relays shall also be designed for assembly to the motor contactors. The Contractor shall include details for each motor contactor, surge suppressor and integral overload relay on the detailed BOM included in the design drawings.

E. Provide loss of voltage relays for the main 3-phase power source. Provide a loss of fan power alarm to indicate a condition that the fan contactor has closed but the fans are not operating. Provide a loss of oil flow alarm to indicate a condition that the pump contactor has closed but no oil is flowing. The Contractor shall include details for each loss of voltage relay on the detailed BOM included in the design drawings.

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19. The control cabinet shall include a transformer monitoring system for transformer SCADA system inputs. The monitoring system shall be integrated with Intelligent Electronic Device (IED) hardware specifically designed for outdoor substation transformer system control and monitoring. The IED shall be installed on a hinged swing-out panel kit (mounted to the Transformer Control Panel front flange or side) to provide access to the rear side of the IED without removing the IED hardware. The swing-out panel kit shall be provided with sealing washers and stainless steel external mounting screws to ensure the NEMA rating of the transformer control panel after installation. The hardware shall be wired to accept digital inputs of oil cooling system status, analog input of oil temperature, analog input of winding temperature, and digital input from all transformer alarms. The integrated IED hardware features shall include but not be limited to:

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A. Programmable CPU unit designed specifically for outdoor utility substation transformer monitoring. The IED-CPU unit shall be able to track thermal conditions and have metering capabilities, monitor tap position, and raise and lower controls, have flexible input and output for transformer status and alarms, log critical data for post-event diagnostics, communicate with SCADA, and have AC metering capabilities.

B. I/O modules for connection to the IED-CPU unit: Analyze system events with inputs and other events timed to the microsecond, perform actions with a processing interval of 2ms, ensure safe operation by using input with logic programmed for local/remote control, LEDs for each I/O point and communication port, have easy integration with SCADA, be designed for harsh outdoor utility substation, and work with the IED-CPU.

C. Thirty two isolated relay outputs (minimum).

D. Thirty two isolated digital inputs (minimum).

E. Eight isolated analog inputs (minimum).

F. 16-bit accuracy at 25-degrees Celsius for analog inputs.

G. SEL 2725 Ethernet switch to network IED devices.

H. E-100BTX-FX-05 Media Converter to connect the fiber optic from the SCADA to the Ethernet switch.

I. Ethernet RJ45 copper connection for local programming and on-line monitoring.

J. Sequential Events Recorder (SER) with critical reporting and logging with capability to store up to 512 sequential events, time dated to nearest millisecond. Provide analysis of SER reports, analog trending, and oscillographic event reports for rapid commissioning, testing, and post-event diagnostics.

K. Communication options that provide integration with a SCADA system RTU with Ethernet, Modbus TCP, DNP3 LAN/WAN, IEC61850, Modbus Serial, and DNP3 RTU protocols.

L. Metering capability with option to measure and record voltage, current, power harmonic voltage and currents to 25th, THD and power factor.

M. Software configurable using off-the-shelf software provided by the IED manufacturer. Programmable functions via the configuration software shall include logic, math, timers, counters, and edge-trigger functions. The IED shall be programmed with the software by the Contractor. The required programming shall include but not be limited to the following:

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a. Logic to process digital inputs and provide alarm outputs equivalent to existing SCADA system alarms.

b. Logic to process analog inputs for temperature and provide alarm

N. Front-panel HMI to provide complete configuration access and displays settings, measurements, and calculated values.

O. Transformer asset monitoring with programmable functionality using input data to provide weekly reports on transformer health. Program the SEL-2414 to include but not be limited to the following.

a. Enable thermal monitoring.

b. Insulation aging acceleration factor limit per recommendation of transformer manufacturer. Program an alarm output and connect alarm to existing SCADA system.

c. Daily rate of loss of life limit per recommendation of transformer manufacturer. Program an alarm output and connect alarm to existing SCADA system.

d. Total accumulated loss of life limit per recommendation of transformer manufacturer. Program an alarm output and connect alarm to existing SCADA system.

e. Top oil temperature limit per recommendation of transformer manufacturer. Program an alarm output and connect alarm to existing SCADA system.

f. Hot spot temperature limit per recommendation of transformer manufacturer. Program an alarm output and connect alarm to existing SCADA system.

P. Power supply designed for DC input voltage of 85 to 275-Volts or AC input voltage of 85 to 264-Volts.

Q. Designed, tested and manufactured for operation in extreme operating environments typical for an electrical utility substations including the following:

a. Temperature operating range of negative 30 degrees Celsius to positive 75 degrees Celsius per IEC 60068-2-1 and 60068-2-2.

b. Relative humidity operating range of 5 to 95 percent.

20. IED logic flow, inputs, and outputs, and addressing shall be similar in design as compared to other AEDC transformers as shown in Appendix E, drawings 5113776-N.

21. Logic programming shall provide detail comments and associated flow chart diagrams for logic scenarios of the different cooling and device operations.

22. Control cabinet drawings, equipment layouts, device locations, labeling shall be similar in design and function as compared to other AEDC transformers and shall be designed and submitted for Government approval.

23. Label all equipment mounted in cabinets on the sub-panel with engraved plastic laminated labels with 0.1875-inch minimum lettering, 0.125-inch thick, black with white center core. Each label shall be attached to the sub-panel with stainless steel pan-head screws. The design drawings for the custom panel design shall include

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a drawing sheet with detailed layout of each component installed on the sub-panel. The drawing shall include a part number reference note to a bill of materials that shall detail the component name or symbol as noted on the wiring drawings, manufacture name, manufacture part number and description specifications.

24. The cabinet enclosure shall be uniquely labeled with a panel identification nameplate on one of the outside doors with a laser imprinted stainless steel nameplate, minimum 2-inches by 6-inches. The nameplates shall be attached with stainless steel pan-head screws with a layer of silicon waterproofing sealant applied to the back nameplate label prior to attaching to the cabinet enclosure to protect the enclosure environment rating. The area around the stainless steel screws shall also be coated with the silicon waterproofing sealant to protect the enclosure environmental rating. The nameplate labels shall have contrasting paint filled lettering for maximum visibility. The nameplates shall have a clear epoxy coating applied to prevent weather damage or fading. The nameplate lettering shall include but not be limited to the following:

    A. Transformer name.

    B. Contractor’s name.

    C. Contractor’s purchase order number and reference drawing numbers for panel layout diagram and panel wiring diagram.

25. The cabinet enclosure shall be labeled with outdoor rated arc flash hazard and electrical voltage hazard labels.

26. All internal wiring shall be continuous between termination points. Splices are not permitted.

27. All mounted equipment shall be readily accessible and installed in such arrangement that maintenance personnel shall have direct access to the equipment without removal of covers or wiring.

28. All equipment shall be mounted to provide segregation of 480VAC or 208VAC three-phase power equipment and lower voltage 120VAC, 125VDC and 24VDC control equipment and wiring.

29. Each cabinet shall be custom designed with Height (H), Width (W) and Depth (D) dimensions sized to provide space for all panel mounted equipment plus minimum 25-percent of spare space for additional future components. Depth shall be 14-inches minimum.

30. Install all equipment in each enclosure to minimize the generation of electromagnetic and radio frequency interference.

31. Replace the existing control power transformer with a NEMA 3R rated control power transformer adequately sized to provide 120VAC-60HZ control power for

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the oil cooling system controls, 120VAC utility power, lights, heater, and control power for the IED hardware. The control power transformer specifications shall include the following:

A. NEMA 3R rain proof construction.

B. Copper magnet wire windings.

C. Encapsulated.

D. UL Class 180-degrees Celsius insulation system, 80-degrees Celsius temperature rise under load.

32. To ensure the integrity of the fiber optic cable, provide a panel mounted fiber optic cable patch panel for fiber optic cable management and cable termination. The fiber optic patch panel shall provide up to 8 ports for cable termination. The enclosure shall provide for sufficient storage room for the fiber service loop. The enclosure shall be compact in design, have 4 duplex SC Port Adapters installed, support multimode configuration, accept up to 8 fusion or mechanical splices, have top, bottom, and rear access, be lockable and rugged plastic for harsh environments.

A. Provide a NEMA 4 rated convenience receptacle mounted on the outside surface of the enclosure to provide programming access to the IED hardware and a 120VAC convenience outlet for programming equipment, as follows:

B. NEMA-4 cast aluminum protective cover and construction.

C. Stainless steel locking mechanism.

D. 3-amp circuit breaker protection on 120VAC convenience outlet.

E. Rubber gasket.

F. Suitable for hose down areas.

G. RJ45 Ethernet programming port with communications cable wired from the programming port to IED hardware programming port.

33. Provide each transformer control cabinet enclosure with a copper grounding strap to provide electrical continuity between the enclosure hinged door and the enclosure body. The grounding strap shall be attached on each end to the grounding stud.

34. Provide each transformer control cabinet enclosure with an integral body grounding stud on the enclosure door and enclosure body.

35. Provide each transformer control cabinet enclosure with an internally mounted (4.5-inch by 8-inch minimum) “Mr. Ouch” safety label.

36. Provide a 0.25-inch clear polycarbonate window kit with a stainless steel frame installed on the control cabinet front panel door to provide a full view of the transformer monitor front panel HMI display. The window kit shall be installed with oil-resistant gaskets to ensure a water-tight seal around the perimeter of the window and frame.

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37. Use din-rail mountable DC-rated circuit breakers for isolation of all DC power circuits and over-current protection. At a minimum, individual circuit breakers shall be provided for each electronic module power supply, internal control panel DC control circuits and DC control circuits for equipment external to the control panel.

38. Provide DC-output power supplies as required for all DC powered electronic hardware. The specifications for the power supplies shall include but not be limited to the following:

A. Designed for DIN-rail mounting.

B. Designed for single-phase 120VAC input.

C. DC on status indicator lights on the front panel.

D. -10 to +70 degrees Celsius rated operating temperature range.

E. Perforated meal covers to provide air flow for external convection cooling.

39. Provide power inverter suitable to operate the entire load of the control cabinet controls. Normal source shall be 120VAC with capability to switch to 125VDC station battery upon loss of 120VAC. For example, the existing power inverters are Majorpower and Majorsine. Notify the Government of the required circuit ampacity with the initial design submittal. The Government will provide the location for the new circuit. Provide all new components, include 2-pole knife switch, fuses, wire, and install the new circuits, as required.

40. Transformer Temperature Cooling System Remote Operator Control Panel: Provide labor and materials to design and fabricate an operator control panel for each of the transformer control cabinet enclosures. The control panels shall be custom designed for each transformer to provide manual and automatic operation of the transformer insulation oil coolers, and mounted as close as possible to the new transformer control cabinet enclosures provided. The control panels shall provide operator controls for the transformer oil cooling system. The design shall include drawings showing all connections schematically and physically to the transformer control cabinet enclosure. Specification details for fabrication and hardware include:

A. Each control panel cabinet shall be 8-inch deep minimum, rugged weatherproof NEMA Type 3R construction, manufactured from at least 14 gauge galvannealed steel. Each cabinet enclosure shall be painted with white powder coated paint over the inside phosphatized surfaces and shall be painted ANSI-70 gray epoxy polyester power coated finish over phosphatized surfaces on the exterior surface.

B. Rain cap to prevent entry of water.

C. Concealed continuous hinged full length door with two stainless steel hand operated quarter turn latches.

D. Memory retaining oil resistant urethane gaskets.

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E. Hinged swing-out panel kit (mounted to the Operator Control Panel front flange or side) to provide access to the rear side of the operator control components without removing the control components. The swing-out panel kit shall have sealing washers and stainless steel external mounting screws to ensure the NEMA rating of the Operator Control Panel after installation of the swing-out panel kit.

F. Operator controls installed on the swing-out panel including:

   a. Selector switches for selection of manual or automatic operation of insulating oil cooling equipment provided for each zone. The selector switches shall be Heavy Duty, NEMA 4X watertight, 30.5mm and provided with finger-safe contact blocks. Each selector switch shall be provided with a custom engraved legend plate fabricated from aluminum material and engraved with black lettering. The selector switch shall be illuminated with a green LED and lens.

   b. LED type lighted pushbuttons on LED type push-to-test indicator lights to indicate cooling pump(s) running and oil flow for each zone. The indicator lights or lighted pushbuttons shall be NEMA 4X watertight, 30mm with finger safe contact blocks. Each indicator shall be provided with a custom engraved legend plate fabricated from aluminum material and engraved with black lettering.

   c. Manual start and stop control pushbuttons for operation for each zone. The pushbuttons shall be NEMA 4X watertight, 30mm, and provided with finger-safe contact blocks. Each pushbutton shall be provided with a custom engraved legend plate fabricated from aluminum material and engraved with black lettering.

   d. All indicator lights, selector switches and pushbuttons shall be mounted on the swing-out panel kit with a 3-inch minimum clearance on all sides.

G. Rigid galvanized steel conduit from the control panel cabinet to the transformer control cabinet enclosures.

H. A 0.25-inch clear polycarbonate window kit with a stainless steel frame shall be installed on the enclosure panel door. The window kit shall provide a full view of all operator panel operator controls. The window kit shall be installed with oil-resistant gaskets to ensure a water-tight seal around the perimeter of the window and frame.

I. Terminate all wiring from the operator devices on the swing-arm panels to a DIN-rail mounted terminal block inside the control panel cabinet. All wiring connections at terminal blocks shall be made with insulated locking fork-type terminals. All control panel cabinet wiring where not installed in channels or ducts, shall be formed into compact bundles, suitably bound together, and properly supported.

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J. All control wiring from the control panel cabinets to the transformer control cabinet enclosures shall be type THHN #14 AWG (minimum) stranded copper conductors. Wire markers shall be installed on each end. Wiring shall be run continuous with no splices. All wiring to the transformer control cabinet enclosures shall be installed in the rigid steel conduit.

END OF APPENDIX C

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APPENDIX D
TRANSFORMER OIL TESTING AND PROCESSING

## 1. SCOPE

1.1. The Contractor shall provide all materials, equipment, and labor to perform hot oil vacuum drying on multiple 161kV transformers, at Arnold Engineering Development Complex (AEDC), Arnold Air Force Base, Tennessee.

1.2. Background: This transformer has an active nitrogen oil preservation system. The Contractor shall confirm that the excess moisture in the transformer insulation has been removed without causing damage by conducting tests at the completion of the drying process.

2. **APPLICABLE DOCUMENTS -** The Contractor shall comply with the latest revisions and provisions of the documents to the extent referenced herein.

2.1. Government Documents: None.

2.2. Non-Government Documents:

2.2.1. ASTM International Standards

| | | |
|---|---|---|
| A. | ASTM D877-19 | Standard Test Method for Dielectric Breakdown Voltage of Insulating Liquids Using Dish Electrodes. |
| B. | ASTM D923-23 | Standard Practices for Sampling Electrical Insulating Liquids. |
| C. | ASTM D924-23 | Standard Test Method for Dissipation Factor (or Power Factor) and Relative Permittivity (Dielectric Constant) of Electrical Insulating Liquids. |
| D. | ASTM D971-20 | Standard Test Method for Interfacial Tension of Oil Against Water by the Ring Method. |
| E. | ASTM D974-22 | Standard Test Method for Acid and Base Number by Color-Indicator Titration. |
| F. | ASTM D1275-24 | Standard Test Method for Corrosive Sulfur in Electrical Insulating Oils. |
| G. | ASTM D1298-23 | Standard Test Method for Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum and |

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Liquid Petroleum Product by Hydrometer Method-API Designation.

| | | |
|---|---|---|
| H. | ASTM D1500-17 | Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale). |
| I. | ASTM D1524-22 | Standard Test Method for Visual Examination of Used Electrical Insulating Oils of Petroleum Origin in the Field. |
| J. | ASTM D1533-20 | Standard Test Method for Water in Insulating Liquids by Coulometric Karl Fischer Titration. |
| K. | ASTM D1816-19 | Standard Test Method for Dielectric Breakdown Voltage of Insulating Oils of Petroleum Origin Using VDE Electrodes. |
| L. | ASTM D2668-13 | Standard Test Method for 2,6-Ditertiary-Butyl Paracresol and 2,6-Ditertiary-Butyl Phenol in Electrical Insulating Oil by Infrared Absorption. |
| M. | ASTM D2945-09 | Standard Test Method for Gas Content of Insulating Oils. |
| N. | ASTM D7151-23 | Standard Test Method for Determination of Elements in Insulating Oil. |

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2.2.2. The Institute of Electrical and Electronics Engineers

| | | |
|---|---|---|
| A. | IEEE C57.93-19 | Guide for Installation and Maintenance of Liquid-Immersed Power Transformers. |
| B. | IEEE C57.106-15 | Guide for Acceptance and Maintenance of Insulating Oil in Equipment. |

## 3. <u>REQUIREMENTS</u>

3.1. <u>Hot Oil Vacuum Processing</u>:

3.1.1. The Contractor shall provide all equipment required for hot oil vacuum processing including oil transfer pump, vacuum pump, oil heater, oil purifier, cold trap, tanker truck, filters, hoses, fittings, thermal insulation, and instrumentation. Oil storage and handling equipment shall be designed for and dedicated to insulating oil handling service and shall be clean and dry. The Contractor shall supply dry gases and any other consumable materials that will be used in the process.

3.1.2. The Contractor shall provide manpower on-site for monitoring of the oil processing equipment at all times.

3.1.3. The Contractor shall provide all electrical power required to run the oil-processing equipment.

3.1.4. The Contractor's procedures for insulation drying shall conform to IEEE C57.93. The procedures shall specify the process that will be used to remove moisture and avoid entrapping gas bubbles. Not less than two weeks prior to deploying equipment to the site, the Contractor shall submit to the Government for approval copies of the insulation drying plans and procedures.

3.1.5. The Contractor's plans and procedures shall include contingency planning for oil spillage.

3.1.6. The Contractor shall sample and test the transformer insulating oil before and after processing. Insulating oil tests shall be in accordance with IEEE C57.106. Oil samples shall be taken as per ASTM D923. Perform tests listed in Table 3. The results after processing shall be in accordance with the values listed in Table 3. The Contractor shall provide test results to the Government Representative.

3.1.7. The Contractor shall perform electrical tests before and after oil processing.

3.1.8. The Contractor shall disconnect all transformer power cables from bushing connections and protect the cables prior to initiating vacuum processing. The Contractor shall also ensure that the transformer tank, fittings, and instrumentation are suitable for vacuum. Instruments and fittings that are not suitable for the vacuum environment shall be valved off or removed. Repair all equipment damaged by negative pressure is the responsibility of the Contractor. The Contractor shall isolate the oil preservation system.

3.1.9. The Contractor shall make piping connections between the transformer and the processing equipment so that oil is introduced to the transformer through the top fill valve and returned to the processor through a hose connected to the bottom drain valve. The fill port shall be on the cover and positioned to distribute the incoming oil over as much core/coil assembly as possible. The top fill valve should be fitted with a spray nozzle or diffuser to evenly spread the oil stream. The Contractor shall pressure test and make all necessary repairs to ensure that there are no leaks prior to subjecting the transformer to vacuum.

3.1.10. The Contractor shall ground and bond together the processing equipment.

3.1.11. The Contractor shall prepare the transformer to retain heat by covering the transformer with thermal insulation blankets and closing cooler valves when the temperature is below 70°F over any 8-hour period.

3.1.12. The Contractor shall not heat the oil to a temperature of more than 85°C at any time. During processing, the Contractor shall control the rate of temperature

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change to prevent thermally shocking the bushings. Once thermal equilibrium is reached, the Contractor shall maintain an inlet temperature of 70°C ± 5°C.

3.1.13. The Contractor shall avoid oil being directed or splashed onto bushings.

3.1.14. The Contractor shall maintain low oil flow rates to prevent foaming or bubble formation.

3.1.15. The Contractor shall drain and store insulating oil from the transformer per the Government approved procedure.

3.1.16. The Contractor shall pull vacuum to a level of 2 Torr or less and maintain the vacuum throughout the dry-out and filling process. See Table 2 for processing requirements of transformers with tanks rated for full vacuum. The Contractor shall close the transformer bottom cooling valves while maintaining vacuum.

3.1.17. After a minimum stand time of 12 hours, the Contractor shall make a vapor pressure (dew point) measurement and determine the moisture equilibrium point with moisture content expressed in percent of dry weight of the insulation. The Contractor shall provide test results to the Government Representative. An acceptable vapor pressure reading of less than 1.0-percent is required prior to final oil filling of the transformer. See Table 4 for reference dew point chart. If a reading of 1.0-percent or less cannot be obtained, then proceed with the cold trap process. If a reading of 1.0-percent or less is obtained, then proceed with the final vacuum and oil fill process.

3.1.18. The Contractor shall circulate oil from the bottom drain valve through the oil processing equipment and heaters and into the top fill valve. Once the oil discharge temperature reaches 50°C, the Contractor shall begin making cold trap measurements every six hours. All cold trap readings shall be documented and charted to plot the progress of the dry-out process as well as the amount of moisture removed from the transformer. The Contractor shall provide charts to the Government Representative. The Contractor shall continue circulation until two consecutive cold trap readings of about 1 oz. /hr. or less over a 6-hour period are achieved. Once this moisture measurement has been obtained, the Contractor shall drain oil from the transformer while maintaining vacuum until the oil level is near the lowest possible level that maintains oil discharge without pump cavitation.

3.1.19. The Contractor shall continue hot oil circulation and charting cold trap readings until two consecutive cold trap readings of about 1 oz. /hr. or less over a 6-hour period are achieved. The Contractor shall provide charts to the Government Representative. Once this moisture measurement has been obtained, the Contractor shall open the bottom cooling valves and drain the remaining oil from the transformer while maintaining vacuum.

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3.1.20. The Contractor shall continue dry vacuum processing and cold trap readings until two consecutive cold trap readings of about 1 oz. /hr. or less over a 6 hour period are achieved. If two consecutive cold trap readings of about 1 oz. /hr. or less over a 6-hour period is not achieved within 48 hours from the start of the dry vacuum process, the Contractor shall repeat the heat up process.

3.1.21. Upon successful completion of hot oil processing, the Contractor shall vacuum fill the transformer. Once filled, the vacuum shall be broken with dry nitrogen.

3.1.22. The Contractor shall remove all vacuum and oil connections installed by Contractor.

3.1.23. The Contractor shall adhere to recommended minimum set time for absorption after final oil filling (IEEE C57.93, Table 4).

3.1.24. The Contractor shall re-run the oil analysis tests and the electrical tests. The Contractor shall provide a detailed report of the test results as part of the transformer test report. The Contractor shall inform the Government Representative following completion of testing of all test values that fall outside the recommended range.

3.1.25. Testing and Inspection Responsibilities: The Contractor shall be responsible for all testing and inspections as listed in the Statement of Work. The Contractor shall supply all interfacing equipment needed to conduct the tests.

A. Notify the Government at least two weeks prior to test dates.

B. The Government reserves the right to witness all tests.

C. The Contractor shall submit all test reports signed by personnel conducting the test and those approving the results.

3.2. Qualifications:

3.2.1. Field Engineer: Submit name and qualifications of individual assigned to project in the role of Field Engineer.

3.3. Acceptance: The Government will accept the work upon completion of the analysis and testing at AEDC, submittal of reports, and demonstration that the requirements of the work have been met.

3.4. Submittals: The Contractor shall submit all documentation in accordance with Appendix A.2 – Submittal Procedure. A summary table of required submittals is listed at the end of this paragraph.

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3.4.1. Submittal Table

| No. | Section Reference | Submittal Description | Due |
|---|---|---|---|
| 1 | Appendix D 3.1.4 | Plans and procedures | Two weeks after award |
| 2 | Appendix D 3.1.6 | Transformer testing before oil processing | No less than 10 days after test |
| 3 | Appendix D 3.1.17 | Vapor pressure (dew point) measurement | Prior to final oil filling |
| 4 | Appendix D 3.1.8 | Electrical testing before oil procession | No less than 10 days after test |
| 5 | Appendix D 3.1.24 | Final oil analysis and electrical testing reports | Prior to release of transformer for re-energizing |
| 6 | Appendix D 3.1.25 | Test reports | Prior to re-energization |
| 7 | Appendix D 3.2.1 | Personnel information assigned to project | Before beginning on-site work |

Table I. Required Submittal List

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## Table 2

## VACUUM PROCESSING AND OIL FILLING OF TRANSFORMERS WITH TANKS RATED FOR FULL VACUUM

| Step | Voltage Class | 69 KV and below | | 115-230 KV | 345-500 KV |
|---|---|---|---|---|---|
| | **Procedure** | Core & Coils exposed out of oil | Core & Coils remain covered in oil | Any internal work | Any internal work |
| 1 | Complete repairs/maintenance | R | R | R | R |
| 2 | Perform dewpoint (12 hours minimum) | R | NR | R | R |
| 3 | If transformer is equipped with LTC, set up for vacuum | R | NR | R | R |
| 4 | Insure core & coils >= 50 F (10 C)<br>Perform hot oil spray/circulation if required<br>Drain all oil | R | NR | R | R |
| 5 | Isolate oil preservation system as required<br>Open all cooler valves<br>Open all pump valves | R | R | R | R |
| 6 | Tank vacuum, absolute pressure, Torr (mm hg) maximum | 2 | NR | 2 | 2 |
| 7 | Hours to hold vacuum | 8 + T/4 | NR | Based on DP | Based on DP |
| 8 | Perform final dewpoint (12 hours minimum)<br>If dewpoint acceptable, proceed with step<br>If dewpoint unacceptable, repeat steps 4-8 | R | NR | R | R |
| 9 | Hours to hold vacuum before final filling | 4 | NR | 8 | 8 |
| 10 | Vacuum during filling, Torr, maximum | 3 | NR | 3 | 2 |
| 11 | Oil quality during filling:<br>Dielectric strength, ASTM D1816 (0.080), KV, minimum<br>Water content, ppm, maximum | <br>50<br>10 | <br>50<br>10 | <br>50<br>10 | <br>50<br>10 |
| 12 | Upon completion of oil fill circulate oil within tank thru vacuum processing equipment, passes, minimum | 2 | 2 | 2 | 2 |
| 13 | Transformers having oil pumps<br>Run one half of pumps - each half, hours<br>Stand time to allow oil absorption, hours, minimum | <br>2<br>16 | <br>2<br>16 | <br>2<br>24 | <br>2<br>48 |
| 14 | Transformers without oil pumps<br>Stand time to allow oil absorption, hours, minimum | 24 | 24 | 48 | 48 |

Notes: R = Required
NR = Not Required
DP = DewPoint
T = Open Time

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**Table 3**

| IN-SERVICE OIL RECLAMATION CHARACTERISTICS AND TESTING REQUIREMENTS | | |
|---|---|---|
| PROPERTY | TEST LIMIT | ASTM TEST METHOD |
| Physical Properties | | |
| Color | ≤3.5 | D1500 |
| Interfacial tension at 25°C (dynes/cm) | ≥32 | D971 |
| Specific gravity at 15°C/15°C | 0.84 min. to 0.91 max. | D1298 |
| Visual examination | Clear and bright | D1524 |
| Metals in oil | | D7151 |
| Electrical Properties | | |
| Dielectric breakdown at 60 Hertz | ≥30 kV | D877 |
| 1.0-mm gap | ≥28 kV | D1816 |
| 2.0-mm gap | ≥47 kV | D1816 |
| Power factor at 60 Hertz, percent, at: | | |
| 25°C | <0.1 | D924 |
| 100°C | <3.0 | D924 |
| Chemical Properties | | |
| Oxidation inhibitor content, percent by weight 2.6 ditertiary-butyl para-cresol | 0.20 min. to 0.30 max. | D2668 |
| Water, percent saturation | <8.0 | D1533 |
| Neutralization number, total acid number mg KOH/g of oil | ≤0.05 | D974 |
| Corrosive sulfur | | D1275 |
| Gas Content | | |
| Dissolved Gas Analysis | 95% Removal of current gasses | D2945 |

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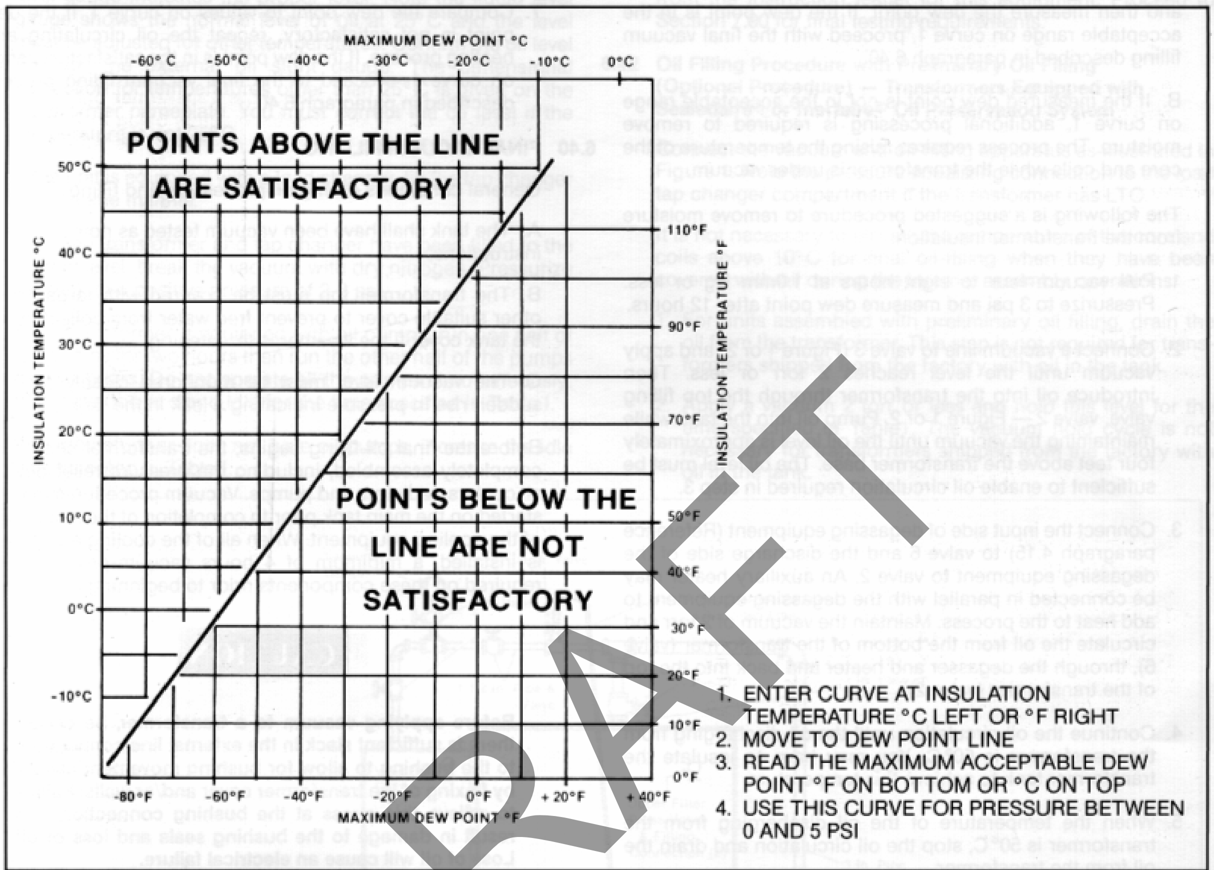
**Table 4**

**MAXIMUM DEW POINT**

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END OF APPENDIX D

APPENDIX E
REFERENCE DRAWINGS

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END OF APPENDIX E

Drawing Approved Electronically
RELEASED FOR O&M
02/05/2013

## NOTES:

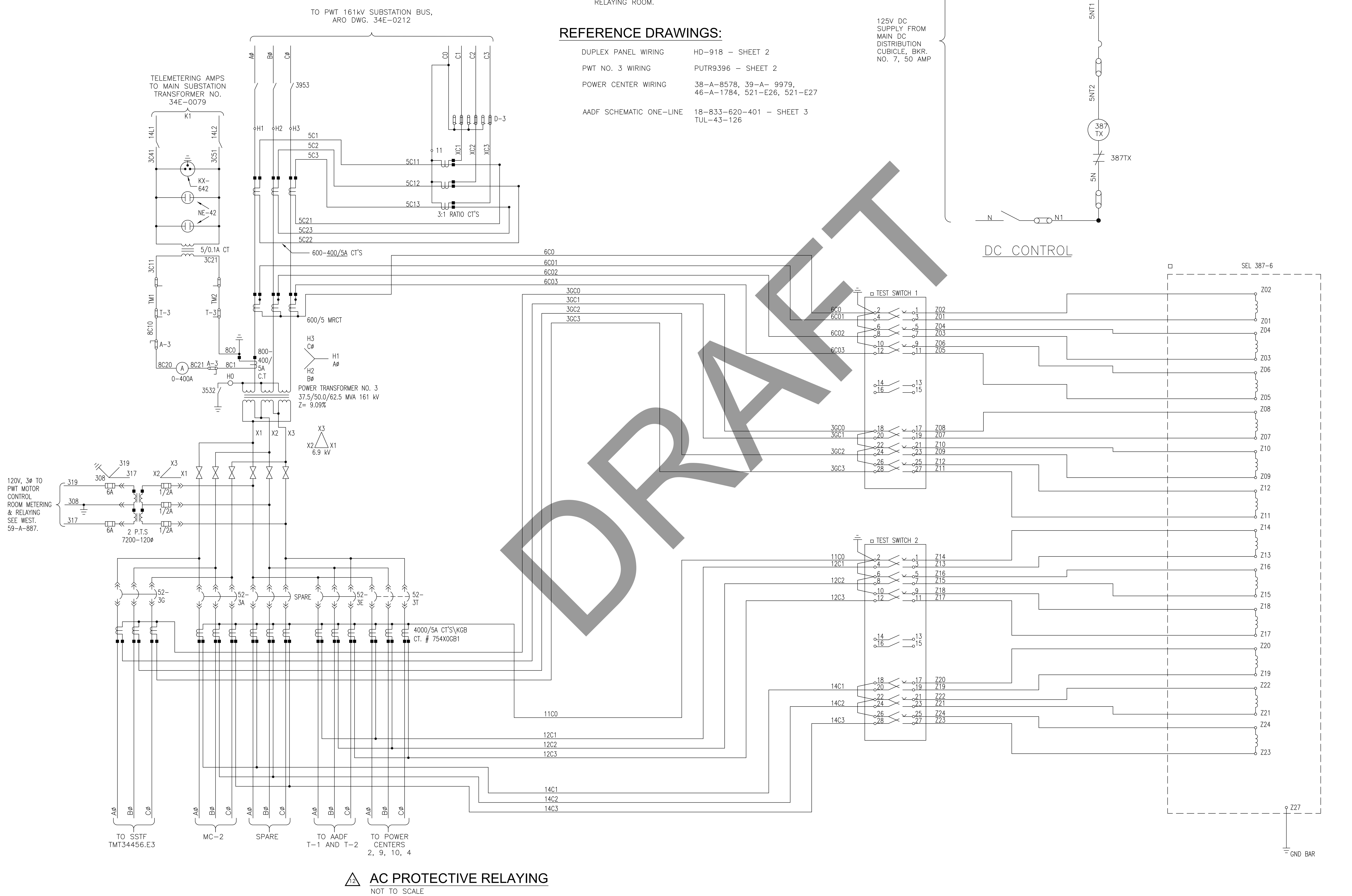
1. SEL387-6 MODEL NO. 0387003X53XX4XX
2. THIS DRAWING SUPERCEDES AC SCHEMATICS FOR PWT NO. 3 TRANSFORMER SHOWN ON DRAWINGS HD-849, AND PYTQ9037.1 SHEET E-6.

## LEGEND:

□ LOCATED IN DUPLEX PANEL 3, PWT PILOT WIRE RELAYING ROOM.

## REFERENCE DRAWINGS:

| | |
|---|---|
| DUPLEX PANEL WIRING | HD-918 – SHEET 2 |
| PWT NO. 3 WIRING | PUTR9396 – SHEET 2 |
| POWER CENTER WIRING | 38-A-8578, 39-A- 9979, 46-A-1784, 521-E26, 521-E27 |
| AADF SCHEMATIC ONE-LINE | 18-833-620-401 – SHEET 3 TUL-43-126 |




## AC PROTECTIVE RELAYING

NOT TO SCALE

THIS DRAWING SUPERCEDES ARO INC. DWG. # 34E-0216 (HAND DRAWN VERSION)

Not Reviewed for Release Outside AEDC

US AIR FORCE / AEDC ARNOLD AFB, TN

DESCRIPTION OF REVISION: 12) SCHEMATIC DIAGRAM REVISED AND REDRAWN PER AS BUILT

| | |
|---|---|
| Job No: | 11889 |
| Building No: | 796 |
| File No: | 0597101 |
| Drawn by: | MELSON-CM |
| Designed by: | BAKER-BC |
| Drawing Check: | VERTREES-VL |
| Engineering Check: | ACKLEN-TR |
| System Engineer: | BAGLEY-TP |
| Approved by: | NORTHCOTT-TW |

SCHEMATIC DIAGRAM
POWER TRANS NO 3 – AC & DC CONTROL
PWT 161 kV SUBSTATION

Drawing Number: 34E-0216 Rev: 12 Sh. 1 of 1

Sheet Reference Number: Sheet 1 of 1

Drawing Approved Electronically
RELEASED FOR O&M
02/05/2013

## NOTES:

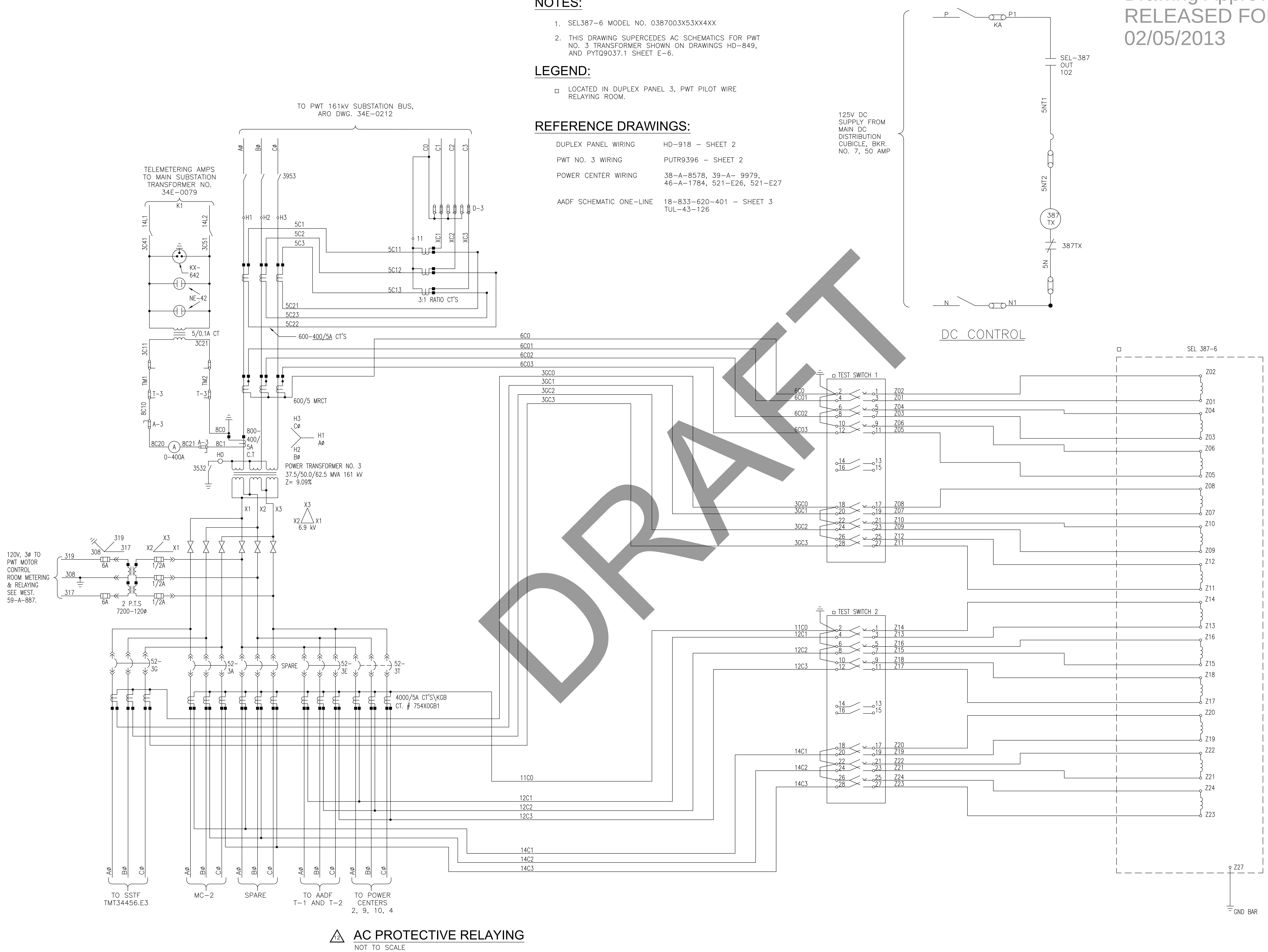
1. SEL387-6 MODEL NO. 0387003X53XX4XX
2. THIS DRAWING SUPERCEDES AC SCHEMATICS FOR PWT NO. 3 TRANSFORMER SHOWN ON DRAWINGS HD-849, AND PYTQ9037.1 SHEET E-6.

## LEGEND:

□ LOCATED IN DUPLEX PANEL 3, PWT PILOT WIRE RELAYING ROOM.

## REFERENCE DRAWINGS:

| | |
|---|---|
| DUPLEX PANEL WIRING | HD-918 – SHEET 2 |
| PWT NO. 3 WIRING | PUTR9396 – SHEET 2 |
| POWER CENTER WIRING | 38-A-8578, 39-A- 9979, 46-A-1784, 521-E26, 521-E27 |
| AADF SCHEMATIC ONE-LINE | 18-833-620-401 – SHEET 3 TUL-43-126 |




⚠12 AC PROTECTIVE RELAYING
NOT TO SCALE

THIS DRAWING SUPERCEDES ARO INC. DWG. # 34E-0216 (HAND DRAWN VERSION)

PLOT SCALE: 1"= 1"

Not Reviewed for Release Outside AEDC

US AIR FORCE / AEDC ARNOLD AFB, TN

DESCRIPTION OF REVISION
12) SCHEMATIC DIAGRAM REVISED AND REDRAWN PER AS BUILT

| | |
|---|---|
| Job No: 11889 | Building No: 796 |
| File No: 0597101 | |
| Drawn by: MELSON-CM | |
| Designed by: BAKER-BC | |
| Drawing Check: VERTREES-VL | |
| Engineering Check: ACKLEN-TR | Approved by: NORTHCOTT-TW |
| System Engineer: BAGLEY-TP | |

SCHEMATIC DIAGRAM
POWER TRANS NO 3 – AC & DC CONTROL
PWT 161 kV SUBSTATION

Drawing Number: 34E-0216 Rev: 12 Sh. 1 of 1

Sheet Reference Number: Sheet 1 of 1

Drawing Approved Electronically
DRAWING OF RECORD
10/08/2012

# REPLACE PWT NO. 3 TRANSFORMER
# ANZY
# JOB NO.

| DESCRIPTION | DRAWING NO. | REV | SHEET NO. |
|---|---|---|---|
| COVER SHEET/DRAWING LIST | PUTR9396 | B | 1 of 4 |
| CONTROL CABINET WIRING DIAGRAM ADDITIONS | PUTR9396 | B | 2 of 4 |
| OA/FA/FOA TRANSFORMER SCHEMATIC WIRING DIAGRAM | PUTR9396 | B | 3 of 4 |
| PARTIAL PLAN – TRANSFORMER BASE | PUTR9396 | B | 4 of 4 |

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PLOT SCALE: 1"= 1"

Not Reviewed for Release Outside AEDC

US AIR FORCE / AEDC
ARNOLD AFB, TN

DESCRIPTION OF REVISION
B) REVISED SHEET 2 OF 4

Drawn by: VERTREES-VL
Designed by: FREDERICK-HM
Drawing Check: MELSON-CM
Engineering Check: BAJAR-KR
System Engineer: BAGLEY-TP
Job No: EP21
File No:
Building No: 732
Approved by: CASTEEL-GL

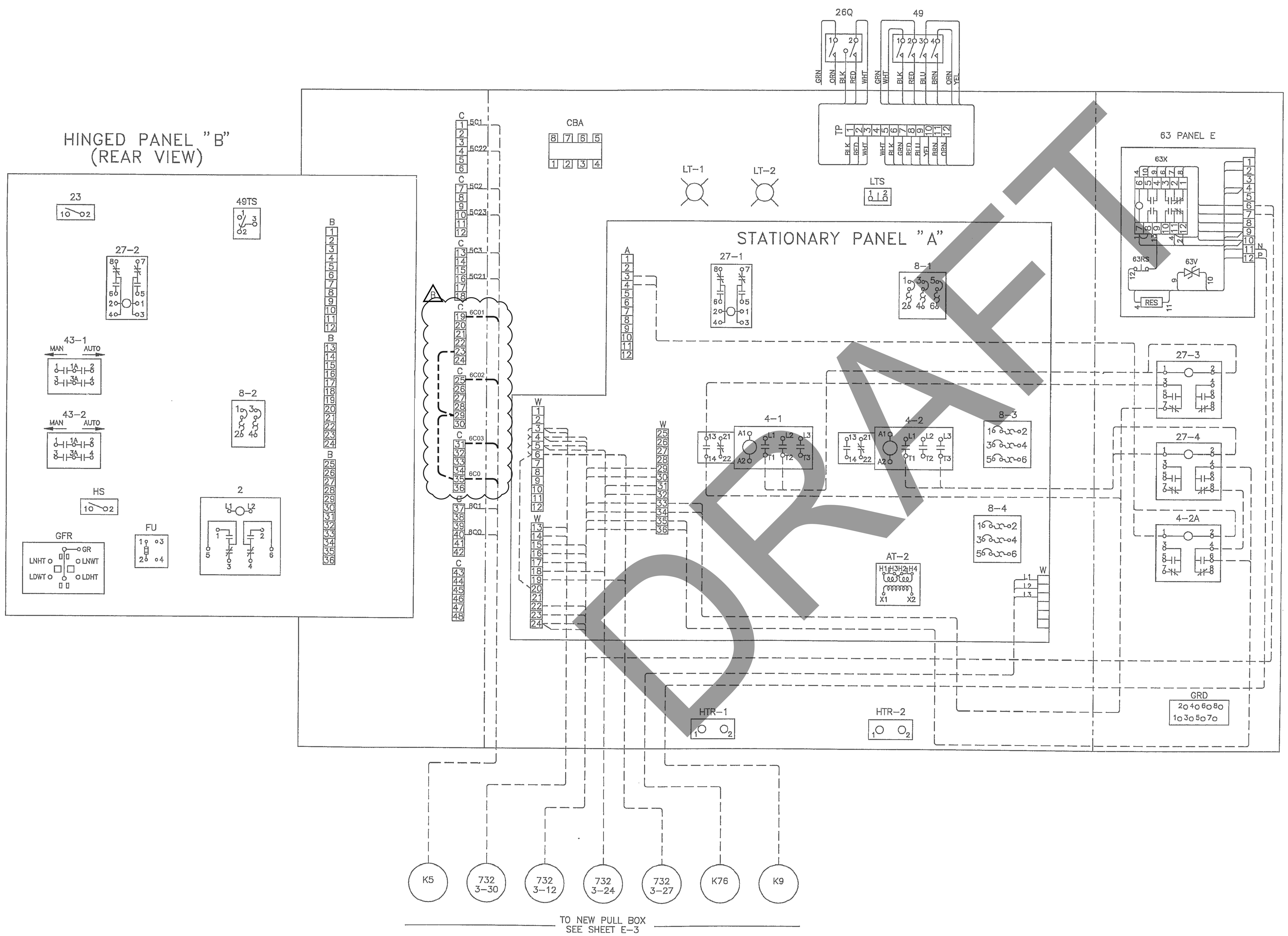
REPLACE PWT NO. 3 TRANSFORMER
CECORS NO. 920141-3

COVER SHEET/DRAWING LIST

Rev: B
Sh. 1 of 4
Drawing Number: PUTR9396

Sheet Reference Number: CS1
Sheet 1 of 4

Drawing Approved Electronically
DRAWING OF RECORD
10/08/2012




## NOTES:

1. SEE ABB DRAWING MNL5845-05 FOR FACTORY WIRING.
2. INSTALL NEW RELAYS 27-3, 27-4, 4-2A AND CONNECT AS SHOWN.

## LEGEND:

------ NEW WIRING

## REF DWGS:

ABB: MNL5845-05 --- CONNECTION WIRING DIAGRAM
S&P: 521-E5 ------ CONTROL DUCTING SECTIONS
521-E24 ------ CONDUIT SCHEDULE
521-E25 ------ CABLE SCHEDULE
521-E26 ------ MAIN SUBSTATION SWBD CONNECTIONS
NELSON ELECTRIC: HD-919 -- WIRING DIAGRAM - SECTION 3 PWT DPLX SWBD
SSI: PUTR9396 SHT E-2 -- XFMR CONTROL PWR SCHEM.
ARO: 34E-0216 -- SCHEMATIC DIAGRAM PWR XFMR #3.

PLOT SCALE: 1" = 1"

Not Reviewed for Release Outside AEDC

US AIR FORCE / AEDC
ARNOLD AFB, TN

DESCRIPTION OF REVISION
B) REVISED WIRING AS SHOWN

| | | |
|---|---|---|
| Drawn by: VERTREES-VL | Job No: EP21 | |
| Designed by: FREDERICK-HM | File No: | Building No: 732 |
| Drawing Check: MELSON-CM | | |
| Engineering Check: BAJAR-KR | | |
| System Engineer: BAGLEY-TP | Approved by: CASTEEL-GL | |

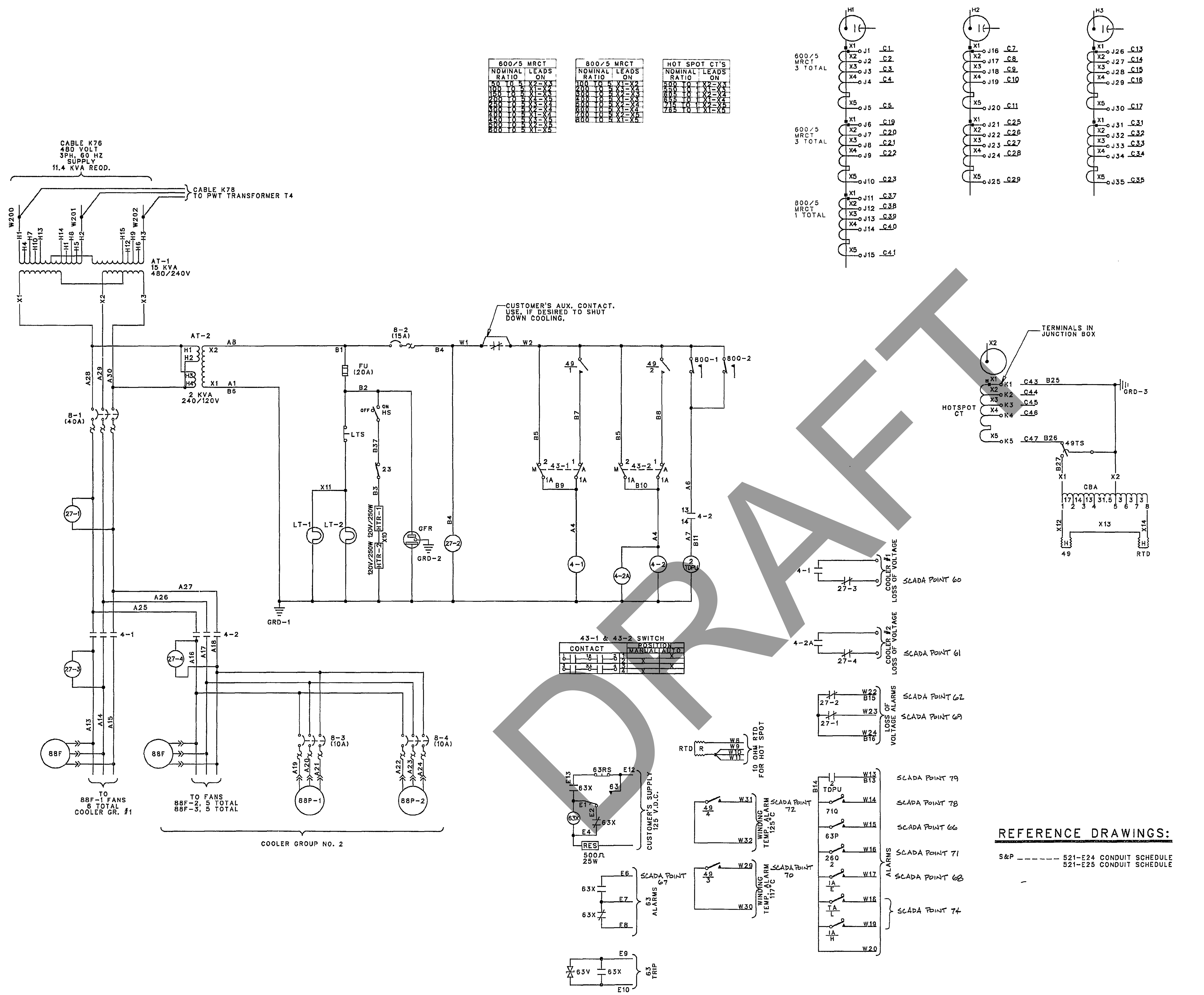
REPLACE PWT NO. 3 TRANSFORMER
CECORS NO. 920141-3

CONTROL CABINET WIRING DIAGRAM ADDITIONS

Drawing Number: PUTR9396
Rev: B
Sh. 2 of 4

Sheet Reference Number: E-1
Sheet 2 of 4

Drawing Approved Electronically
DRAWING OF RECORD
10/08/2012




## LEGEND:

| | |
|---|---|
| 2 | OIL FLOW TIME DELAY RELAY |
| | TDPU TIME DELAY PICK-UP |
| 4-1.2 | COOLER CONTACTOR |
| 8-1.2 | CIRCUIT BREAKER |
| 23 | HEATER THERMOSTAT |
| 26Q | HOT OIL THERMOMETER |
| 1 | ALARM CONTACT (60°C) NOT USED |
| 2 | ALARM CONTACT (90°C) |
| 27-1.2 | LOSS OF VOLTAGE RELAY |
| 43-1.2 | COOLER CONTROL SWITCH |
| 49 | THERMAL DEVICE |
| 1 | FAN CONTACT (70°C) |
| 2 | FAN CONTACT (75°C) |
| 3 | ALARM CONTACT (117°C.) |
| 4 | ALARM CONTACT (125°C.) |
| H | HEATER ELEMENT |
| 49TS | THERMAL DEVICE CURRENT TEST SWITCH |
| 63 | SUDDEN PRESSURE RELAY |
| RS | RESET SWITCH |
| V | VOLTRAP |
| X | SEAL-IN RELAY |
| 63P | MECHANICAL RELIEF DEVICE |
| 71Q | LIQUID LEVEL GAUGE |
| 80Q-1.2 | OIL FLOW GAUGE |
| 88F-1.2.3 | FAN MOTOR WITH THERMOGAURD |
| 88P-1.2 | PUMP MOTOR |
| AT-1 | AUXILIARY TRANSFORMER (480/240V.) |
| AT-2 | AUXILIARY TRANSFORMER (240/120V.) |
| CBA | CURRENT BALANCING AUTOTRANSFORMER |
| FU | FUSE |
| GFR | GROUND FAULT RECEPTACLE |
| HS | HEATER SWITCH |
| HTR-1.2 | COMPARTMENT HEATER |
| IA | INERTAIRE |
| E | EMPTY CYLINDER |
| H | HIGH PRESSURE |
| L | LOW PRESSURE |
| LT | COMPARTMENT LIGHT |
| LTS | COMPARTMENT LIGHT SWITCH |
| RTD | REMOTE TEMPERATURE DETECTOR |
| H | HEATER ELEMENT |
| RC | RESISTANCE COIL |
| 4-24 | AUXILIARY SG RELAY |

CAPACITOR
NORMALLY OPEN RELAY CONTACT
NORMALLY CLOSED RELAY CONTACT
THERMAL TRIP ELEMENT
CIRCUIT BREAKER
DISCONNECT DEVICE
OPERATING COIL
MOMENTARY CONTACT SWITCH
MAINTAINED CONTACT SWITCH
PUSH BUTTON SWITCH
RESISTOR
THERMAL DEVICE TEST SWITCH

## REFERENCE DRAWINGS:

S&P ------ 521-E24 CONDUIT SCHEDULE
521-E25 CONDUIT SCHEDULE

PLOT SCALE: 1"= 1"

Not Reviewed for Release Outside AEDC

US AIR FORCE / AEDC
ARNOLD AFB, TN

DESCRIPTION OF REVISION

| | | |
|---|---|---|
| Drawn by: VERTREES-VL | Job No: EP21 | |
| Designed by: FREDERICK-HM | File No: | Building No: 732 |
| Drawing Check: MELSON-CM | | |
| Engineering Check: BAJAR-KR | | |
| System Engineer: BAGLEY-TP | Approved by: CASTEEL-GL | |

REPLACE PWT NO. 3 TRANSFORMER
CECORS NO. 920141-3
OA/FA/FOA TRANSFORMER
SCHEMATIC WIRING DIAGRAM

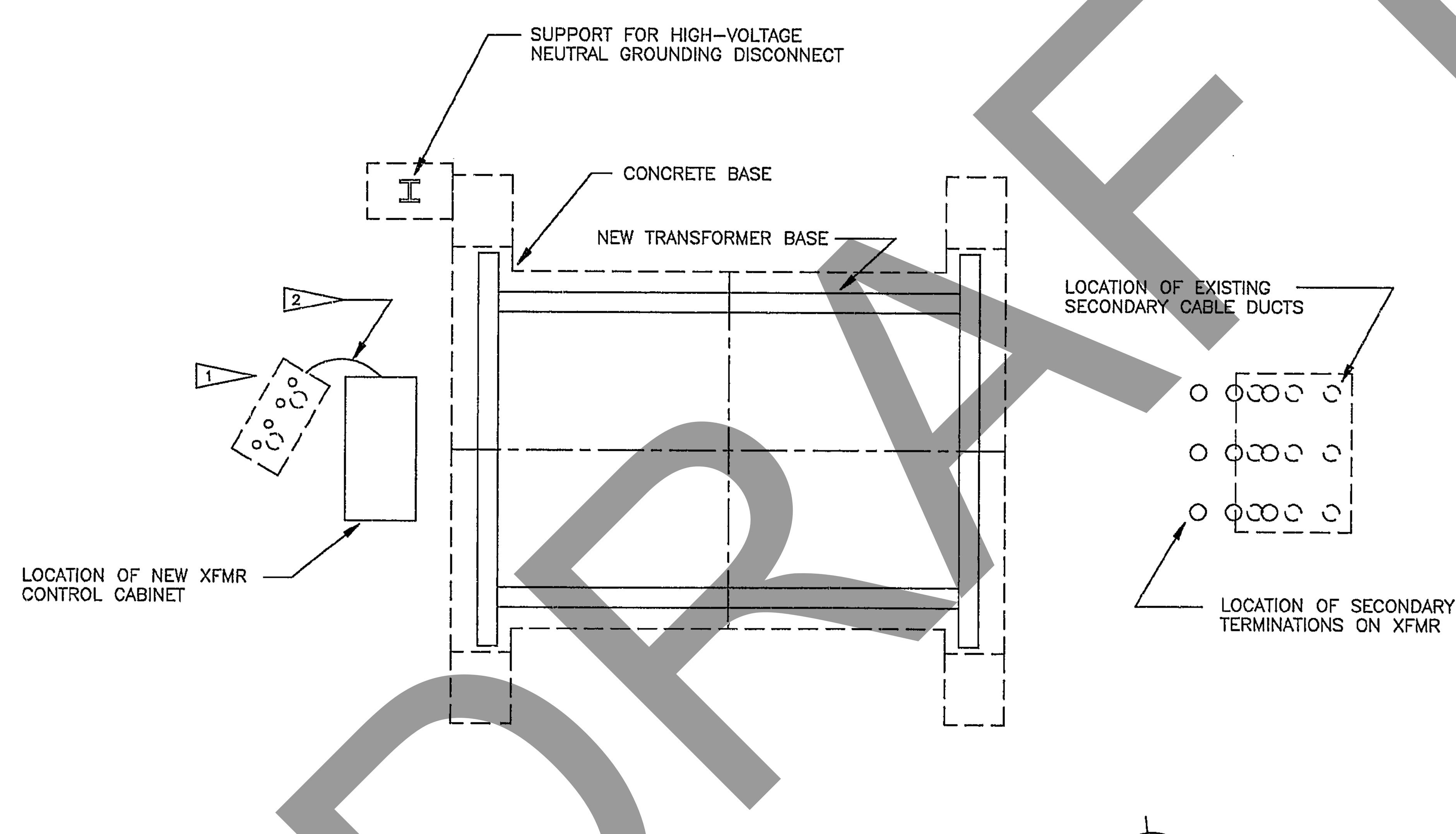
Drawing Number: PUTR9396
Rev: B
Sh. 3 of 4

Sheet Reference Number: E-2
Sheet 3 of 4

Drawing Approved Electronically
DRAWING OF RECORD
10/08/2012

## NOTES:

1. INSTALL A NEW PULL BOX ONTO EXISTING CONDUIT STUB-UPS
2. INSTALL TWO FOUR-INCH SEALTITE CONDUITS BETWEEN THE NEW PULL BOX AND THE NEW XFMR CONTROL CABINET




DRAFT

PLAN VIEW: TRANSFORMER BASE
SCALE: 1" = 2'- 0"

PLOT SCALE: 1" = 1"

Not Reviewed for Release Outside AEDC

US AIR FORCE / AEDC
ARNOLD AFB, TN

DESCRIPTION OF REVISION

| | |
|---|---|
| Drawn by: VERTREES-VL | Job No: EP21 |
| Designed by: FREDERICK-HM | File No: |
| Drawing Check: MELSON-CM | Building No: 732 |
| Engineering Check: BAJAR-KR | |
| System Engineer: BAGLEY-TP | Approved by: CASTEEL-GL |

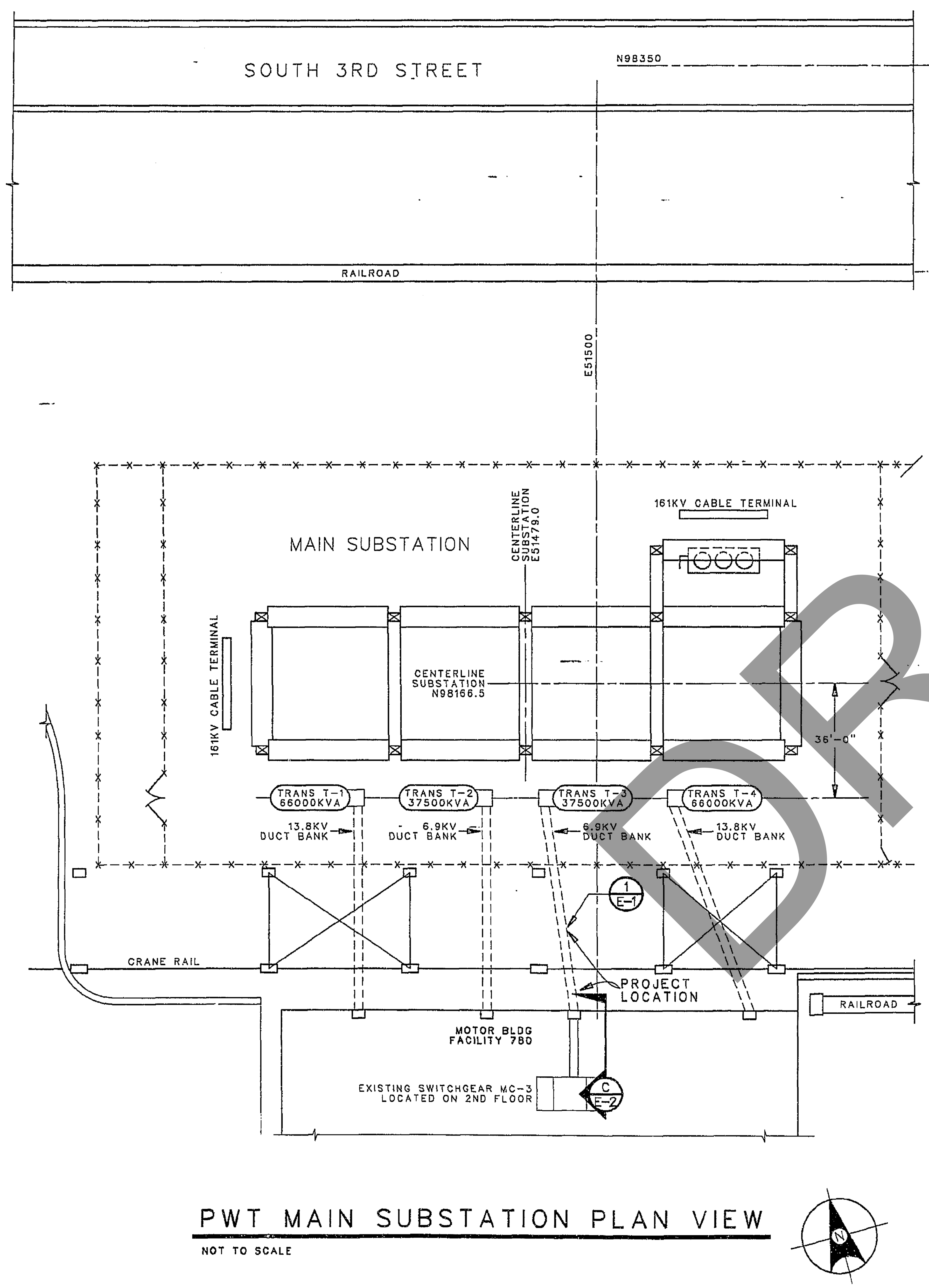
REPLACE PWT NO. 3 TRANSFORMER
CECORS NO. 920141-3

PARTIAL PLAN - TRANSFORMER BASE

Drawing Number: PUTR9396
Rev: B
Sh. 4 of 4

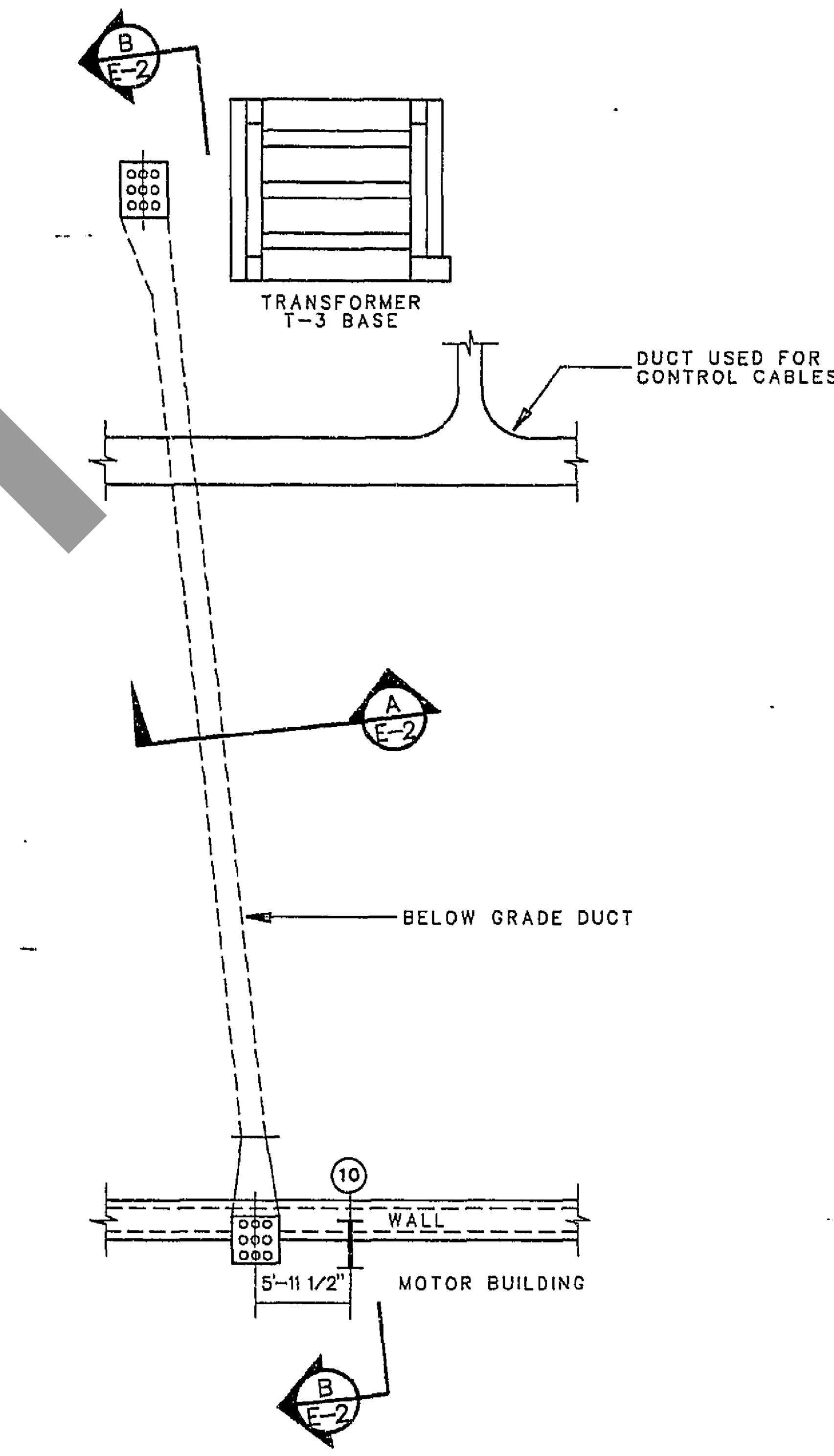
Sheet Reference Number: E-3
Sheet 4 of 4

| LETTER | REVISION | DATE | BY | APVD |
|---|---|---|---|---|




PWT MAIN SUBSTATION PLAN VIEW

NOT TO SCALE




1 E-1 DETAIL

NOT TO SCALE

DRAFT

| PYTQ9037.1 | SH 2 |
|---|---|

AEDC CAD/CAM SYSTEM

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BY ELECTRONIC MEASURES

OUTSTANDING CHANGE ORDERS

PREPARED FOR: US AIR FORCE/AEDC,ARNOLD AFB,TN

PREPARED BY: SSI SERVICES, INC.

PROJECT TITLE: REPLACE PWT NO. 3 TRANSFORMER AND POWER CABLES CECORS NO. 920141-2

DRAWING TITLE: PROPULSION WIND TUNNEL MAIN SUBSTATION LOCATION PLAN

| SYSTEM NO | FILE NO | |
|---|---|---|
| 02300007 | PYTQ9037 | |
| AF PROJ NO: EP21 | BOM NO | SCALE: NTS |
| PLANNER | PHONE | SHOP JOB NO |
| JOB CONTACT | PHONE | COST CODE |

| | NAME | DATE | APVD | NAME | DATE |
|---|---|---|---|---|---|
| DRAWN | JD MCFADDEN | 10JUN92 | GB13 | B.L. Mayberry | 7.23.92 |
| CHECKED | T.R.Aikle | 22 JUL 92 | | | |
| DESIGN | W.H. Barnwood | 22 JUL 92 | | | |
| STRESS | N/A | | | | |

| SIZE | DRAWING NO | REV | REF NO | SHEET NO |
|---|---|---|---|---|
| SR | PYTQ9037.1 | A | E-1 | 2 of 10 |

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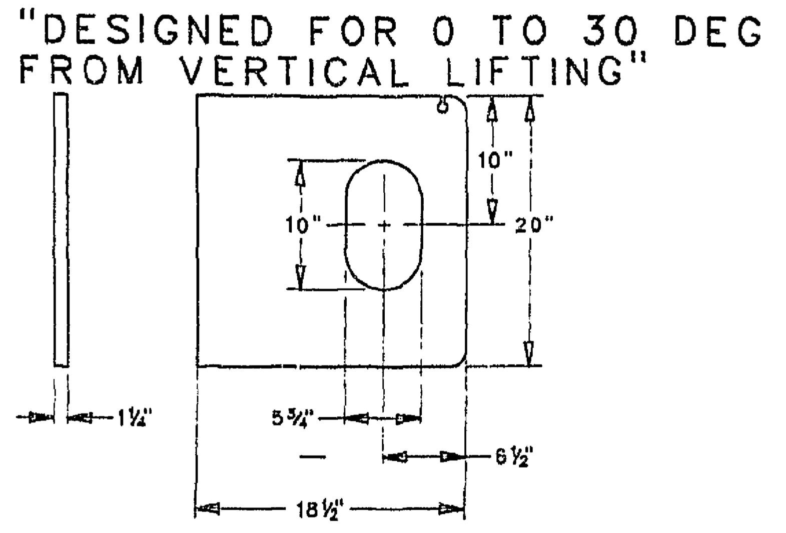
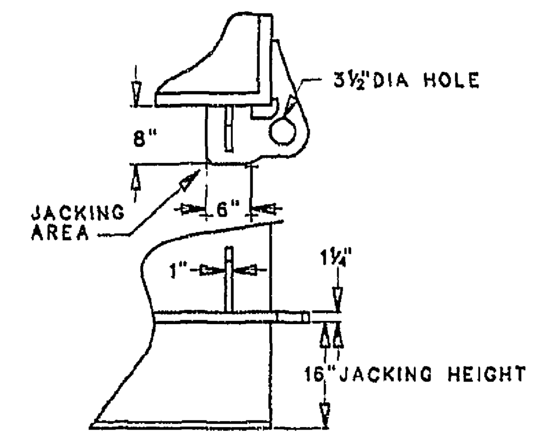
DRW1 P9037_IE01 1 01JUL92

| LETTER | REVISION | DATE | BY | APVD |
|---|---|---|---|---|
| | | | | |

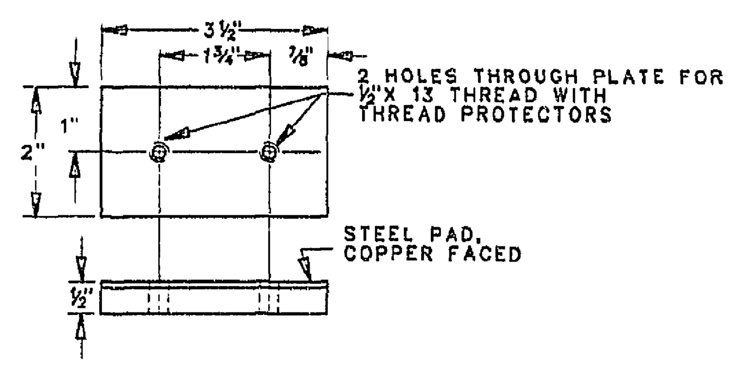
"DESIGNED FOR 0 TO 30 DEG FROM VERTICAL LIFTING"



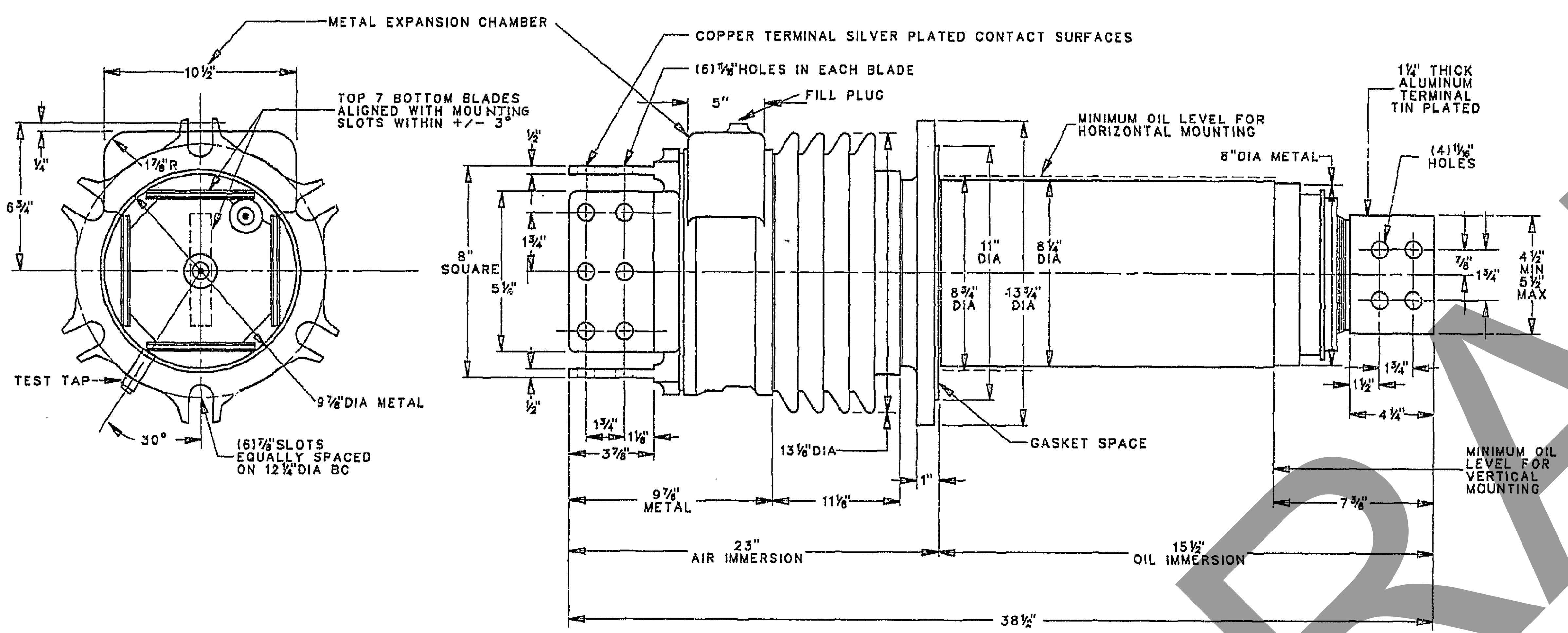

DETAIL OF ITEM 19




DETAIL OF ITEM 30

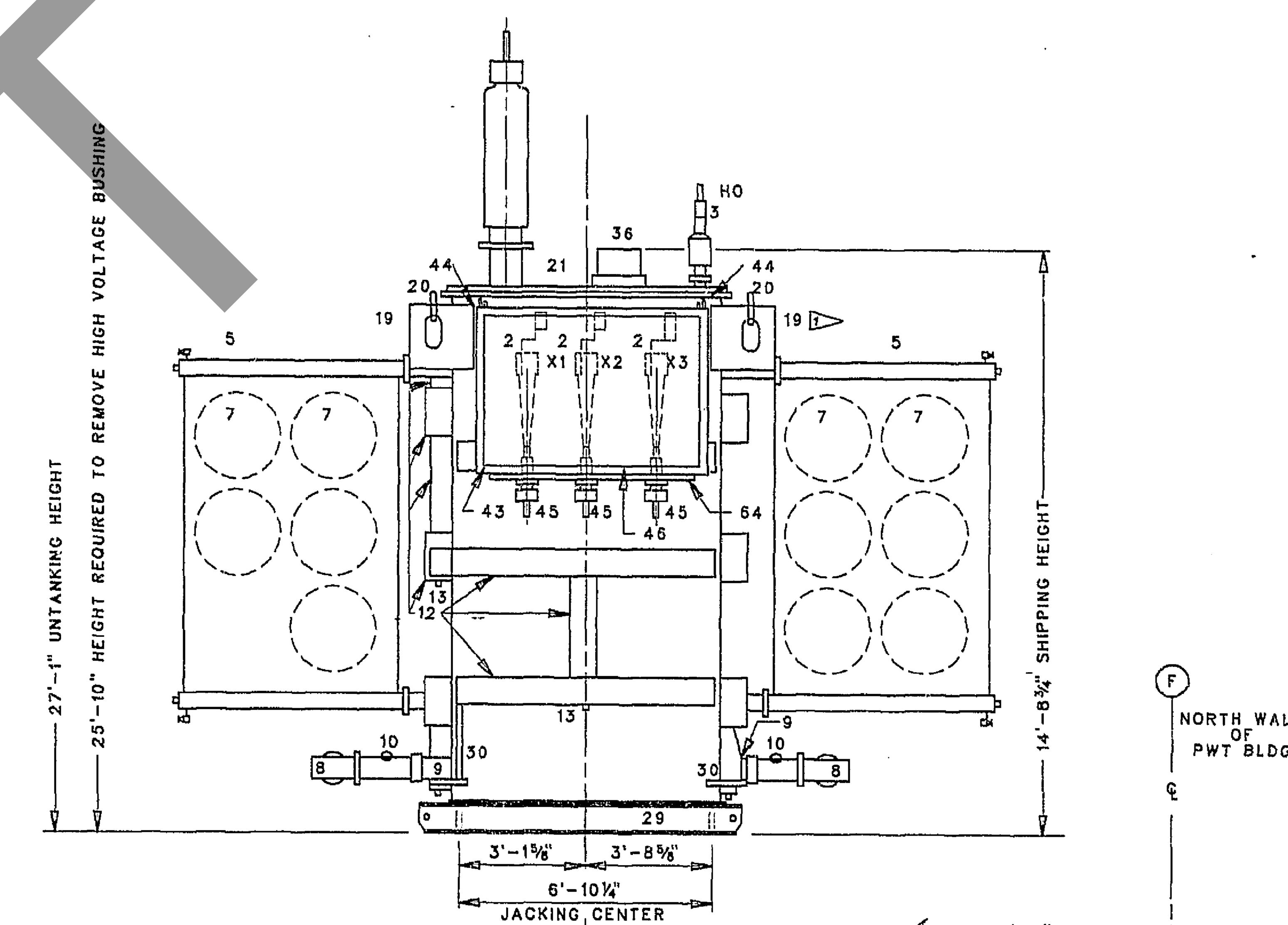



DETAIL OF ITEM 24



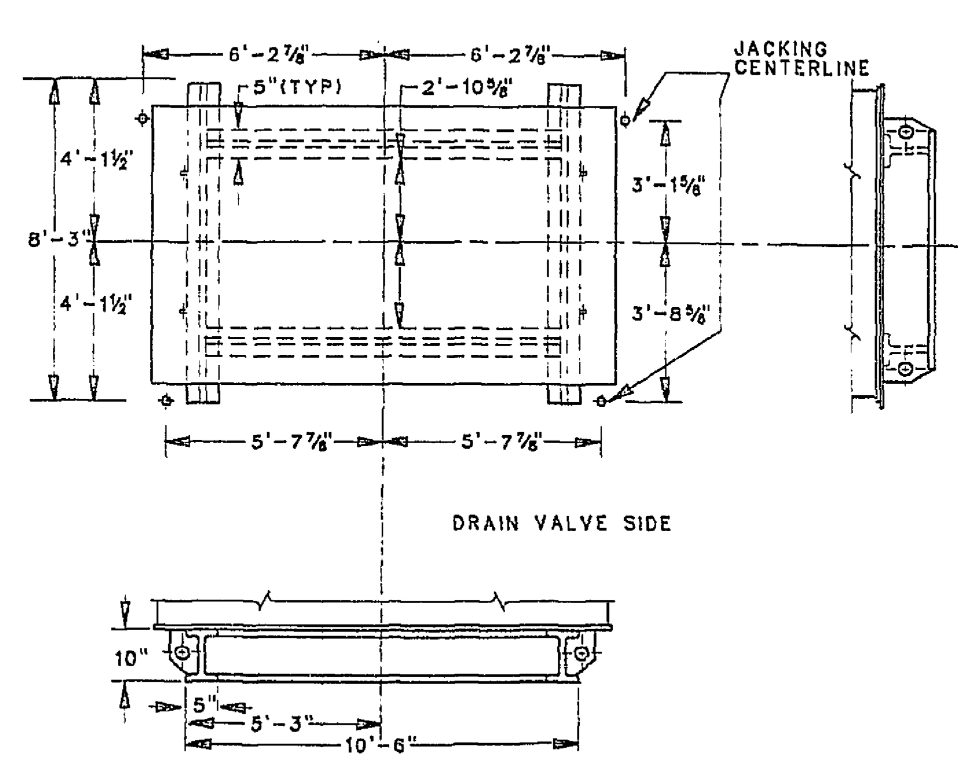

LOW VOLTAGE TRANSFORMER BUSHING

TYPE T-CLASS 25KV-RATED AMPS 6000

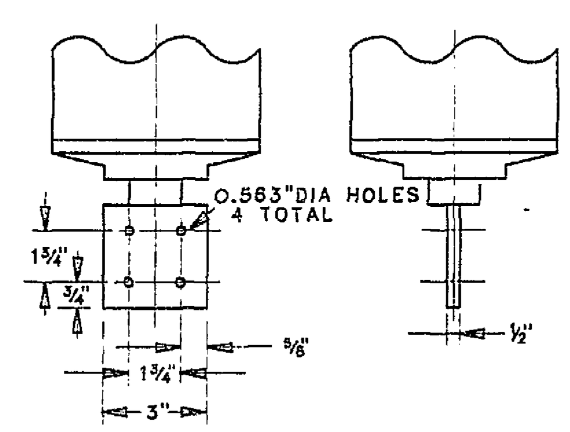



PWT TRANSFORMER NO. 3

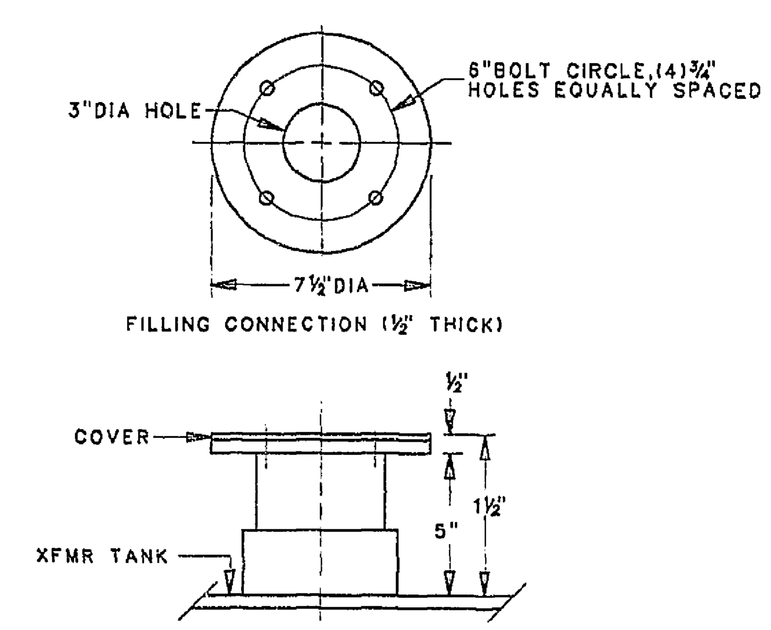
WEST ELEVATION




DETAIL OF ITEM 29




DETAIL OF ITEM 45




DETAIL OF ITEM 23

NOTES:
1. FOR PLAN VIEW AND DESCRIPTION OF ITEMS,SEE SHEET E-4..
2. CGI=COMPLETE CENTER OF GRAVITY.
3. EXISTING FOUNDATION PIERS TO BE USED FOR NEW TRANSFORMER. CENTERLINE OF TRANSFORMER TO BE 66'0" FROM CENTERLINE OF BUILDING NORTH WALL.

PYTQ9037.1 SH 4

AEDC CAD/CAM SYSTEM

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OUTSTANDING CHANGE ORDERS

PREPARED FOR US AIR FORCE/AEDC,ARNOLD AFB,TN — PREPARED BY SSI SERVICES, INC.

PROJECT TITLE: REPLACE PWT NO. 3 TRANSFORMER AND POWER CABLES CECORS NO. 920141-2

DRAWING TITLE: PWT TRANSFORMER NO.3 WEST ELEVATION AND LOW VOLTAGE BUSHING DETAILS

| SYSTEM NO | FILE NO | | NAME | DATE | APVD | NAME | DATE |
|---|---|---|---|---|---|---|---|
| 02300007 | PYT09037 | DRAWN | D.G.GATES | 06MAY92 | | | |
| AF PROJ NO: EP21 | BOM NO | CHECKED | | 22 JUL 92 | | | |
| | SCALE: NTS | DESIGN | | 22 JUL 92 | | | |
| PLANNER | PHONE | STRESS | N/A | | | | |
| JOB CONTACT | PHONE / COST CODE | SIZE: SR | DRAWING NO: PYTQ9037.1 | REV: A | REF NO: E-3 | SHEET NO: 4 of 10 | |

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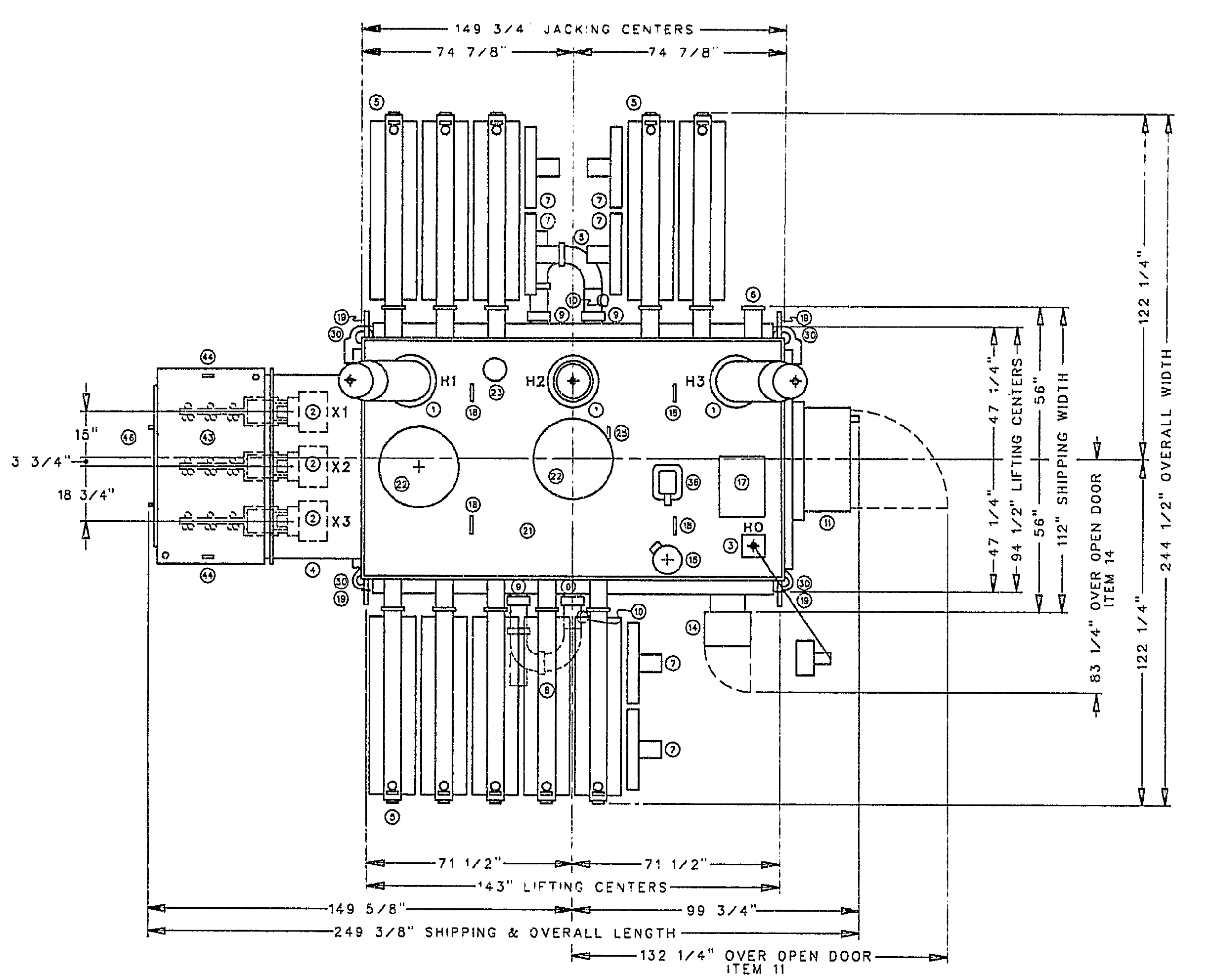
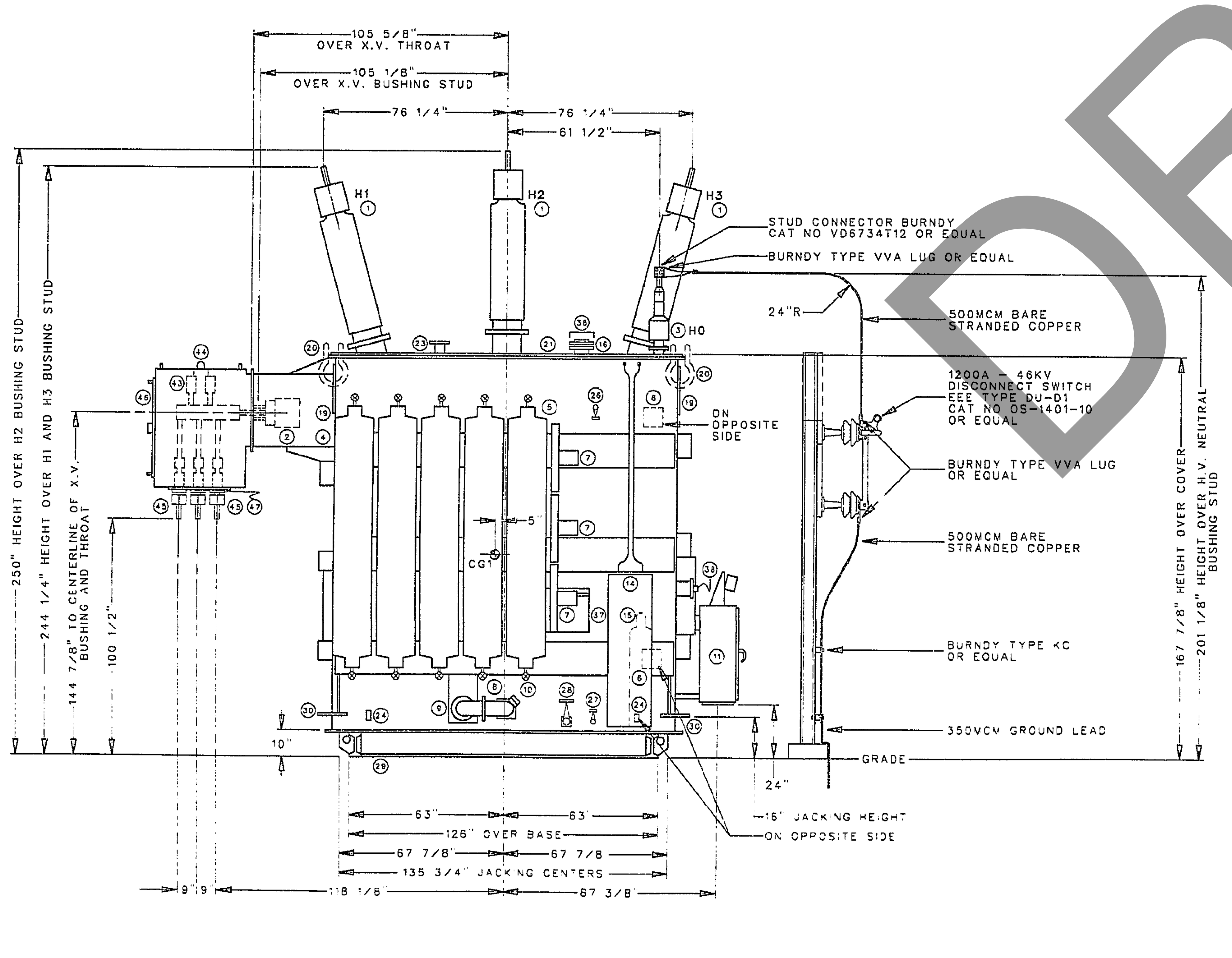
| LETTER | REVISION | DATE | BY | APVD | | | |
|---|---|---|---|---|---|---|---|
| A | ADDED REF. DWG. FOR AADF | 9/24/01 | RN | TRA | 9/24/01 | KN | 9/24/01 |




TOP VIEW




SOUTH ELEVATION

NOT TO SCALE

| ITEM NO. | DESCRIPTION |
|---|---|
| 1 | H.V. BUSHING - 3 TOTAL - STUD SIZE 1.5 INCH DIAMETER - 12 THREAD X 2 INCHES LONG - INTERNAL TOP CONNECTED - DRAW THROUGH LEAD. |
| 2 | X.V. BUSHING - 3 TOTAL - EXTERNAL SIX HOLE SPADE TYPE TERMINAL - 4 BLADES - INTERNAL BOTTOM CONNECTED - BOLTED TERMINAL. |
| 3 | H.V. NEUTRAL BUSHING - 1 TOTAL - WITH STUD SIZE 1.5 INCH DIAMETER - 12 THREAD X 2.125 INCHES LONG - INTERNAL TOP CONNECTED - DRAW THROUGH LEAD. |
| 4 | X.V. THROAT |
| 5 | RADIATORS - WITH 0.75 INCH DRAIN AND VENT VALVE - CONNECTED AT TOP AND BOTTOM WITH SHUT-OFF VALVES ON CASE. |
| 6 | SPARE RADIATOR VALVES - 2 TOTAL. |
| 7 | FORCED-AIR COOLING FANS - FAN MOTORS ARE THREE PHASE - 60 HERTZ - 230 VOLTS - WITH OSHA FAN GUARDS. |
| 8 | FORCED OIL PUMP WITH DRAIN AND VENT PLUGS - 2 TOTAL - MOTOR IS THREE PHASE - 60 CYCLES - 230 VOLTS. |
| 9 | SHUT-OFF VALVES - 4 TOTAL. |
| 10 | OIL FLOW INDICATOR WITH ALARM CONTACTS - 2 TOTAL. |
| 11 | CONTROL CABINET FOR FAN, PUMPS, ALARM, AND CURRENT TRANSFORMER LEADS - WITH UNDRILLED PLATE OVER 10 X 26.5 OPENING FOR CONNECTION WITH PROVISION FOR PADLOCKING. |
| 12 | TANK BRACES USED FOR ADDITONAL GAS SPACE. DO NOT DRILL THROUGH THESE BRACES. |
| 13 | DRAIN PLUG - CONTRUCTION TO OPEN, DRAIN OIL, AND REPLACE PLUG - 2 TOTAL. |
| 14 | INERTAIRE CABINET WITH ALARM CONTACTS - HINGED CABINET DOOR WITH PROVISION FOR PADLOCKING. |
| 15 | NITROGEN CYLINDER. |
| 16 | AUTOMATIC RESET MECHANICAL RELIEF DEVICE WITH SEMAPHORE AND ALARM CONTACTS FOR INDICATING RELIEF DEVICE TRIP - WITH YELLOW FLAG ASSEMBLY. |
| 17 | JUNCTION BOX FOR CURRENT TRANSFORMER #10 WIRE LEADS. |
| 18 | LIFTING LOOPS - 4 TOTAL - FOR LIFTING COVER ONLY. |
| 19 | LIFTING HOOKS - 4 TOTAL - FOR LIFTING COMPLETE UNIT - SEE DETAIL ON E-3. |
| 20 | ANCHOR SHACKLE - FURNISHED BY FIELD. |
| 21 | CASE AND COVER - WELDED. |
| 22 | MANHOLE 24 INCH DIAMETER OPENING - 2 TOTAL. |
| 23 | 3 INCH FLANGE FOR FILLING CONNECTION - SEE DETAIL ON E-3. |
| 24 | GROUND PAD ON CASE WALL - 2 TOTAL - SEE DETAIL ON E-3. |
| 25 | MAIN TRANSFORMER CORE GROUND PAD - WITH CAPTIVE BOLTS. |
| 26 | VALVE - 1 INCH - FOR UPPER FILTER PRESS CONNECTION. |
| 27 | VALVE - 1 INCH GLOBE - FOR LOWER FILTER PRESS CONNECTION. |
| 28 | VALVE - 2 INCH - FOR COMBINATION LOWER FILTER PRESS CONNECTION AND COMPLETE DRAIN - WITH 0.375 SAMPLING DEVICE. |
| 29 | BASE - 10 INCH I-BEAM - WITH VENTILATION PORTS - SEE DETAIL ON E-3. |
| 30 | COMBINATION JACKING AND PULLING LUG FOR COMPLETE UNIT - 4 TOTAL - SEE DETAIL ON E-3. |
| 31 | HAND HYDRAULIC TYPE JACKS - 50 TON - WITH PRESSURE GAUGE AND PRESSURE RELIEF DEVICE - 4 TOTAL - SHIPPED DIRECT TO ARNOLD AFB TN. FROM SUPPLIER. |
| 32 | LIQUID TEMPERATURE INDICATOR DIAL TYPE WITH CAPILLARY TUBE WITH MAXIMUM INDICATING HAND AND ALARM CLOSING CONTACTS. |
| 33 | HOT SPOT TEMPERATURE INDICATOR DIAL TYPE WITH CAPILLARY TUBE WITH MAXIMUM INDICATING HAND AND ALARM CLOSING CONTACTS. |
| 34 | HOT SPOT TEMPERATURE DETECTOR - COPPER RESISTANCE COIL OF 10 OHMS NOT SHIELDED AT 25°C, FOR SWITCHBOARD TEMPERATURE INDICATOR. |
| 35 | MAGNETIC LIQUID LEVEL GAUGE WITH LOW LIQUID LEVEL ALARM CONTACTS. |
| 36 | SUDDEN PRESSURE RELAY. |

| ITEM NO. | DESCRIPTION |
|---|---|
| 37 | EPT AUXILIARY TRANSFORMER - 15 KVA - 3 PHASE - 60 HERTZ - 480 TO 240 VOLTS. |
| 38 | DE-ENERGIZED TAP CHANGER OPERATING MECHANISM WITH PROVISION FOR PADLOCKING. |
| 39 | DE-ENERGIZED TAP CHANGER WARNING NAMEPLATE. |
| 40 | PERMANENT MARKINGS TO INDICATE 25°C OIL LEVEL. |
| 41 | DIAGRAM INSTRUCTION PLATE. |
| 42 | DRESS NAMEPLATE. |
| 43 | X.V. 110 KV BIL AIR FILLED TERMINAL CHAMBER - BOLTED TO TRANSFORMER. |
| 44 | LIFTING LOOPS - 2 TOTAL - FOR LIFTING X.V. AIR FILLED TERMINAL CHAMBER ONLY. |
| 45 | TERMINAL CHAMBER BUSHING FOR SECONDARY CABLE CONNECTION - 9 TOTAL - SEE DETAIL ON E-3. |
| 46 | BOLTED ON COVER OVER 37 X 52 INCH OPENING FOR ACCESS TO X.V. CONNECTIONS. |
| 47 | NON MAG PLATE OVER 25 X 44.5 INCH OPENING. |
| | CGI = COMPLETE CENTER OF GRAVITY. |

## NOTES:

1. SEE ABB INSTRUCTION BOOK MNL5845-12 FOR INSTALLATION AND MANTENANCE.
2. OUTLINES SHOWN ARE TO BE USED FOR ERECTION PURPOSES.
3. BEFORE BREAKING THE SEAL ON ANY OPENING, BE SURE THE LIQUID IS LOWERED BELOW THE OPENING.
4. SEE ABB INSTRUCTION BOOK BEFORE FILLING THE CASE WITH LIQUID.
5. **DO NOT** LIFT TRANSFORMER AND CASE TOGETHER UNLESS THE COVER IS FASTENED IN PLACE.
6. THE CASE IS BRACED FOR FILLING UNDER FULL VACUUM.
7. OPERATION OF TAP CHANGER MECHANISM MUST BE CHECKED BEFORE SEALING TRANSFORMER.
8. DO NOT ENTER UNIT WITHOUT CHECKING OXYGEN CONTENT FOR SAFETY.
9. FIELD TO LOCATE AND INSTALL A 20"X 24"X 8" JUNCTION BOX TO INTERCEPT THE EXISTING CONDUITS AND CONTROL CABLES ENTERING THE TRANSFORMER CONTROL CABINET. FABRICATE MOUNTING BRACKETS TO ATTACH THE JUNCTION BOX TO THE TRANSFORMER USING P1000 UNISTRUT. USE LIQUID-TIGHT FLEXIBLE CONDUIT AS REQUIRED FROM THE EXISTING CONDUIT STUB-UPS TO THE JUNCTION BOX. INSTALL TERMINAL BLOCKS IN THE JUNCTION BOX TO TERMINATE THE EXISTING CABLES. USE WIRING FROM THE TERMINAL BLOCKS TO THE TRANSFORMER CONTROL CABINET COMPATIBLE WITH EXISITNG CIRCUITS.

## REFERENCE DRAWINGS:

PWT MAIN 161KV SUBSTATION

ARO, INC.————————34E-0144, 34E-0212

PENNSYLVANIA TRANSFORMER CO.————60938, 61562

SVERDRUP & PARCEL, INC.————SK-500-E065, SK-521-E07, SK-521-E10, SK-521-E20, SK-521-S182, SK-521-E15

A BLACK & VEATCH————E-501

PYTQ9037.1 SH 5

AEDC CAD/CAM SYSTEM

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OUTSTANDING CHANGE ORDERS

PREPARED FOR US AIR FORCE/AEDC,ARNOLD AFB,TN

PREPARED BY SSI SERVICES, INC.

PROJECT TITLE

REPLACE PWT NO. 3 TRANSFORMER AND POWER CABLES

CECORS NO. 920141-2

DRAWING TITLE

PWT TRANSFORMER #3 SOUTH ELEVATION PLAN AND COMPONENTS DESCRIPTION

| SYSTEM NO | FILE NO | |
|---|---|---|
| 02300007 | PYTQ9037 | |
| AF PROJ NO: EP21 | BOM NO | SCALE: NTS |
| PLANNER | PHONE | SHOP JOB NO |
| JOB CONTACT | PHONE | COST CODE |

| | NAME | DATE | APVD | NAME | DATE |
|---|---|---|---|---|---|
| DRAWN | JD MCFADDEN | 07MAY92 | GB13 | BR Mungrem | 7-23-92 |
| CHECKED | TRAcklen | 22 JUL 92 | | | |
| DESIGN | W.H. Bainwood | 22 JUL 92 | | | |
| STRESS | N/A | | | | |

| SIZE | DRAWING NO | REV | REF NO | SHEET NO |
|---|---|---|---|---|
| SR | PYTQ9037.1 | A | E-4 | 5 OF 10 |

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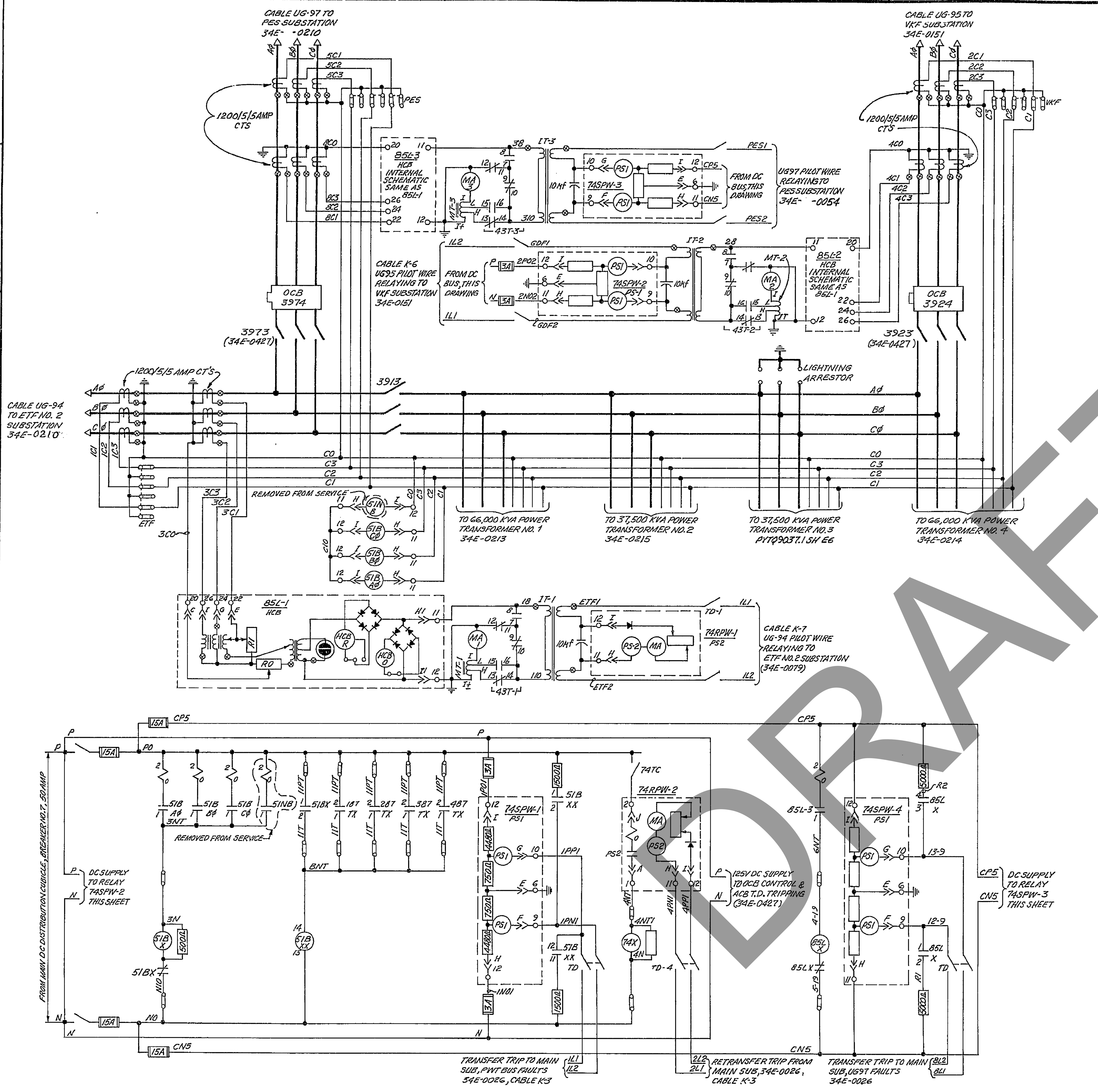
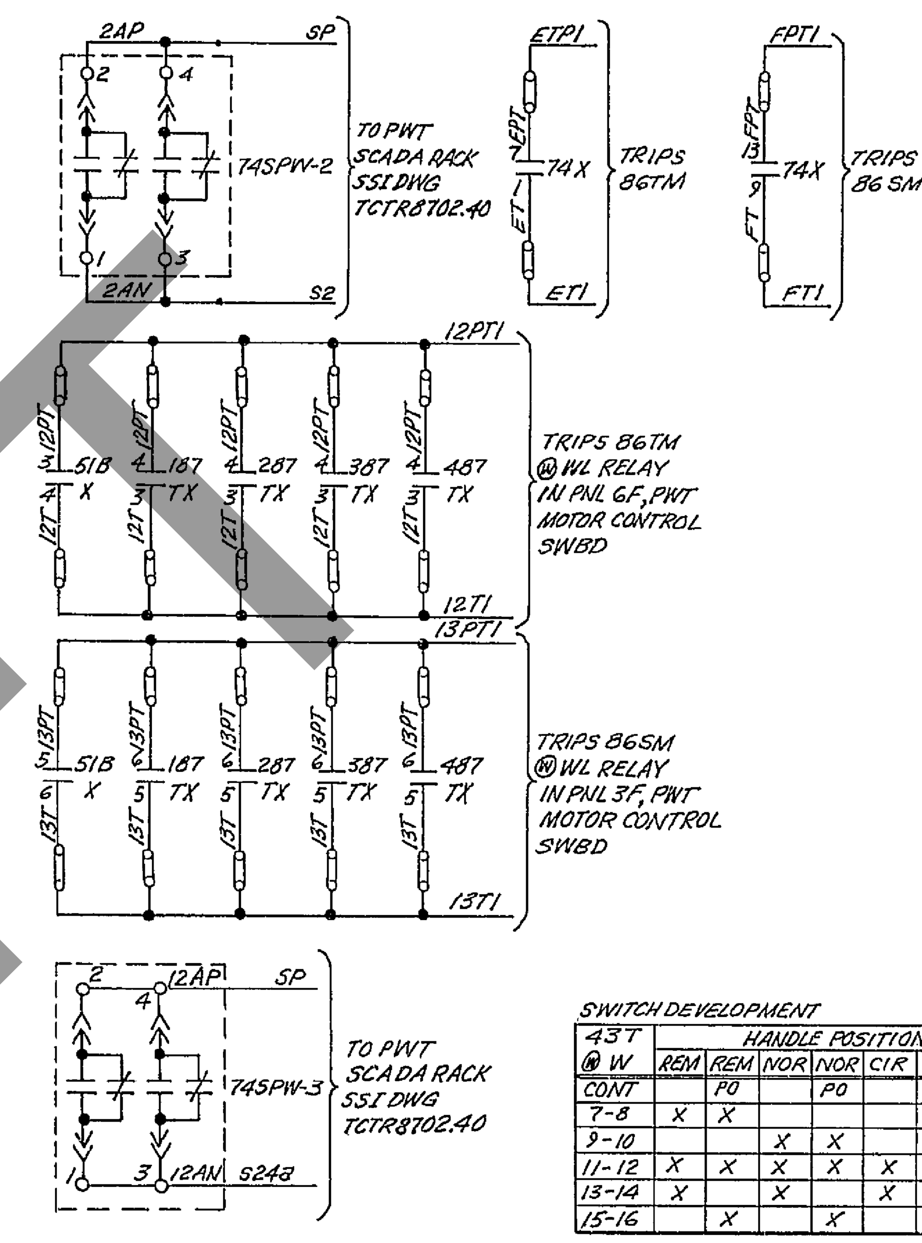
| LETTER | REVISION | DATE | BY | APVD |
|---|---|---|---|---|




| DEVICE NO. | MFG TYPE | DEVICE IDENTIFICATION |
|---|---|---|
| 43T | Ⓦ W | PILOT WIRE AC CURRENT TEST SWITCH |
| 51B | Ⓦ COH | BUS DIFF AC OVERCURRENT RELAY |
| 51NB | Ⓦ COH | BUS DIFF AC GROUND OVERCURRENT RELAY |
| 51BX | Ⓦ WL | AUX LOCKOUT RLY TO DEVICES 51B, 51NB; HAND RESET |
| 51BXX | Ⓖ HFA | AUX RLY TO DEVICES 51BX, 187TX, 287TX, 387TX & 487TX |
| 74RPW | Ⓦ PS2 | PILOT WIRE SUPERVISORY RECEIVING RLY |
| 74SPW | Ⓦ PS1 | PILOT WIRE SUPERVISORY SENDING RLY |
| 74TC | TS1A | RETRANSFER TRIP CUTOFF SWITCH |
| 74X | Ⓦ MG-6 | AUX RELAY TO DEVICE 74RPW-2 |
| 86TM | Ⓦ WL | MASTER LOCKOUT RLY; TRIPS INCOMING TRANSONIC BKRS |
| 86SM | Ⓦ WL | MASTER LOCKOUT RLY; TRIPS INCOMING SUPERSONIC BKRS |
| 85L | Ⓦ HCB | DIFF OVERCURRENT PILOT WIRE RELAY |
| 85LX | Ⓦ WL | AUX LOCKOUT RLY TO DEVICE 85L; HAND RESET |
| 87TX | Ⓦ WL | XFMR FAULT LOCKOUT RLY; HAND RESET |
| | | PREFIX DENOTES XFMR NO. |
| IT | | PILOT WIRE INSULATING XFMR |
| MT | | PILOT WIRE AC CURRENT METERING XFMR |




SWITCH DEVELOPMENT

| 43T Ⓦ W | HANDLE POSITION | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | REM | REM | NOR | NOR | CIR | CIR | LOC | LOC |
| CONT | | PO | | PO | | PO | | PO |
| 7-8 | X | X | | | | | X | X |
| 9-10 | | | X | X | | | X | X |
| 11-12 | X | X | X | X | X | X | | |
| 13-14 | X | | X | | X | | X | |
| 15-16 | | X | | X | | X | | X |

REFERENCE DRAWINGS:

SCHEMATIC DIAGRAM-AC & DC CONTROL
- POWER XFMR NO.1_34E-0213
- POWER XFMR NO.4_34E-0214
- POWER XFMR NO.2_34E-0215
- POWER XFMR NO.3_PYTQ9037.1 SH E6

PYTQ9037.1 SH 6

DRAFT

FIRST ISSUED BY ARO AS 34E-0212. LAST REVISION NO. 4.

PREPARED FOR: US AIR FORCE/AEDC, ARNOLD AFB, TN

PREPARED BY: SSI SERVICES, INC

OUTSTANDING CHANGE ORDERS

PROJECT TITLE: REPLACE PWT NO. 3 TRANSFORMER AND POWER CABLES CECORS NO. 920141-2

DRAWING TITLE: SCHEMATIC DIAGRAM BUS DIFFERENTIAL & TRANSFER TRIPPING PWT 161 KV SUBSTATION

| | NAME | DATE | APVD | NAME | DATE |
|---|---|---|---|---|---|
| DRAWN | DG GATES | | GB13 | B R Mayberry | 7-23-92 |
| CHECKED | TRAchley | 22 JULY 92 | | | |
| DESIGN | W. H. Darnwood | 22 JUL 92 | | | |
| STRESS | N/A | | | | |

| SYSTEM NO | FILE NO | AF PROJ NO | BOM NO | SCALE |
|---|---|---|---|---|
| 02300007 | PYTQ9037 | EP21 | | NTS |

| PLANNER | PHONE | SHOP JOB NO |
|---|---|---|
| JOB CONTACT | PHONE | COST CODE |

| SIZE | DRAWING NO | REV | REF NO | SHEET NO |
|---|---|---|---|---|
| SR | PYTQ9037.1 | A | E-5 | 6 OF 10 |

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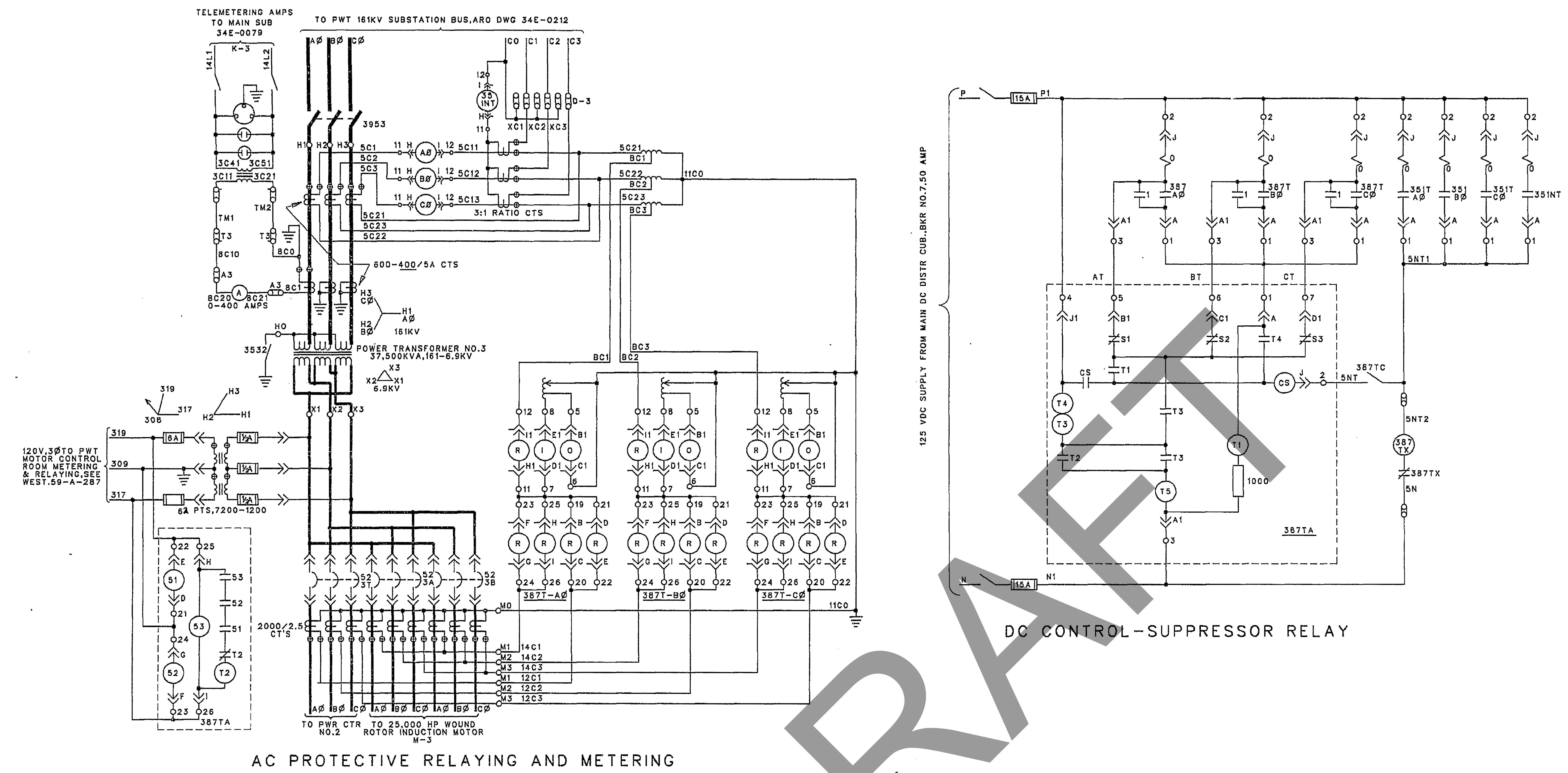
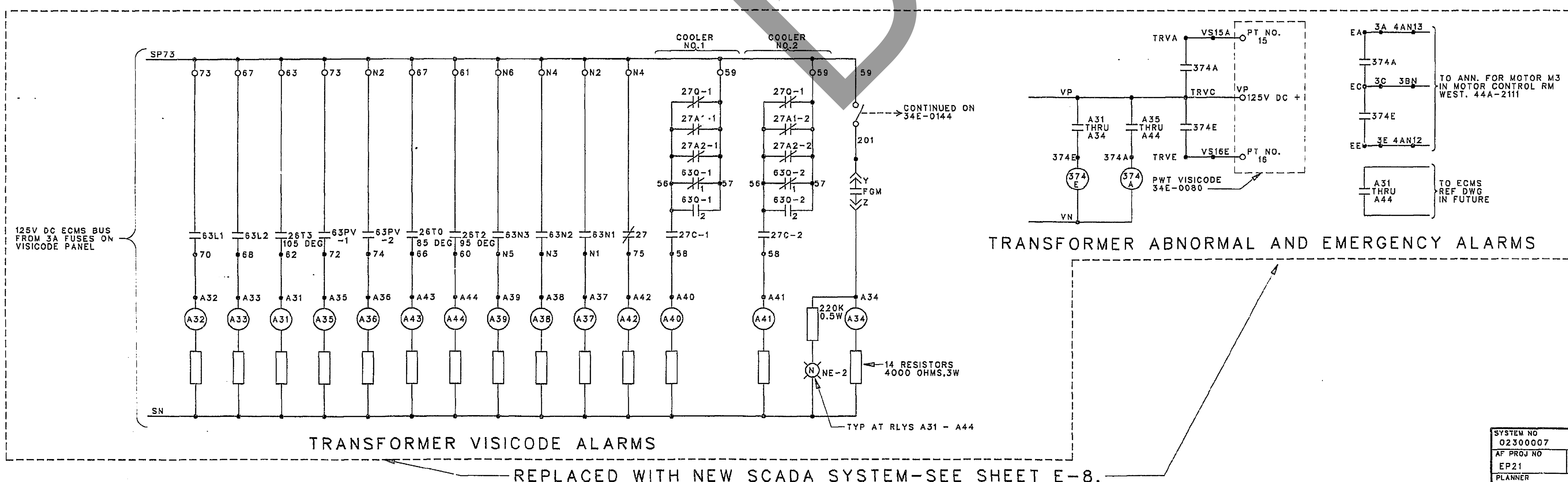



| DEVICE NO | MFRR NO. | DEVICE DISCRIPTION |
|---|---|---|
| 26T3 | TR3 | TEMP RELAY 105 DEG |
| 26T2 | TR2 | TEMP RELAY 95 DEG |
| 26TO | P165 | DIAL TYPE HOT OIL THERMOMETER 85 DEG |
| 27 | W-SG | COOLERS UNDERVOLTAGE ALARM RELAY |
| 27A1 | UV2 | FAN NO.1 UNDERVOLTAGE ALARM RELAY |
| 27A2 | UV4 | FAN NO.2 UNDERVOLTAGE ALARM RELAY |
| 27C | UV1 | COOLER UNDERVOLTAGE ALARM RELAY |
| 27O | UV3 | PUMP UNDERVOLTAGE ALARM RELAY |
| 63L | | MAGNETIC TYPE OIL LEVEL GAUGE |
| 63N-1 | N1-N2 | N GAS PRESS.SW-BOT PRESS.200PSI |
| 63N-2 | N5-N6 | N GAS PRESS.SW-HIGH 5.5PSI |
| 63N-3 | N3-N4 | N GAS PRESS.SW-LOW 0.5PSI |
| 63PV | | XFMR TANK PRESSURE SAFETY VALVE |
| FGM | | FAULT GAS MONITOR (PORTABLE UNIT) |
| 43AT | | ANNUNCIATOR TEST TOGGLE SWITCH |
| A31-A44 | | ALARM RELAYS IN ECMS RELAY RACK |
| 351T | W-CO | XFMR PHASE OVERCURRENT RELAY |
| 351NT | W-CO | XFMR GROUND OVERCURRENT RELAY |
| 374A | W-SG | XFMR ABNORMAL ALARMS RELAY |
| 374E | W-SG | XFMR EMERGENCY ALARMS RELAY |
| 387T | W-CA | XFMR DIFFERENTIAL OVERCURRENT RELAY |
| 387TA | W-TS1 | MAGNETIZING INRUSH TRIPPING SUPR RELAY |
| 387T | | TRIP CUTOFF SWITCH -XFMR DIFF OVERCURRENT |
| 387TX | W-WL | AUX LOCKOUT TO DEVICES |
| NE-42 | GE | NEON LAMP SURGE PROTECTOR |
| KX-642 | W | ARGON LAMP SURGE PROTECTOR |
| 63O | | OIL PRESS. SW-CONT 1 CLOSE ON ZERO PRESS. |
| | | OIL PRESS. SW-CONT 2 CLOSE ON HIGH PRESS. |

REFERENCE DRAWINGS:
SCHEMATIC DIAGRAM:POWER TRANSFORMERS 2 & 3 COOLERS---34E-0144
SCHEMATIC DIAGRAM:BUS DIFF & TRANSFER TRIPPING--PYTQ9037.1,SH E-5
WIRING DIAGRAM:ECMS INTERFACE RELAYS---ARO DWG LYT05570.83,SH 13
WIRING DIAGRAM:XFMR CONTROL CABINENTS---S & P DWG 521-E20
WEST. ELECT. CORP. ----41-A-2364
44-A-2111
59-A-287




REPLACED WITH NEW SCADA SYSTEM-SEE SHEET E-8.

PYTQ9037.1 SH 7

AEDC CAD/CAM SYSTEM

*DO NOT MANUALLY AS BUILT*
THIS FILE CONTROLLED
BY ELECTRONIC MEASURES

FIRST ISSUED BY ARO AS 34E-0216.LAST REVISION NO.6

OUTSTANDING CHANGE ORDERS

PREPARED FOR: US AIR FORCE/AEDC,ARNOLD AFB,TN
PREPARED BY: SSI SERVICES, INC.

PROJECT TITLE: REPLACE PWT NO. 3 TRANSFORMER AND POWER CABLES CECORS NO. 920141-2

DRAWING TITLE: SCHEMATIC DIAGRAM-AC & DC CONTROL PWT 161KV SUBSTATION

| SYSTEM NO | FILE NO | | |
|---|---|---|---|
| 02300007 | PYTQ9037 | | |
| AF PROJ NO: EP21 | BOM NO | SCALE: NTS | |
| PLANNER | PHONE | SHOP JOB NO | |
| JOB CONTACT | PHONE | COST CODE | |

| | NAME | DATE | APVD | NAME | DATE |
|---|---|---|---|---|---|
| DRAWN | DG GATES | 18FEB92 | GB13 | BK Mayberry | 7-23-92 |
| CHECKED | TPAchler | 22 JULY 92 | | | |
| DESIGN | W.H. Dainwood | 22 JUL 92 | | | |
| STRESS | N/A | | | | |

| SIZE | DRAWING NO | REV | REF NO | SHEET NO |
|---|---|---|---|---|
| SR | PYTQ9037.1 | A | E-6 | 7 OF 10 |

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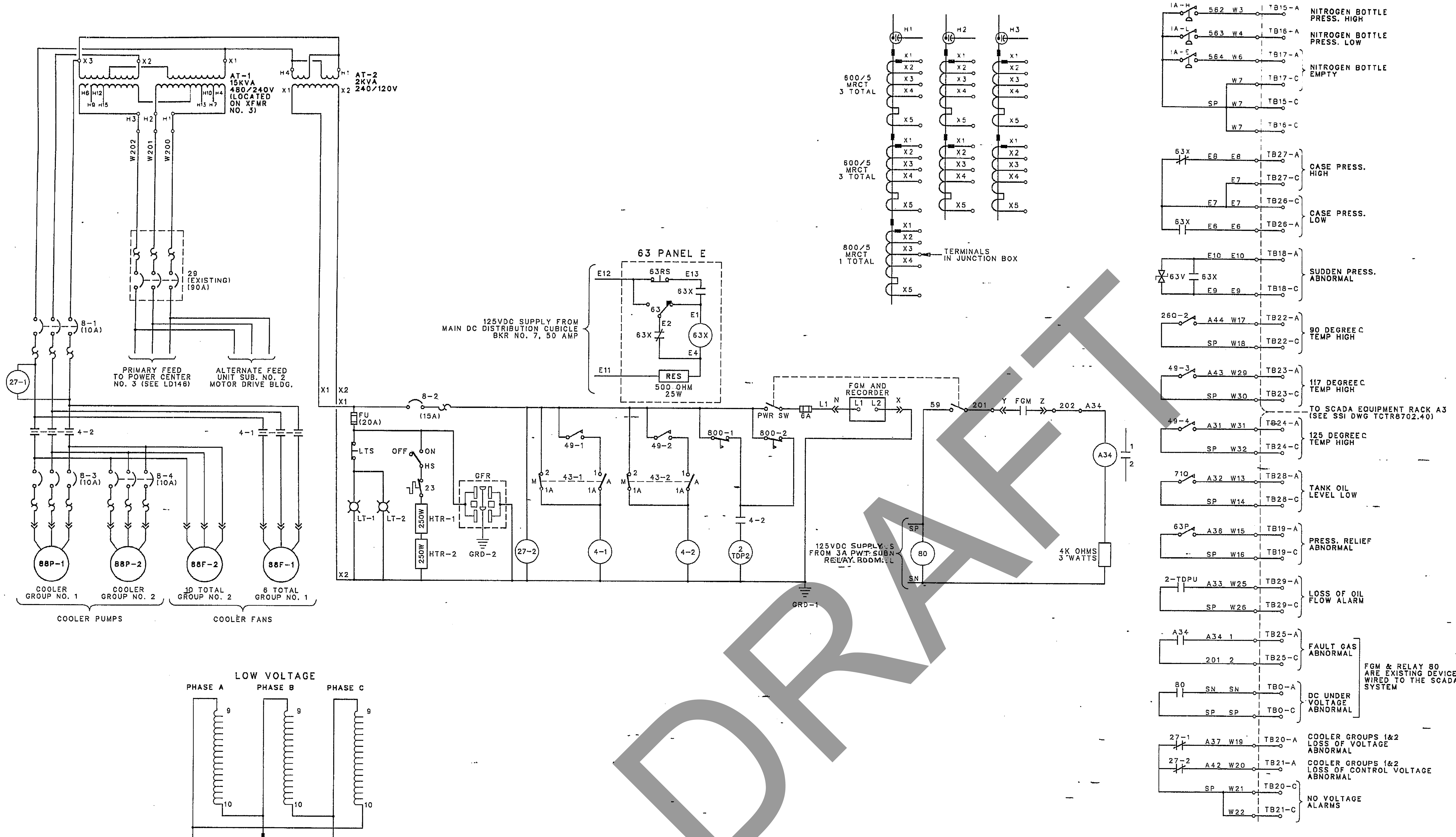


| 43-1 AND 43-2 SWITCHES | | | |
|---|---|---|---|
| CONTACTS | | POSITIONS | |
| | | MAN | AUTO |
| 1 —2 | 1 | | X |
| | 2 | X | |
| 3 —4 | 3 | | X |
| | 4 | X | |

X = CLOSED CONTACT

NOTES:

1. FOR REFERENCE DRAWINGS: SEE SHEET E-9.

DEVICE LIST (APPARATUS SHOWN DE-ENERGIZED)

- CBA ——— CURRENT BALANCING AUTOTRANSFORMER
- AT-1 ——— AUX TRANSFORMER (480/240V)
- AT-2 ——— AUX TRANSFORMER (240/120V)
- FU ——— FUSE
- GFR ——— GROUND FAULT RECEPTACLE (NEMA 5-20R)
- HS ——— HEATER SWITCH
- IA ——— INERTAIRE ALARM
- IA-E ——— EMPTY CYLINDER
- IA-H ——— HIGH PRESSURE
- IA-L ——— LOW PRESSURE
- LT-1&2 ——— COMPARTMENT LIGHTS
- LTS ——— COMPARTMENT LIGHT SWITCH
- 2-TDPU ——— OIL FLOW TIME DELAY RELAY TIME DELAY PICK UP.
- 4-1&2 ——— COOLER CONTACTORS
- 8-1&2 ——— CIRCUIT BREAKERS
- 23 ——— HEATER THERMOSTAT
- 26Q ——— HOT OIL THERMOMETER
- 26Q-1 ——— FAN AND PUMP ALARM CONTACT (60°C) NOT USED
- 26Q-2 ——— ALARM CONTACT (90°C)
- 27-1&2 ——— LOSS-OF-VOLTAGE RELAYS
- 29 ——— MAIN CIRCUIT BREAKER (EXISTING)
- 43-1&2 ——— COOLER CONTROL SWITCHES (GE TYPE SB-1)
- 49 ——— THERMAL DEVICE
- 49-1 ——— FAN CONTACT (70°C)
- 49-2 ——— FAN CONTACT (75°C)
- 49-3 ——— ALARM CONTACT (117°C)
- 49-4 ——— ALARM CONTACT (125°C)
- 49-H ——— HEATER ELEMENT
- 49TS ——— THERMAL DEVICE CURRENT TEST SWITCH
- 63 ——— SUDDEN PRESSURE RELAY
- 63RS ——— RESET SWITCH
- 63V ——— SURGE SURPRESSOR
- 63X ——— SEAL-IN RELAY
- 63P ——— MECHANICAL RELIEF DEVICE
- 71Q ——— LIQUID LEVEL GAUGE
- 80Q-1&2 ——— OIL FLOW GAUGES
- 88F-1&2 ——— FAN MOTORS WITH THERMOGUARD
- 88P-1&2 ——— PUMP MOTORS
- FGM ——— FAULT GAS MONITOR (PORTABLE UNIT)
- 80 ——— UNDERVOLTAGE DC ALARM RELAY
- 43TI ——— TERMPERATURE INDICATOR SELECTOR SWITCH
- HTR-1&2 ——— COMPARTMENT HEATER
- RTD ——— RESISTANCE TEMPERATURE DETECTOR FOR HOT SPOT.
- RTD-H ——— HEATER ELEMENT
- RTD-RC ——— RESISTIVE ELEMENT

LEGEND:

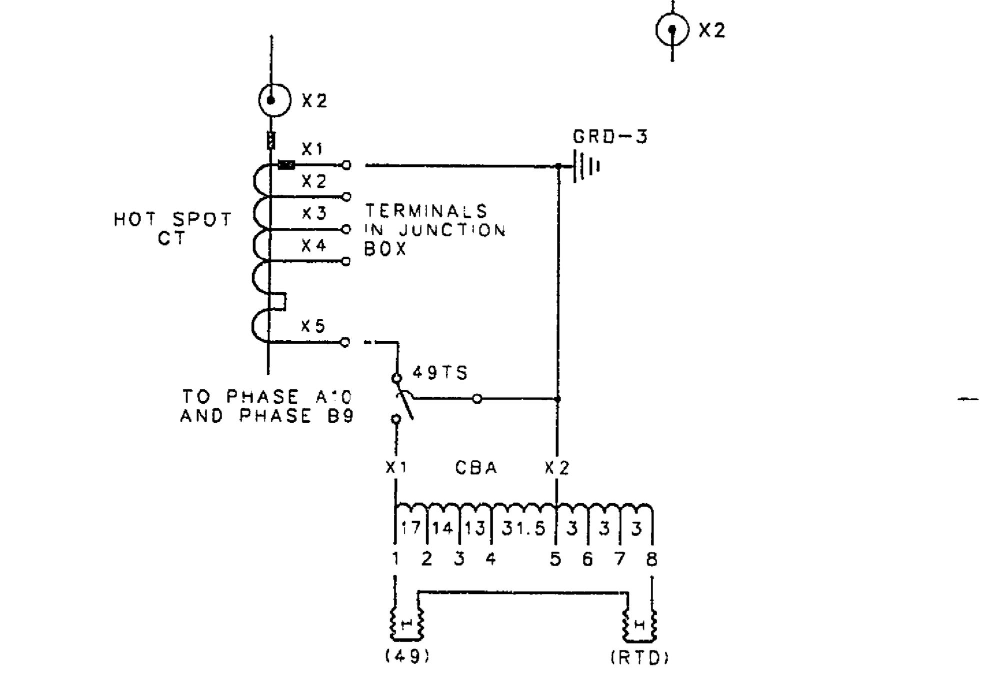
- CAPACITOR
- NORMALLY OPEN RELAY CONTACT
- NORMALLY CLOSED RELAY CONTACT
- THERMAL TRIP ELEMENT
- CIRCUIT BREAKER
- DISCONNECT DEVICE
- OPERATING COIL
- LIGHT
- MOMENTARY CONTACT SWITCH
- MAINTAINED CONTACT SWITCH
- PUSHBUTTON SWITCH
- FUSE
- THERMOSTAT
- THERMAL DEVICE TEST SWITCH

| LETTER | REVISION | DATE | BY | APVD |
|---|---|---|---|---|
| | | | | |

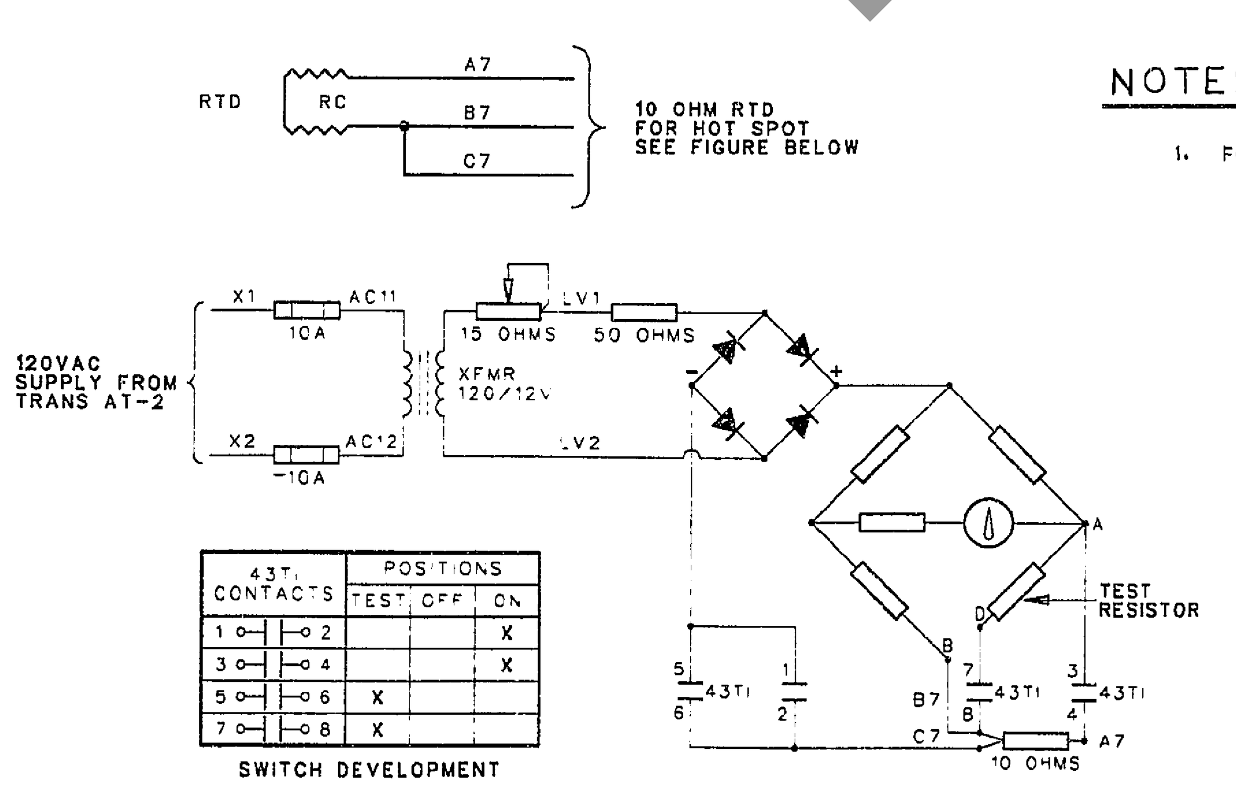
PYTQ9037.1 SH 9

AEDC CAD/CAM SYSTEM

*DO NOT MANUALLY AS BUILT* THIS FILE CONTROLLED BY ELECTRONIC MEASURES



## XFMR INTERNAL OIL AND RTD PROTECTION



| 43TI CONTACTS | POSITIONS | | |
|---|---|---|---|
| | TEST | OFF | ON |
| 1 — 2 | | | X |
| 3 — 4 | | | X |
| 5 — 6 | X | | |
| 7 — 8 | X | | |

SWITCH DEVELOPMENT

## RESISTANCE TEMPERATURE DETECTOR

| 600/5 MRCT | |
|---|---|
| NOMINAL RATIO | LEADS ON |
| 50 TO 5 | X2-X3 |
| 100 TO 5 | X1-X2 |
| 150 TO 5 | X1-X3 |
| 200 TO 5 | X4-X5 |
| 250 TO 5 | X3-X4 |
| 300 TO 5 | X2-X4 |
| 400 TO 5 | X1-X4 |
| 450 TO 5 | X3-X5 |
| 500 TO 5 | X2-X5 |
| 600 TO 5 | X1-X5 |

| 800/5 MRCT | |
|---|---|
| NOMINAL RATIO | LEADS ON |
| 100 TO 5 | X1-X2 |
| 200 TO 5 | X3-X4 |
| 300 TO 5 | X2-X3 |
| 400 TO 5 | X1-X3 |
| 500 TO 5 | X2-X4 |
| 600 TO 5 | X1-X4 |
| 700 TO 5 | X2-X5 |
| 800 TO 5 | X1-X5 |

| HOT SPOT CT'S | |
|---|---|
| NOMINAL RATIO | LEADS ON |
| 500 TO 1 | X2-X3 |
| 550 TO 1 | X1-X3 |
| 605 TO 1 | X2-X4 |
| 655 TO 1 | X1-X4 |
| 715 TO 1 | X2-X5 |
| 765 TO 1 | X1-X5 |

## CT MULTI-RATIO CONNECTIONS

(FOR WINDING CONNECTIONS SEE NAMEPLATE DIAGRAM ABB DWG MNL5845-10)

OUTSTANDING CHANGE ORDERS

PREPARED FOR US AIR FORCE/AEDC, ARNOLD AFB, TN — PREPARED BY SSI SERVICES, INC.

PROJECT TITLE: REPLACE PWT NO. 3 TRANSFORMER AND POWER CABLES

CECORS NO. 920141-2

DRAWING TITLE: PWT TRANSFORMER #3, 161KV SUBSTATION COOLERS & TEMP INDICATION, SCADA INTERFACE SCHEMATIC DIAGRAM

| SYSTEM NO | FILE NO | | NAME | DATE | APVD | NAME | DATE |
|---|---|---|---|---|---|---|---|
| 02300007 | PYTQ9037 | DRAWN | JD MCFADDEN | 22MAY92 | | | 7-23-92 |
| AF PROJ NO EP21 | BOM NO | CHECKED | | 22 JULY 92 | | | |
| SCALE NTS | | DESIGN | W.H. | 22 JUL 92 | | | |
| PLANNER | PHONE | SHOP JOB NO | STRESS | N/A | | | |
| JOB CONTACT | PHONE | COST CODE | | | | | |

SIZE: SR — DRAWING NO: PYTQ9037.1 — REV: A — REF NO: E-8 — SHEET NO: 9 of 10

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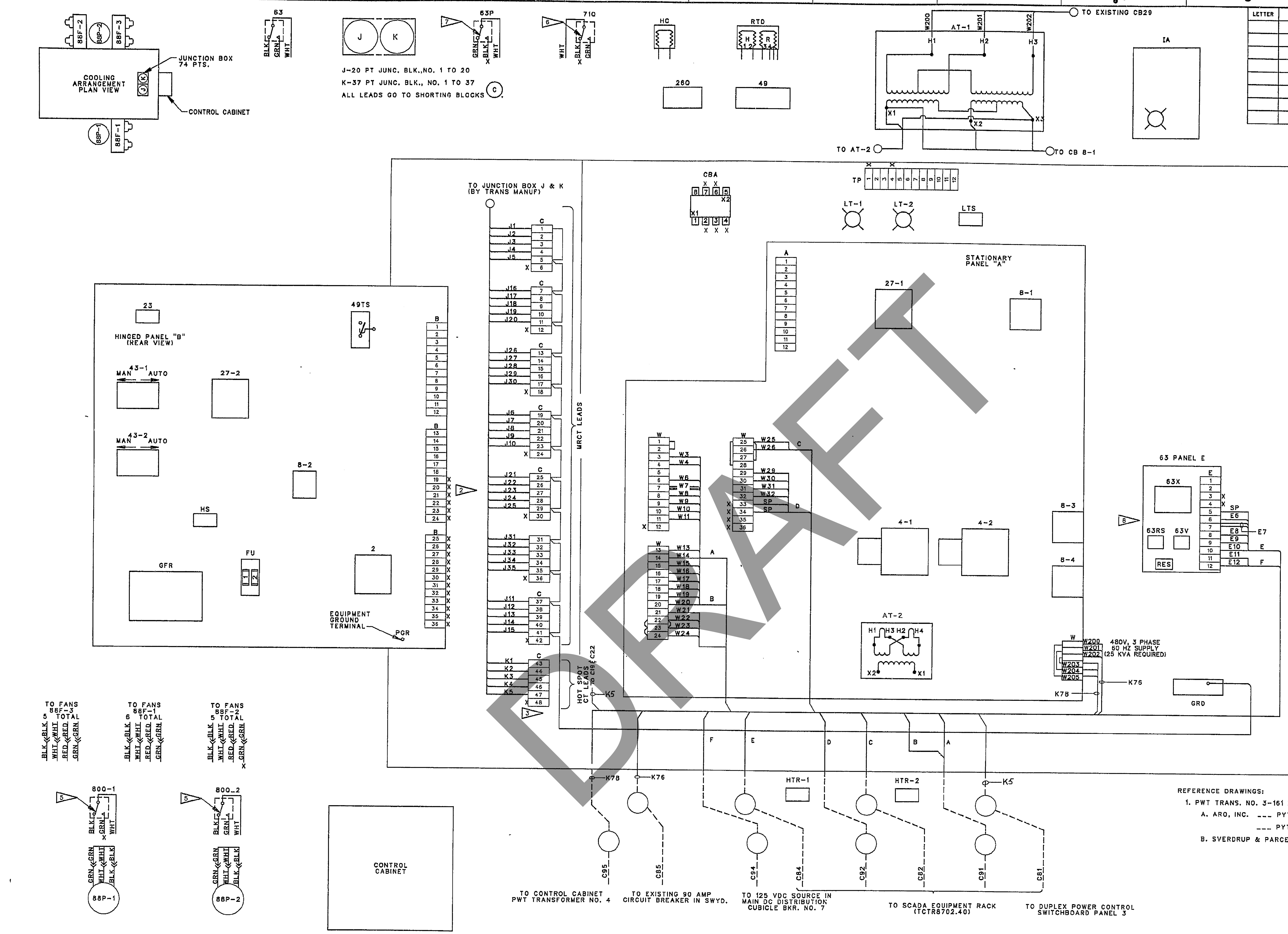
16332-24-001

| LETTER | REVISION | DATE | BY | APVD |
| --- | --- | --- | --- | --- |
| | | | | |




REFERENCE DRAWINGS:

1. PWT TRANS. NO. 3-161 KV SUBSTATION
   - A. ARO, INC. --- PYTQ9037.1 SH. E-5
   - --- PYTQ9037.1 SH. E-6
   - B. SVERDRUP & PARCEL, INC. --- 521-E3, 521-E4, 521-E5, 521-E10, 521-E20, 521-E22, 521-E26, 521-E27

PYTQ9037.1 SH 10

AEDC CAD/CAM SYSTEM

*DO NOT MANUALLY AS BUILT* THIS FILE CONTROLLED BY ELECTRONIC MEASURES

## NOTES:

1. FOR DESCRIPTION OF DEVICE NUMBER SEE SHEET E-8.
2. AN "X" PLACED AT A DEVICE TERMINAL INDICATES AN UNUSED TERMINAL.
3. SHORTING BLOCKS
4. STATIONARY PANEL SHOWN WITH CABINET DOOR OPEN
5. CONTACT CLOSES ON NO OIL.
6. CONTACT CLOSES ON LOW OIL.
7. CONTACT MUST BE MANUALLY RESET AFTER TRIPPING.
8. ALL DEVICES ON THIS DRAWING ARE PREWIRED BY THE TRANSFORMER MANUFACTURER.

OUTSTANDING CHANGE ORDERS

PREPARED FOR US AIR FORCE/AEDC, ARNOLD AFB, TN — PREPARED BY SSI SERVICES, INC.

PROJECT TITLE: REPLACE PWT NO.3 TRANSFORMER AND POWER CABLES

CECORS NO. 920141-2

DRAWING TITLE: PWT TRANSFORMER NO.3, 161-KV SUBSTATION CONTROL CABINET CONNECTION WIRING DIAGRAM

| SYSTEM NO | FILE NO | | | NAME | DATE | APVD | NAME | DATE |
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 02300007 | PYTQ9037 | | DRAWN | VL VERTREES | 15JUN92 | | | 7-23-92 |
| AF PROJ NO: EP21 | BOM NO | SCALE: NTS | CHECKED | | 22 JUL 92 | | | |
| PLANNER | PHONE | SHOP JOB NO | DESIGN | W.H. Burwood | 22 Jul 92 | | | |
| JOB CONTACT | PHONE | COST CODE | STRESS | N/A | | | | |

| SIZE | DRAWING NO | REV | REF NO | SHEET NO |
| --- | --- | --- | --- | --- |
| SR | PYTQ9037.1 | A | E-9 | 10 OF 10 |

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APPENDIX F
NAMEPLATE DATA

DRAFT

END OF APPENDIX F

EQID 008084

Page 70

PWT #3

ABB
ASEA BROWN BOVERI

ABB POWER T & D COMPANY INC.

THREE PHASE
60 HERTZ
TYPE SL
TRANSFORMER
CLASS OA / FA / FOA
INSULDUR INSULATION

WINDING
161000 GRD.Y/92953 VOLTS
6900 VOLTS

65° C. AVG. RISE
37500/50000/62500 KVA
37500/50000/62500 KVA

L-SPEC MNL5845-08
WIRING DIAGRAM MNL5845-04

SERIAL MNL5845-01

GALLONS OIL, TRANS. TANK 6845

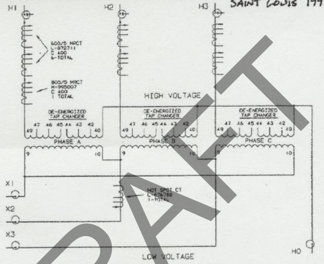
IMPEDANCE 9.09 % AT 37500 KVA, 161000 TO 6900 VOLTS

FULL WAVE IMPULSE TEST LEVEL: HIGH VOLTAGE 650 KV. LOW VOLTAGE 110 KV.
HIGH VOLTAGE NEUTRAL AND BUSHING 250 KV.

APPROX. WEIGHT IN LBS. CORE AND COILS 73,200 CASE 40,300 OIL 51,338 TOTAL 164,838

CAUTION : DO NOT ATTEMPT TO HANDLE, INSTALL, USE OR SERVICE THIS TRANSFORMER BEFORE READING INSTRUCTION BOOK MNL5845-12. TO DO SO MAY LEAD TO BODILY INJURY OR PROPERTY DAMAGE OR BOTH.

SAINT LOUIS 1992




**CONNECTIONS**

| WINDING | VOLTAGE | AMPERES AT 62500 KVA | DE-ENERGIZED TAP CHANGER POS. | CONNECTS IN EACH PHASE |
|---|---|---|---|---|
| H.V. WYE | 169050 | 213 | 1 | 44 TO 45 |
| | 165025 | 219 | 2 | 43 TO 45 |
| | 161000 | 224 | 3 | 43 TO 46 |
| | 156975 | 230 | 4 | 42 TO 46 |
| | 152950 | 236 | 5 | 42 TO 47 |
| X.V. DELTA | 6900 | 5230 | | |

THE TRANSFORMER IS DESIGNED FOR OPERATION BETWEEN PRESSURE LIMITS OF 5.0 POUNDS PER SQUARE INCH POSITIVE AND 0.5 POUNDS PER SQUARE INCH POSITIVE.

THE HIGH VOLTAGE WINDING NEUTRAL MUST BE PERMANENTLY GROUNDED EITHER DIRECTLY OR THROUGH A LOW IMPEDANCE. IF AN IMPEDANCE IS PLACED IN THE GROUND CIRCUIT, THE VOLTAGE FROM NEUTRAL TO GROUND DURING A FAULT MUST NOT EXCEED 46000 VOLTS.

THE TRANSFORMER MUST NOT BE ENERGIZED FROM ANY VOLTAGE SOURCE WHEN DE-ENERGIZED TAP CHANGERS ARE OPERATED.

THE 25°C LIQUID LEVEL IS 13.50 INCHES BELOW TOP OF HIGHEST MANHOLE FLANGE. LIQUID LEVEL CHANGES 0.950 INCHES FOR EACH 10°C CHANGE IN AVERAGE LIQUID TEMPERATURE.

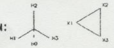
UNTANKING WEIGHT (HEAVIEST PIECE) 72600 LBS.

THE TRANSFORMER TANK IS DESIGNED TO WITHSTAND COMPLETE VACUUM AND AN INTERNAL PRESSURE OF 10 POUNDS PER SQUARE INCH.

CONDUCTOR MATERIALS: H.V. CU., L.V. CU.

* .30% OF DOPC.

PHASOR DIAGRAM:




MADE IN U.S.A. NP# MNL5845-10 SUB B

.032 STAINLESS STEEL #6240-SATIN FINISH-ETCHED-FILLED WITH BLACK ENAMEL.
REDUCE TO 7-1/2 IN. X 15 IN. 1/4 IN. HOLE (4 TOTAL)
PROPORTION 2:1 14-7/16 IN. BETWEEN CENTERS OF HOLES ON LONG EDGE OF PLATE.
ASSEMB. SEC. ST. LOUIS 6-15/16 IN. BETWEEN CENTERS OF HOLES ON SHORT EDGE OF PLATE.

SUB B
N.P. DRAFT. M. KOESTER DATE 1-22-92
CHKD. JOE BEES 1-22-92
ENGINEER C. BAKER 1-22-92

ABB POWER T&D COMPANY INC.
APPARATUS TRANSFORMER INSTRUCTION NAMEPLATE
DIMENSIONS IN INCHES-SCALE NTS SUB B
MNL5845-10
POWER TRANSFORMER DIVISION MUNCIE, IN./ ST. LOUIS, MO


Photo 1 – PWT 3 Transformer PWT Substation

DRAFT

APPENDIX H
SHE STANDARDS

Attached Separately

DRAFT

END OF APPENDIX H AND SOW